QUALIFICATION STANDARD
AND PLAN (QSP)

CSQ - Naval Warfare Officer (NWO) II

MITE ID

NWO II:

NWO II Mod 1 - Academic - 123898
NWO II Mod 1 - NTO Academic - 123935
NETP Mod 2 - Perform as Member of the FP Team - 118532
NETP Mod 3 - Perform as Member of the DCO - 118533
NETP Mod 4 - Carry Out Survival Drills - 118534
NWO II Mod 5 – Sea Environment Training (OJT) - 123899

Regular Force: 00207-05 NWO, 00213-01 INT, 00328-04 LOG, 00344-01 CSEO, 00345-01 MSEO
Primary Reserve: 00207-13 NWO, 00213-09 INT, 00328-06 LOG
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): Commanding Officer *HMCS VENTURE*

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and / or L.

Revision	Change Description	Date	Authority
1.0	Established separate NWO II QSP		QS – CNTG TP – CO HMCS VENTURE

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

TABLE OF CONTENTS

RECORD OF AMENDMENTS……2
TABLE OF CONTENTS……..3
FOREWORD……6
QUALIFICATION STANDARD……7
CHAPTER 1 – QUALIFICATION STANDARD……1
AIM……...1
GRANTING OF NQUAL……1
QUALIFICATION……1
PREREQUISITES……1
CHAPTER 2 – PERFORMANCE OBJECTIVES……2
PO 001 – Execute Administrative and Leadership Duties of a Divisional Officer……..2
PO 002 – Conduct Navigation……...4
PO 003 – Apply Regulations for Preventing Collisions at Sea……6
PO 004 – Conduct Shiphandling and Seamanship……8
PO 005 – Perform Bridge Watch Administration……...10
PO 006 – Direct the Operation of Onboard Engineering Systems……..11
PO 007 – Employ Naval Communications……12
PO 008 – Direct Action and Emergency Procedures……..13
PO 009 – Support Operations Other Than Warfare……14
PO 010 – Support Air Operations……...16
TRAINING PLAN……..17
CHAPTER 3 – TRAINING STRATEGY……1
TRAINING DURATION……1
REQUIRED COURSE STAFFING……1
TRAINING LIMITATION……3
COURSE CAPACITY……..3
DESIGNATED SCHEDULING AND COURSE LOADING……3
QUALIFICATION AND ASSESSMENT……....3
UNSATISFACTORY PROGRESS……..4
REMOVAL FROM TRAINING……...5
TRAINING REVIEW BOARD……5
RECERTIFICATION REQUIREMENTS……6
LANGUAGE OF INSTRUCTION……...6
PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)……6
POST COURSE REVIEWS (PCR)……...6
COURSE REPORTING……6
CHAPTER 4 - LESSON SPECIFICATIONS……...7
PO 001 - Execute Administrative and Leadership Duties of a Divisional Officer……7
EO 001.01 - Perform Naval Drill and Ceremonial Procedures……7
EO 001.02 - Apply Practical Leadership Skills……...10
EO 001.03 - Manage People……19
EO 001.04 - Describe the Functional Organization……21
EO 001.05 - Describe the Managerial Environment of the RCN……23
EO 001.06 - Draft RCN Correspondence……25

PO 002 - Conduct Navigation…….27
EO 002.01 - Explain the Effects of Meteorological Conditions on Shiphandling……..27
EO 002.02 - Apply Nautical Measurements……30
EO 002.03 - Identify Aids to Navigation…….33
EO 002.04 - Calculate Time and Time Zones…….37
EO 002.05 - Employ Mental Math Techniques……...40
EO 002.06 - Identify Features of Nautical Charts……...43
EO 002.07 - Conduct Visual Fix and Dead Reckoning Techniques……...47
PO 003 - Apply Regulations for Preventing Collision at Sea…….54
EO 003.01– Apply the Regulations for Preventing Collisions at Sea…….54
EO 003.02 – Apply Rules on Lights, Shapes, Sound Signals Under Rules 24-26……..59
EO 003.03 –Apply Regs for Vessels with Restricted Maneuverability Under Rules 27-31…...62
EO 003.04 – Identify and Interpret Signals Used by Vessels in Sight of One Another IAW 32-37:……65
EO 003.05 – Apply Technical Details for Lights, Shapes, and Distress Signals in Annex II & IV…….68
PO 004 - Conduct Shiphandling and Seamanship……..71
EO 004.01 - Perform Basic Rope Work…….71
EO 004.02 - Identify Upper Deck Fittings…….78
EO 004.03 - Con the Ship……..80
EO 004.04 - Act as Part Ship Officer in Seamanship Evolutions……..83
EO 004.05 - Act as OOW for Berthing and Slipping…….85
EO 004.06 - Perform the Duties of a Small Boat Coxswain……..90
PO 005 - Perform Bridge Watch Administration……94
EO 005.01 – Describe the duties and responsibilities of the Watch on Deck…….94
EO 005.02 - Implement Ship’s Routines……98
PO 006 - Direct the Operation of Onboard Engineering Systems……101
EO 006.01 - Describe the Key Components of Marine and Combat Systems Engineering….101
PO 007 - Use Naval Communications……..105
EO 007.01 - Use Flags and Pennants……105
PO 008 - Direct Action and Emergency Procedures…….108
EO 008.01 - Describe the Action and Emergency Organization……..108
EO 008.02 - Act as OOW in Response to Emergencies……110
EO 008.03 - Explain the Ship’s Fighting Organization…….113
PO 009 - Support Operations other than Warfare…….116
EO 009.01 - Describe Key Events in the History of the RCN……..116
EO 009.02 - Practice Customs and Traditions of the RCN……...125
EO 009.03 - Apply Naval Terminology……127
EO 009.04 - Conduct Ceremonial Procedures……...131
EO 009.05 - Employ Personal Safety Procedures……135
EO 009.06 - Explain Naval Doctrine ……...139
EO 009.07 - Identify General Ship Types……141
EO 009.08 - Identify RCN Ships and Supporting Aircraft……144
PO 0010 - Support Air Operations……147
EO 010.01 - Describe Basic Safety Procedures for Shipbourne Air Operations……...147
QUALIFICATION STANDARD ANNEXES……149

ANNEX A – MAIN REFERENCES……1
ANNEX B – TASK LIST……1
ANNEX C – QS SCALAR……...1
ANNEX D – RECORD OF DECISIONS QSWB……1
RECORD OF AMENDMENT & AMPLIFICATION……4
TRAINING PLAN ANNEXES……5
ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES……6
ANNEX F – COSTING INFORMATION……1
ANNEX G – SUGGESTED SEQUENCING AND TIMINGS……1
OPTIMUM TRAINING SCHEDULE – NWO II Mod 1……2
OPTIMUM TRAINING SCHEDULE – NWO II Mod 1 – NTO……7
ANNEX H – ASSESSMENT PLAN……....1
ANNEX I – ON JOB PERFORMANCE RECORD (OJPR)……1
ANNEX J – TT PLAN……..1
ANNEX K – RECORD OF DECISIONS TPWB……1
RECORD OF AMENDMENT & AMPLIFICATION……3
ANNEX L – ACRONYMS AND ABBREVIATIONS……1
ANNEX M – ACADEMIC CAUTION……1
ANNEX N – PERFORMANCE REGRESSION……..1
ANNEX O – PROFESSIONAL LEADERSHIP QUALITIES (PLQ) CHIT……...1
ANNEX P – NAVIGATION AND BRIDGE SIMULATOR HANDBOOK……...1

FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the Officer General Specification (2020-11-20) and the Occupational Specification, Naval Warfare Officer - MOS ID 00207, published 2018-02-05.

3. This QSP publication is effective on receipt and supersedes any previous QSP. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened December 2023 at HMCS VENTURE working in collaboration with Subject Matter Experts (SME) drawn from participating organizations. The subsequent Training Plan (TP) at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened at HMCS VENTURE working in collaboration with SMEs drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A_P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NTORD 4500-XX is required.

QUALIFICATION STANDARD

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. As Commander of Naval Training Group, it is my intention for the NWO II QSP and training to serve as initial occupational training for Naval Warfare Officers as well as Initial Environmental Training (IET) for Naval Officer occupations.

2. The overall aim of the formal in-house portion of this training is to produce an officer with the requisite IET to progress their training in a safe and professional manner and provide the foundational introduction to Naval Warfare Officer training. When combined with NWO III and IV, and practical consolidation in the fleet, the NWO will have the skills to achieve the initial qualification levels defined in the occupational specification: Officer of the Day qualification (OOD), Bridge Watchkeeping (BWK) qualification and Naval Warfare Officer Professional Qualification (NWOPQ).

3. The NWO II course is intended as the first incremental training step and is not intended to produce an occupationally functional graduate. This course will impart the fundamental knowledge, skills, and attitudes required for a junior officer to progress to their next phase of training and safely serve as a member of a ship's crew.

GRANTING OF NQUAL

4. A trainee will achieve the qualification XXXX when the standards listed in the Performance Objectives (POs) have been met and the following MITE codes have been achieved:

NWO II: 123898, 118532, 118533, 118534, 123899

QUALIFICATION

5. This QSP is combined with the NETP QSP to ensure junior officers have the required skills to operate safely in the sea environment. The chart below indicates the required training for junior officers, based on their occupation, to complete initial environmental training for their respective occupation:

Required training by occupation:		
NWO	**NTO**	**Log O/Int O**
NWO II Mod 1 (123898)	NWO II Mod 1 NTO (123935)	NETP Mod 1 (118531)
NETP Mod 2 (118532)	NETP Mod 2 (118532)	NETP Mod 2 (118532)
NETP Mod 3 (118533)	NETP Mod 3 (118533)	NETP Mod 3 (118533)
NETP Mod 4 (118534)	NETP Mod 4 (118534)	NETP Mod 4 (118534)
NWO II Mod 5 – OJT (123899)	At-sea training completed in other NTO training.	NWO II Mod 5 – OJT (123899)

PREREQUISITES

6. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. Completion of BMOQ;

b. Meet the minimum operational standards IAW DAOD 5023-1; and

c. Successful completion of the FORCE Evaluation IAW DAOD 5023-2.

7. Modules 1-4 are pre-requisites for Module 5. All other modules may be completed out of sequence if required, though the preference is to complete them in order of sequence.

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001 – Execute Administrative and Leadership Duties of a Divisional Officer

1. Performance: Execute administrative and leadership duties of a Divisional Officer.

2. Conditions:

    a. Given:

        (1) a division within a ship's company;

        (2) support, to include:

            (a) Divisional PO / CPO;

            (b) Executive Officer / Department head; and

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (1) on board a ship at sea or alongside;

        (2) any time of day and night; and

        (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall execute the administrative and leadership duties of a divisional officer, IAW refs, to include:

    a. performing naval drill and ceremonial procedures;

    b. applying practical leadership skills;

    c. managing people;

    d. describing the functional organization;

    e. describing the managerial organization of the RCN; and

    f. drafting RCN correspondence.

4. References: A1, A3, A5, A23, A24, A28, A30, A35, A43, A48, A52, A53, A63, A78, B45, C6, C37.

5. Task Numbers: See Annex C.

6. Training Limitations:

7. Remarks : Nil

PO 002 – Conduct Navigation

1. Performance: Conduct navigation.

2. Conditions:

    a. Given:

        (1) ship at sea or alongside;

        (2) support, to include:

            (a) bridge team;

            (b) on-watch personnel; and

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (4) on board a ship at sea or alongside;

        (5) any time of day and night; and

        (6) any conditions of weather and visibility.

3. Standard: The Naval Officer shall conduct navigation, IAW references, by:

    a. explaining the effects of varying meteorological conditions;

    b. applying nautical measurements;

    c. identifying aids to navigation;

    d. calculating time and time zones;

    e. interpreting tide and current data;

    f. employing mental math techniques;

    g. identifying features of nautical charts; and

    h. conducting visual fix and dead-reckoning techniques.

4. References: A16, A17, B7, B24, B25, B36, B37, B39, B43, B46, B54, B55

5. Task Numbers: See Annex C.

6. Training Limitations: Nil.

7. Remarks:

   a. This is an introduction to navigation planning and presents the foundational knowledge for students to use in NWO III and NWO IV when panning coastal and pilotage navigation passages.

   b. This performance objective is not delivered during the NTO abbreviated academic phase.

PO 003 – Apply Regulations for Preventing Collisions at Sea

1. Performance: Apply regulations for preventing collisions at sea.

2. Conditions:

    a. Given:

        (1) ship at sea or alongside;

        (2) support, to include:

            (a) bridge team; and

            (b) on-watch personnel.

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (1) on board a ship at sea or alongside;

        (2) any time of day and night; and

        (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall apply regulations for preventing collisions at sea, IAW references, by:

    a. defining key terminology and core concepts outlines in Rules 3 & 20-23;

    b. applying rules on lights, shapes, sound signals, and restricted visibility under 24-26;

    c. applying regulations for vessels with restricted maneuverability under Rules 27-31;

    d. identifying and interpreting signals used by vessels in sight of one another IAW with Rules 32-37; and

    e. applying technical details for lights, shapes, and distress signals in Annex II & IV.

4. References: A7, A16, A65, B23, B27, B35, B36, B38, B39, B57, C2, C3, C13, C17, C18, C19, C30, C32, C35, C39, C45, D1.

5. Task Numbers: See Annex

6. Training Limitations: Nil.

7. Remarks: This performance objective is not delivered during the NTO abbreviated academic phase.

PO 004 – Conduct Shiphandling and Seamanship

1. Performance: Conduct shiphandling and seamanship.

2. Conditions:

   a. Given:

      (1) ship at sea or alongside;

      (2) support, to include:

         (a) bridge team; and

         (b) on-watch personnel.

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) on board a ship at sea; or alongside

      (2) any time of day or night; and

      (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall conduct shiphandling and seamanship skills, IAW references, to include:

   a. performing basic rope work;

   b. identifying upper deck fittings;

   c. conning the ship;

   d. acting as part ship officer in seamanship evolutions;

   e. acting as OOW / OOD for berthing and slipping; and

   f. performing the duties of a small boat coxswain.

4. References: A7, A10, A16, A17, A39, A53, A72, A73, B29, B30, B31, B34.

5. Task Numbers: See Annex C.

6. Training Limitations: Nil

7. Remarks: The following objectives will not be covered during the NTO abbreviated academic phase:

a. Conning the ship

b. Performing duties of a small boat coxswain.

PO 005 – Perform Watch Administration

1. Performance: Perform watch administration.

2. Conditions:

    a. Given:

        (1) ship at sea or alongside;

        (2) support, to include:

            (a) bridge team; and

            (b) on-watch personnel.

        (1) references.

    b. Denied: Nil.

    c. Environment:

        (1) on board a ship at sea or alongside;

        (2) any time of day or night; and

        (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall perform watch administration, IAW refs, by:

    a. describing the duties and responsibilities of the Watch on Deck/Alongside Duty Watch; and

    b. implementing ship's routine.

4. References: A7, A16, A17, A53, B29, B30, B31, C34.

5. Task Numbers: See Annex C.

6. Training Limitations: Nil.

7. Remarks: Nil.

PO 006 – Direct the Operation of Onboard Engineering Systems

1. Performance: Direct the operation of onboard engineering systems.

2. Conditions:

   a. Given:

      (1) ship at sea or alongside;

      (2) support, to include:

         (a) bridge team; and

         (b) on-watch personnel.

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) on board a warship at sea or alongside;

      (2) any time of day and night; and

      (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall direct the operation of onboard engineering systems, by describing the key components of marine and combat systems engineering.

4. References: A17, A51, A54, A55, A67.

5. Task Numbers: See Annex C.

6. Training Limitations: Nil.

7. Remarks: Nil.

PO 007 – Employ Naval Communications

1. Performance: Employ naval communications.

2. Conditions:

    a. Given:

        (1) ship at sea or alongside;

        (2) support, to include:

            (a) bridge team; and

            (b) on-watch personnel.

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (1) on board a ship at sea or alongside;

        (2) any time of day and night; and

        (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall employ naval communications, by interpreting signal flags and pennants.

4. References: A10, A39, A73, B24, B25, B26, C23.

5. Task Numbers: See Annex C.

6. Training Limitations: Nil.

7. Remarks: Nil.

PO 008 – Direct Action and Emergency Procedures

1. Performance: Direct action and emergency procedures.

2. Conditions:

    a. Given:

        (1) ship at sea or alongside;

        (2) support, to include:

            (a) bridge team; and

            (b) on-watch personnel.

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (1) on board a ship at sea or alongside;

        (2) any time of day and night; and

        (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall direct action and emergency procedures, IAW refs, by:

    a. describing the action and emergency organization;

    b. act as OOW/OOD in response to emergencies; and

    c. explaining the ship's fighting organization.

4. References: A51, A53, A55, A60, A61, A73.

5. Task Numbers: See Annex C.

6. Training Limitations: Nil.

7. Remarks: Nil.

PO 009 – Support Operations Other Than Warfare.

1. Performance: Support operations other than warfare.

2. Conditions:

   a. Given:

      (1) ship at sea or alongside;

      (2) support, to include:

         (a) bridge team; and

         (b) on-watch personnel.

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) on board a ship at sea or alongside;

      (2) any time of day and night; and

      (3) any conditions of weather and visibility.

3. Standard: the Naval Officer shall conduct operations other than warfare, IAW refs, by:

   a. describing key events in the History of the RCN;

   b. practicing customs and traditions of the RCN;

   c. applying naval terminology, to include:

      (1) conducting shipboard ceremonial procedures, to include:

         a. colours;

         b. sunset; and

         c. dressing ship.

   d. employing personal safety procedures

   e. explaining the principal elements of naval doctrine;

   g. identifying general ship types to include:

   h. identifying RCN ships and supporting aircraft.

4. References: A10, A23, A39, A53, A73, B29, B30, B31, B34, C29.

5. Task Numbers: See Annex C.

6. Training Limitations: Nil.

7. Remarks: Nil.

PO 010 – Support Air Operations

1. Performance: Support air operations.

2. Conditions:

   a. Given:

      (1) ship at sea;

      (2) support, to include:

         (a) bridge team; and

         (b) on-watch personnel.

      (3) references.

   d. Denied: Nil.

   e. Environment:

      (1) on board a ship at sea or alongside;

      (2) any time of day or night; and

      (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall support air operations IAW refs, by:

   a. describing safety procedures during air operations, to include:

      (1) conducting FOD rounds;

      (2) wearing safety gear;

      (3) identifying out of bounds areas for avoidance;

      (4) identifying flight safety procedures to be followed when working on the flight deck;

      (5) explaining safety procedures when operating in high-risk areas; and

      (6) explaining procedures for operating in our of bounds areas.

4. References: A72, A73.

5. Task Numbers: See Annex C.

6. Training Limitations: Nil.

7. Remarks: Nil.

TRAINING PLAN

CHAPTER 3 – TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. classroom lectures;

   b. demonstration & practice; and

   c. OJT.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 24 training days for NWO II module 1, 10 training days for NWO II module 1 (NTO), and 5 training days for NWO II module 5.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Training Officer				
Rank	MOSID	Qualifications Required	Qty	PO / EO
Lt(N)	00207-05/13	NWOPQ Qualified (ResF = Lt(N) with BWK and 6 months at sea experience) AHCH – Instructional Techniques Basic AIMU – Alternate Instructional Techniques	2	All

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
PO2 - PO1	00105	Drill Instructor	1	EO 001.01
PO2 – PO1	00105	Boat Coxswain Course	2	EO 004.06
PO2 – PO1	00105	Seamanship Instructor	1	EO 004.01 004.02
MS – PO2	00299	Naval Communicator	1	EO 007.01
Lt(N)	00207 – 05/13	NWOPQ Qualified (ResF = Lt(N) with BWK and 6 months at sea experience)	2*	NABS Serial support

Guest Lecturers / Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

Unit CPO1	1	Leadership	001.02	2
Marine Science Instructor (Civilian)	2	Collision Regulations Navigation	002.05 002.06 002.07 003.01	42
Panel of Junior and senior leaders in the fleet	4-6	Leadership Panel	001.02	4

*Note: Based on 4-cubicle availability (5 trainees per cubicle). If fewer cubicles, available, time commitment increases. Max of 7 trainees per cubicle.

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel, or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

a. Due to limitations on platform and simulator availability, practical experience aboard ships and in the simulator may vary for trainees; and

b. Weather conditions and/or ship's programs may affect the timings and ability to deliver all training as per the recommended course schedule.

COURSE CAPACITY

5. The capacity per session is based on available classroom & simulator resources and guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 21;

b. Maximum – 24; and

c. Minimum – 14.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NTORD 4540-2).

7. The authority to load members onto this course will be IAW NTORD 4540-43. NTG HQ to be included as an info addressee on all loading messages. Course loading will be accomplished by the occupation BTL manager.

QUALIFICATION AND ASSESSMENT

8. This is a pass / fail course. Students must successfully complete all POs as indicated to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NTORD 4500-51. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirements for a TRB IAW NTORD 4500-54.

11. Students shall be provided feedback throughout the academic and practical modules of training. One of the following administrative measures may be used to notify a student of unsatisfactory training progress:

a. Academic Caution (Annex M);

b. Performance Regression (Annex N); and

c. Professional Leadership Qualities (PLQ) Chit (Annex O).

12. Academic Caution: To be issued when a student does not meet the required standard on a CEC IAW Annex H. All academic cautions are to be recorded in the student's training file. There are two levels of academic caution:

a. Level One: Issued after the initial failure of any CEC; and

b. Level Two: Issued after the subsequent failure of a previously failed CEC re-write.

13. Performance Regression: To be issued when a trainee fails to maintain a previously achieved standard during NWO training through formal evaluation. A trainee that receives two Performance Regressions for the same subject, or a total of three Performance Regressions, shall be referred to a Training Review Board (TRB). The process is as follows:

a. Initial Performance Regression is issued to alert the student to their poor performance. A divisional interview must be conducted to provide the trainee with a plan to correct their performance deficiency IAW Annex N;

b. Second Performance Regression is issued. A divisional interview must be conducted to provide the student with an update on their performance and further direction on how to correct their performance deficiency IAW Annex N. A second Performance Regression for the same deficiency requires a referral to a TRB IAW Annex N; and

c. Third Performance Regression is issued if the student fails to correct their performance IAW Annex N. The issuance of a third Performance Regression shall result in a TRB IAW Annex N.

14. Professional Leadership Qualities (PLQ) Chit: To be issued when a student fails to display the leadership qualities of a naval officer IAW Annex P during NWO training. The process is as follows:

a. Issue initial negative PLQ Chit to alert the student to their poor attributes IAW Annex O. A divisional interview must be conducted to provide the student with a plan to correct their deficiency;

b. Issue a second PLQ Chit to alert the student to their poor attributes IAW Annex O. A second divisional interview shall be conducted to provide the student with a plan to correct their deficiency. A second negative PLQ Chit for the same deficiency may result in remedial measures (initial counselling); and

c. Issue a third negative PLQ Chit to alert the student to their poor attributes IAW Annex O. A third divisional interview shall be conducted to initiate the remedial measures (initial counselling) process. For severe contraventions of the PLQs, a divisional interview shall be conducted and initiate the remedial measures (initial counselling) process.

REMOVAL FROM TRAINING

15. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance / conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NTORD 4500-29 or 4500-54.

TRAINING REVIEW BOARD

16. Training review board (TRB) shall be conducted IAW NTORD 4500-54. Every effort must be made to assist a student having difficulty before it becomes necessary to convene a TRB.

17. A TRB shall be convened for:

a. failure of a supplemental CEC (re-test, level two caution);

b. cumulative failure of three (3) CECs;

c. failure to complete the NWO II Mod 5 PTR package;

d. academic misconduct;

e. if a trainee is absent for more than two (2) training days during the academic phase or 16 training hours during the practical phase; and

f. any exceptional circumstance that may impact the trainee's training.

RECERTIFICATION REQUIREMENTS

14. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

15. The language of instruction shall be IAW the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

16. PLARs shall be conducted IAW DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

17. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

18. A Canadian Forces Course Report, Form CF 377 and CF 289 Certificate of Military Achievement, shall be prepared and distributed IAW DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 - LESSON SPECIFICATIONS

PO 001 - Execute Administrative and Leadership Duties of a Divisional Officer

EO 001.01 - Perform Naval Drill and Ceremonial Procedures

1. Performance: Perform naval drill and ceremonial procedures.

2. Conditions:

    a. Given:

        (1) drill deck;

        (2) naval sword; and

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall perform naval drill IAW refs, to include:

    a. standard drill; and

    b. naval sword drill.

4. Teaching points: see table below.

5 Total EO Time: 5 x 50-minute periods.

6. References: A3.

7. Training and learning aids.

8 Test Details: Practical testing to occur during scheduled drill periods.

9. Remarks:

    a. This training is to be conducted during the regularly scheduled drill periods;

    b. CTO or student lead dress inspections to occur during the drill periods; and

    c. Parade practices are outside of this requirement.

Period #	Description	Instructional Strategies	Time	References/Remarks
1	Standing Drill Refresher to include: 1. Engage in the Formation of a Squad 2. Demonstrate the position of Attention. 3. Demonstrate Standing at Ease 4. Stand at Ease to/from attention. 5. Demonstrate dressing a Squad 6. Remove/ Replace Headdress 7. Salute to • Front • Left • Right	**Method:** EDI **Media:** Instructor **Environment:** Drill hall/ parade square	1 x 50 min	CAF Manual of Drill and Ceremonial, CAF Dress Manual
2	Refresher Drill (Part 2) 1. Turn to and incline from the Right/left 2. Close to • Right and • Left 3. Open order/close order March 4. Ranks • Falling out of Ranks • Falling in • Sizing in Ranks • Sizing in Single Rank • Forming two Ranks from Three	**Method:** EDI **Media:** Instructor **Environment:** Drill hall/ parade square	1 x 50 min	CAF Manual of Drill and Ceremonial, CAF Dress Manual

3	Introduction to the naval sword 1. Demonstrate proper care and maintenance of • sword, • scabbard, • Belt 2. Explain the general rules of wearing and carrying a sword, to include: • How the belt is worn; • When swords are drawn; • Hooking and unhooking the scabbard; (The only exception to the scabbard being hooked up when the sword is sheathed is the Colour Officer who "wears the sword through the pocket"); and When sitting, the lion on the pommel always faces forward.)	**Method:** EDI **Media:** Instructor **Environment:** Drill hall/parade square	1X 50 min	CAF Manual of Drill and Ceremonial, CAF Dress Manual
4	Explain and demonstrate the following with the sword sheathed: 1. Draw and return the sword; 2. Position of attention; 3. Position of stand at ease; 4. Position of stand easy; 5. The quick march; 6. The halt; 7. Salute at the halt; and 8.Salute on the march.	**Method:** EDI **Media:** Instructor **Environment:** Drill hall/ parade square	1X50 min	CAF Manual of Drill and Ceremonial, CAF Dress Manual
5	Explain and demonstrate the following with the sword drawn: 1. Position of attention; 2. Position of stand at ease; 3. Position of stand easy; 4. The quick march; 5. The halt; 6. Salute at the halt; and 7. Salute on the march.	**Method:** EDI **Media:** Instructor **Environment:** Drill hall/ parade square	1X50 min	CAF Manual of Drill and Ceremonial, CAF Dress Manual

EO 001.02 - Apply Practical Leadership Skills

1. Performance: Apply practical leadership skills.

2. Conditions:

    a. Given:

        (1) ship; and

        (2) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall apply practical leadership skills, IAW references to include:

    a. describing the professional leadership qualities in a naval environment;

    b. explaining the expectations for an officer in the CAF;

    c. discussing structure of groups and teams;

    d. explaining team characteristics;

    e. describing team building;

    f. discussing group development;

    g. discussing factors affecting group / team performance;

    h. discussing time management techniques;

    i. explaining the impact of bias on leadership; and

    j. explaining the attributes of leadership in the CAF

4. Total EO Time: 11 x 50-minute periods as follows:

    a. Lecture / Guided Discussion, 7 x 50 minutes; and

    b. Leadership panel, 4 x 50 minutes

5. Total EO time: 10 x 50 mins.

6. References: A5, A30, A35, A78, Are You Aware of Your Biases? (hbr.org) , C6, C20, C78.

7. Training Aids:

8. Test Details: Assessed throughout the NWO training with PLQ s and PD.

9. Remarks:

   a. Portions of the EO shall be taught early in the course to lay the foundation for the PLQ throughout the course and NWO training; and

   b. Time management as a general concept and not bridge time management specifically.

10. Teaching Points:

L = Lecture, D = Demonstration, P = Practical Exercise, C =CBT, DL = Distributed Learning

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Explain the Expectations for an officer in the CAF. 1. Describe professional leadership qualities (PLQs) in a naval environment; 2. Define officer like qualities. 3. Explain PLQs 4. Explain VEN/TE expectations for dress and deportment. 5. Maintain good order and discipline. 6. Explain the balance responsibilities of leadership and followership	**Method:** Lecture **Media:** PPT **Environment** Classroom	1 x50min	Queen's Regulations and Orders (QR&Os) Canadian Forces Administrative Orders (CFAOs) To be scheduled on the first day.

Period #	Description	Instructional Strategies	Time	References/ Remarks
2-5	Leadership Panel 1. Discuss the current state of the occupation 2. Discuss career challenges 3. Discuss leadership challenges of a junior officer.	**Method:** Panel Discussion **Media:** **Environment:** Classroom or Theatre	4 x50 min	Invite guests to explain and discuss the leadership challenges faced by new officers under training in a ship. Encourage the discussion of experiences to demonstrate the uniqueness of leadership within the naval environment
6	Display Officer-Like Qualities 1. Discuss what it means to be an officer 2. Discuss dynamics of Officer NCM relations • Officers and junior NCMs • Officers and NCOs • Officers and senior NCOs 3.Discuss expectations of a junior officer on ship 4. Describe what it means to be an officer 5. Discuss taking initiative by proactively addressing tasks and challenges on the ship 6. Discuss earning the respect of followers. 7. Recognize and appreciate the skills and knowledge of NCMs 8. Take on responsibility in the ship • Phase 1 – doing rounds • Phase 2 – CIWS • Phase 3 – launching boats	**Method:** Guided Discussion **Media:** PPT or whiteboard to highlight key concepts and questions **Environment:** Classroom	1 x50 min	Led by CTO or Senior Unit Chief

Period #	Description	Instructional Strategies	Time	References/ Remarks
7	Leadership Discussion 1. Discuss the most important qualities of a leader. 2. Describe how a leader builds trust in their team. 3. Discuss failures of leadership 4. Compare Civilian and Military Leadership 5. Discuss impact of life experiences on leadership. • Overcoming adversity • Cross cultural interactions • Personal challenges 6. Discuss how leaders can balance authority and empathy.	**Method:** Self-reflection followed by in class discussion. **Media:** Homework assignment. **Environment:** Classroom	1 x50 min	Led by snr instructor or Lt(N). Students will be asked to answer questions prior to class, then they will discuss their answers as a group.
8	Understanding Leadership in the CAF 1. Explain the characteristics of leadership. (Integrity, courage, loyalty, respect, selflessness, responsibility, accountability, teamwork, professionalism, initiative, judgement, physical and mental fitness) 2. Discuss different perspectives on leadership. • Compare autocratic, democratic, and transformational leadership style. 3. Explain the impact of nature vs nurture. 4 Discuss diversity in the CAF.	**Method:** Guided Discussion **Media:** PPT or whiteboard to highlight key concepts and questions, printed handouts **Environment:** Classroom	1 x50 min	Led by snr Instructor or Lt(N) *Fundamentals of Naval Leadership (US Naval Academy) Behaviour In Organizations (Cdn Edition) (Prentice Hall Canada, Inc.)

8	5. Explain the principles of leadership and describe their importance*: • Know yourself and seek self-improvement • Be technically and tactically proficient • Seek responsibility and take responsibility for your actions • Make sound and timely decisions • Set the example • Know your people and look out for their welfare • Keep your people informed • Develop a sense of responsibility in your subordinates • Ensure assigned tasks are understood, supervised and accomplished • Train your people and as team • Employ your command in accordance with its capabilities. 6. Describe the ripple effect of behaviour • How leaders' actions and behaviour influence their team 7. Discuss building trust. • Importance of trust in a relationship • Building and maintaining trust within a team. 8.Discuss the Mainguy report • Significance • Impact on leadership policies in the CAF 9.Discuss considerations for your career as an officer: • Balancing family life with military duties			

Period #	Description	Instructional Strategies	Time	References/ Remarks
9	Characteristics of Teams and Understanding Bias and Its Effect on Leadership 1. Define bias and its various forms. • Conscious • Unconscious • Confirmation bias 2. Consider personal beliefs and assumptions • Examine core beliefs. Reflect on interactions with people from diverse 3. Explain the influence of previous experience on leadership. • learning from success and failures • building resilience • developing emotional intelligence • cultivating empathy • informs decision making • influence leadership style • reflect on ability to empathize with other people. 4. Challenge long held beliefs and values • allow others to challenge your assumptions. 5. Accept feedback 6. Embrace diverse opinions	**Method:** Guided Discussion Case studies Self-reflection exercise **Media:** PPT or whiteboard to highlight key concepts and questions, printed handouts, case study materials **Environment:** Classroom	1x50 min	Led by snr instructor or Lt(N). Guide to the Divisional System Case studies will be used to demonstrate the impact of bias in leadership. *Trusted to Serve Are You Aware of Your Biases? (hbr.org) *Fundamentals of Naval Leadership (US Naval Academy) Behaviour In Organizations (Cdn Edition) (Prentice Hall Canada, Inc.)

	7. Discuss the characteristics of effective teams*. • Clear goals • Defined roles • Open communication • Mutual trust • Diversity and inclusion • Effective leadership • Accountability and responsibility • Flexibility and adaptability • Continuous improvement • Support the welfare of members. • Ethical conduct 8. Discuss structure of groups and teams. • Informal • Formal • Hierarchical • Functional • Cross-functional • Virtual 9.Discuss factors affecting group and team performance. • Group cohesion and trust • Organizational support • Goal clarity • Team composition • Communication 10.Describe variables influencing group and team performance. 11.Discuss strategies for developing teams. • Training and development • Defining roles are responsibilities. • Encouraging open communication • Establishing clear goals • Building trust 12. Discuss the impact of bias on leadership			

Period #	Description	Instructional Strategies	Time	References/ Remarks
	• Team dynamics • Decision making 13. Describe how bias can affect relationship dynamics. • Decision making • Power dynamics • Conflict resolution • Trust • Emotional expression			
10	Apply Time Management Techniques 1. Discuss time management techniques • Setting SMART goals • Prioritization ○ Urgent-important matrix • Working in focused intervals (Pomodoro technique) • Time blocking 2. Manage Time wisely • Setting boundaries • Delegating tasks • Adapting to change • Self-discipline 3. Discuss overcoming procrastination 4. Planning and scheduling • Daily, weekly and long-term planning 5. Discuss Work/Life balance • Maintaining perspective • Forming healthy habits 6. Describe contingency planning. 7. Discuss adapting to change	**Method:** Discussion, reflection, case study **Media:** PPT or whiteboard to highlight key concepts and questions, printed handouts, case study materials **Environment:** Classroom	1 x Guided 50 min	Personal Time Management 3rd Edition (Marion E. Haynes)

Period #	Description	Instructional Strategies	Time	References/ Remarks
11 Confirm number of periods.	Character based leadership 1. Explain the origins and definitions of CBL 2. Describe the foundations of character • Virtues • Personality traits • Values 3. Discuss the Character framework 4. Discuss drive- deficiency/excess 5. Discuss creating new habits 6. Discuss vulnerability and transcendence 7. Display emotional intelligence 8. Discuss discovering your 'why'.	**Method:** Guided Discussion **Media:** PPT or whiteboard to highlight key concepts and questions, printed handouts, case study materials **Environment:** Classroom	Confirm	Led by LCC Chief or CTO Ivey Institute for Leadership Hand-outs Prereading - Trusted to Serve Fighting Spirit: The Profession of Arms in Canada

EO 001.03 - Manage People

1. Performance: Manage people.

2. Conditions:

    a. Given:

        (1) division; and

        (2) references. .

    b. Denied: assistance.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall manage people, by:

    a. describing the divisional system, to include:

        (1) origins of the divisional system;

        (2) shipboard organization; and

        (3) individual divisional responsibilities.

    b. discussing the roles of the divisional officer.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: A5, A23, A30, A53, A73.

7. Training Aids: Nil.

8. Test Details:

    a. CEC General Naval Knowledge – Part II; and

    b. Must achieve a minimum of 70% on the CEC.

9. Remarks: This EO should be taught early in the course as trainees will be expected to use the divisional system to deal with problems and requests throughout the training.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Manage People 1.Explain the divisional system • Origins • Mainguy report • Principles • Shipboard Organization • Functional vs divisional organizations 2. Describe the responsibilities of members of the Chain of Command: • CO • XO • HOD • CHOD • Div O • Div PO • Members of the division 3. Explain the role of a Divisional Officer • Duties • Assets 4. Explain administration for the following: • Injuries • Leave • Conduct • Performance • Feedback notes • Remedial measures	**Method:** Lecture **Media:** Instructor PPT **Environment:** Classroom	1x 50 min	NAVORDS, SSOs, Guide to divisional systems, CFAOs, DAODS

EO 001.04 - Describe the Functional Organization

1. Performance: Describe the functional organization.

2. Conditions:

    a. Given:

        (1) ship; and

        (2) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall describe the functional organization by:

    a. explaining the functional organization of an HMC ship; and

    b. identifying the primary duties and responsibilities of key personnel IAW ship's standing orders and the watch and station bill.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: A73.

7. Training and Learning Aids: Nil.

8. Test Details:

    a. CEC General Naval Knowledge – Part II; and

    b. Must achieve a minimum of 70% on the CEC.

9. Remarks: This EO should be scheduled at the beginning of the course as it sets the context for the remainder of the course.

Period	Description	Instructional Strategies	Time	References/ Remarks
1	Describe the Functional Organization 1. Compare the functional vs. fighting organization 2. Explain the ship's functional environment 3. Explain the roles of key personnel command team • CO • XO • Cox'n • HODs/CHODS/divisional heads 4. Explain the departmental organization • Operations Department ○ Key personnel ○ Duties ○ Responsibilities • Combat Systems Engineering Department ○ Key personnel ○ Duties ○ Responsibilities • Marine Engineering Department ○ Key personnel ○ Duties ○ Responsibilities • Logistics Department ○ Key personnel ○ Duties ○ Responsibilities • Air Department ○ Key personnel ○ Duties ○ Responsibilities	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	SSOs

EO 001.05 - Describe the Managerial Environment of the RCN

1. Performance: Describe the managerial environment of the RCN.

2. Conditions:

    a. Given: naval environment.

    b. Denied:

        (1) references; and

        (2) assistance.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall explain the managerial environment of the RCN to include:

    a. describing the mission of the Royal Canadian Navy (RCN);

    b. describing the role and objectives of the RCN;

    c. describing organization of the RCN;

    d. describing the responsibilities of formations to include:

        (1) MARPAC;

        (2) MARLANT; and

        (3) the Naval Reserves;

    e. stating the hierarchy between policies.

4. Teaching Point: see table below.

5. Total EO Time: 2 x 50-minute periods.

6. References: A5, A7, A23, A27, A35, A53, A41, A42, A73, B44, B56, C31.

7. Training and Learning Aids: Nil.

8. Test Details:

a. CEC General Naval Knowledge – Part II; and

b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Fleet, formation and HQ Organization 1. Explain the RCN Mission Statement 2. Explain the responsibilities of RCN 3. Explain CAF Organization 4. Explain Force Employment:: CJOC, CANSOFCOM, NORAD 5.Explain Force Generation: RCN. RCAF, Cdn Army 6. Discuss international Responsibilities: NATO, UN. 7. Describe the structure of the RCN: RCN NDHQ, MARPAC/MARLANT, NAVRES	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1 x50 min	QR&Os, CFAOs, DAODs, NAVORDs, MARPACORDs, MARLANTORDs, SSOs, CSOs, Environmental Management System (EMS), Fishery Act, Transportation of Dangerous Goods Act. Mariner's handbook 7th edition.
2	Explain the regulations and Orders governing the RCN to include: 1. NDA 2. QR&Os (to be KR&Os) 3. DAODs 4. NAVORDS 5. MARPACORDs 6. SSOs 7. CSOs 8. SEMS 9. Discuss international maritime law and the RCN's obligations	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1 x50 min	QR&Os, CFAOs, DAODs, NAVORDs, MARPACORDs, MARLANTORDs, SSOs, CSOs, environmental management system (EMS), fishery act, transportation of dangerous goods act. Mariner's handbook 7th edition.

EO 001.06 - Draft RCN Correspondence

1. Performance: Draft RCN correspondence.

2. Conditions:

    a. Given: references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall draft military correspondence IAW the military writing manual, to include:

    a. preparing the following correspondence utilizing the appropriate military writing formats:

        (1) internal and external letters;

        (2) memorandums;

        (3) briefing notes; and

        (4) minute sheets.

    b. reviewing common errors made in drafting military correspondence; and

    c. preparing folders military correspondence for submission.

4. Teaching Points: See table below.

5. Total EO Time: 2 x 50-minute periods.

6. References: A10, A73.

7. Training and Learning Aids: Nil.

8. Test Details:

    a. CEC General Naval Knowledge – Part II; and

    b. Must achieve a minimum of 70% on the CE.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Draft RCN Correspondence 1. Draft RCN correspondence to include: • Letters • Memorandums • Briefing notes • Minute sheets 2. Use formatting rules in correspondence 3. Describe the characteristics of the following components: • Heading • Body • close 3. Write a minute on a memorandum	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	SSOs, CFCD 105
2	Review a memo from a subordinate 1. Identify common formatting errors 2. Identify common grammatical mistakes 3. Provide writing guidance to subordinates. 4. Prepare a folder for submission of military correspondence.	**Method:** Exercise **Media:** **Environment:** Classroom	1x50 min	SSOs, CFCD 105

PO 002 - Conduct Navigation

EO 002.01 - Explain the Effects of Meteorological Conditions on Shiphandling

1. Performance: Explain the effects of meteorological conditions on shiphandling.

2. Conditions:

   a. Given:

      (1) ship;

      (2) surface analysis map;

      (3) surface prognosis map; and

      (4) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall explain the effect of meteorological conditions on shiphandling, navigation and operations, to include:

   (a) describing the effects of atmospheric pressure systems and wind;

   (b) describing the effects of air masses and frontal systems as they pertain to a marine environment; and

   (c) describing wave dynamics.

4. Teaching Points: See table below.

5. Total EO Time: 2 x 50-minute periods.

6. References: B46.

7. Training and Learning Aids: Deployable meteorological instruments.

8. Test Details: Tested in NWO III Meteorology Exam.

9. Remarks: This EO is an introduction to the concepts. They will be reviewed, expanded upon, and tested in NWO III.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1 and 2	Explain the Effects of Meteorological Conditions on Shiphandling 1. Discuss basic weather concepts 2. Describe wind • How it is measured • Terminology ○ Veer ○ Back 3. Describe causes of wind • Atmospheric pressure variations • Local effects 4. Explain frontal systems • cold front • frontal waves • occulated fronts • warm front 5. Explain clouds and precipitation • strati form • cumuliform • nimbo form • chiriform • dew point • lifting process ○ orographic ○ frontal ○ convection ○ Convergence • Describe characteristics of fog ○ Type ○ Humidity ○ Locality ○ Dissipation	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	2 x 50 mins	Manual of Marime Weather Observations (MANMAR)

	6. Explain Tropical cyclones • Coriolis force • Water temperature 7. Explain Buy Ballots law • How it works • using buy ballots law 8. Explain sea breeze vs land breeze 9. Explain katabatic winds • How they are formed • resulting weather 10. Describe sea and swell • definition • formation 11. Explain generation of wind waves • 3 basic limiting factors • wind speed • duration • fetch 12. Explain hurricane/typhoon warning systems 13. Describe factors affecting waves height • sea and swell • windspeed • duration • fetch 14. Define significant wave height 15. Define rogue waves 16. Explain the Beaufort scale 17. Describe wind force and knots • gale • storm • hurricane 18.Explain ice formation and effects of freezing spray 19. Estimate wind speed and direction.			

EO 002.02 - Apply Nautical Measurements

1. Performance: Apply nautical measurements.

2. Conditions:

    a. Given:

        (1) ship;

        (2) chart;

        (3) navigational instruments; and

        (4) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall apply nautical measurements to include:

    a. angular units;

    b. linear distance measurements;

    c. speed; and

    d. depth measurements.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: B24, B25.

7. Training and Learning Aids: CBT Module.

8. Test Details:

    a. CEC Navigation Fundamentals; and

    b. Must achieve a minimum of 70% on the EC.

9. Remarks: (1) This EO introduces geodesy and demonstrates need for students to be comfortable with conversions of different units of measurements.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Apply Nautical Measurements to include: 1. Define basic concepts related to geodesy, to include: • Oblate spheroid • Equatorial radius • Polar radius • Axis of rotation • Great circle • Small circle 2. Define meridians of longitude, to include: • Prime meridian • International dateline • Parallels of latitude 3. Explain measurement of latitude and longitude, to include: • angular units • degrees • minutes • seconds 4. Explain linear distance measurement, to include: • international nautical mile • sea mile • tactical mile • cable • yards 5. Convert between distance units, to include: • degrees of latitude • nautical mile • cables • yards • feet	**Method:** Lecture **Media:** PPT **Environment:** Classroom	1 x50 min	Admiralty Manual of Navigation Vol 1 Admiralty Manual of Navigation Vol 2

	6. Explain depth measurements and convert between measurements, to include: • meters • fathoms • feet • shackles 7. Explain speed measurement.			

EO 002.03 - Identify Aids to Navigation

1. Performance: Identify aids to navigation.

2. Conditions:

   a. Given:

      (1) ship;

      (2) NABS or NPTT; and

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall identify aids to navigation to include:

   a. recognizing the following:

      (1) lateral buoys;

      (2) cardinal buoys;

      (3) special buoys;

      (4) day beacons;

      (5) range markers;

      (6) lighted fixed aids; and

      (7) AIS aids to navigation.

   b. describing the features of navigational aids to include but not limited to:

      (1) color;

      (2) shape;

      (3) top mark(s); and

      (4) purpose;

c. stating the light characteristics of all fixed and floating aids to navigation to include:

(1) timing; and

(2) patterns.

d. explaining navigation aids on a navigation chart, to include:

(1) name;

(2) characteristics;

(3) vertical height; and

(4) range of visibility.

e. recognizing visual aids to navigation when executing coast navigation by day and night.

4. Teaching Points: See table below.

5. Total EO Time: 4 x 50-minute periods as follows:

a. 2 x 50-minute lecture; and

b. 2x 50-minute practical.

6. References: A17, B37, B39, B43, B54, B55.

7. Training and Learning Aids.

8. Test Details:

a. CEC Navigation Fundamentals;

b. Must achieve a minimum of 70% on CEC; and

c. NWO II Mod 5 PTR Package.

9. Remarks: Concepts also practiced during the NABS phase, see Annex P.

Period #	**Description**	**Instructional Strategies**	**Time**	**References / Remarks**

1 and 2	Identify Aids to Navigation 1. Describe the shapes, colours, top markings, and numbering systems of the following: a. Lateral Buoys • Starboard hand • Port Bifurcation • STBD Bifurcation • Fairway buoys • Isolated danger buoys b. Cardinal Buoys 2. Describe the shape, colour, and function of the following: a. Special Buoys b. Day beacons c. Fixed Aids to Navigation • Major Lighted Aids • Minor Lighted Aids • Range Markers d. AIS Aids to Navigation • Physical AIS A to N • Virtual AIS A to N • Monitored Synthetic AIS A to N • Predicted Synthetic AIS A to N • AIS aids to Navigation 3. Describe the difference between Fixed aid to navigation, buoy, and AIS aid to navigation and how to distinguish between them. 4. Define the upstream direction. 5. Describe both IALA systems and their significance for navigation.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	2 x 50 mins	- BWK Manual for the RCN, - CCG List of Lights, Buoys and Fog Signals, - Radio Aids to Marine Navigation, - Chart 1, - Quick Reference Guide to Canada's Navigation Aids, - Canadian Aids to Navigation System (2011)

	6. Describe how Canadian Aids to Navigation are used; what they mark, what the navigation significance is in Canadian waters.			
3 and 4	NPTT Exercise – Identify Aids to Navigation • Lesson 1 – Halifax buoy transit • Lesson 2 – buoy slalom	**Method:** Computer based training **Media:** NPTT **Environment:** Classroom	2 x 50 mins	

EO 002.04 - Calculate Time and Time Zones

1. Performance: Calculate time and time zones.

2. Conditions:

    a. Given: references.

    b. Denied: Nil.

    c. Environment: classroom

3. Standard: The Naval Officer shall calculate time and time zone IAW refs to include:

    a. identifying the significance of longitude as it relates to:

        (1) the Prime Meridian;

        (2) the International Date Line (IDL);

        (3) time, and

        (4) time zones.

    b. identifying the relationship between Greenwich Mean Time (GMT) Coordinated Universal Time (UTC) and Zulu Time;

    c. applying naming conventions of times zones, to include the use of:

        (1) ‘+’ or ‘integer’; and

        (2) letter designations.

    d. taking account of Local time; and

    e. applying Daylight Savings Time (DST).

4. Teaching Points: See table below.

5. Total EO Time: 2 x 50-minute periods.

6. References: A25, B7, B8, C8.

7. Training and Learning Aids:

    a. PowerPoint supplement; and

b. Handout of the time zones of the world (marine convention).

8. Test Details:

a. CEC Navigation Fundamentals; and

b. Must achieve a minimum of 70% on the EC

9. Remarks: Nil.

Period#	Description	Instructional Strategies	Time	References/ Remarks

1	Calculate Time and Time Zones
	
	1. Describe theory of time and time zones
	• Meridians of longitude
	• GMT & UTC
	• International Date Line
	
	2. Calculate time and time zone problems:
	• Time zone naming conventions
	  ○ + or -
	  ○ Letter designations
	• Marine conventions
	• Political conventions
	• Local time
	• Daylight savings time (DST)
	  ○ How it is applied
	  ○ Where and when it is in effect in Canada
	  ○ Its relationship with UTC
	• Ship's Clocks
		Method:
		Lecture
		
		Media:
		Powerpoint
		
		Environment:
		Classroom
			1x50 min	DIP 100, ADMIRALTY Total tides software, ANP 1 – NATO Master Navigation Supplement, Bridge Master 180/250 Radar Series Ship's Manual Vol II, Bridge Procedure Guide

2	Perform Calculations Calculate the following: • Time Zone from Longitude • Time Zone from Longitude with DST • UTC from local time • Local Time from UTC • Date and Time when crossing the IDL-	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	DIP 100, ADMIRALTY Total tides software, ANP 1 – NATO Master Navigation Supplement, Bridge Procedure Guide

EO 002.05 - Employ Mental Math Techniques

1. Performance: Employ mental math techniques.

2. Conditions:

   a. Given:

      (1) supervision.

   b. Denied:

      (1) references; and

      (2) assistance.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall employ mental math to include:

   a. applying the radian rule;

   b. applying the sine rule;

   c. applying the mathematical relationship between speed, time and distance;

   d. using the radian / sine rule to determine:

      (1) the CPA of a stationary contact or object;

      (2) the distance off track given a bearing and a distance to a leadmarks;

      (3) what course to steer to regain track by certain time or by a certain point in the water; and

      (4) the amount of safe water to port/stbd based on clearing bearings.

   e. using the 6-3-1 rule to solve speed, time, and distance problems.

4. Teaching points: See table below.

5. Total EO Time: 7 x 50-minute periods as follows:

6. References:

7. Training and Learning Aids:

   a. CBT Module.

8. Test Details:

   a. CEC Navigation Fundamentals; and

   b. Must achieve a minimum of 70% on the CEC.

9. Remarks:

   a. The mental math that will be taught in this EO, will follow the Naval Officer through their training.

   b. There should be an additional three periods of review/practice/quizzes.

   c. Each lecture should be spaced out by a minimum of one day to give learners time to digest each lesson before introducing them to new material.

   d. There should be daily MM quizzes through the course. CTO lead, and/or student lead.

Period#	Description	Instructional Strategies	Time	References/ Remarks

1	Perform Mental Math (Introduction) 1. Describe the Radian Rule • Concept of the radian • Circumference of a circle • Arc length = radius 2. Describe the Golden Rule 3. Describe the Sine Rule • Small rule approximation	**Method:** Lecture/ guided discussion **Media:** Powerpoint**,** quizzes, assignments **Environment:** Classroom	1x50 min	Instructor should illustrate points by using examples.
2	Calculate Speed/Time/Distance 1. Describe the use Nautical measurements vs metric 2.Use 6-3-1 rule 3.Explain the concept of knots	**Method:** Lecture **Media:** Powerpoint **,** quizzes and assignments **Environment:** Classroom	1x50 min	
3	Calculate Regain 1. Calculate Angle to regain 2. Explain the distances of the parts of the triangle. 3. Describe the application and consolidation of regain	**Method:** Lecture/ guided discussion **Media:** Powerpoint, quizzes and assignments **Environment:** Classroom	1x50 min	
4-7	Review periods	**Method:** Guided discussion **Media:** Quizzes, assignments **Environment:** Classroom	4x 50	Periods do not need to be scheduled together.

EO 002.06 - Identify Features of Nautical Charts

1. Performance: Identify features of nautical charts.

2. Conditions:

   a. Given:

      (1) ship;

      (2) electronic or paper chart;

      (3) navigation tools; and

      (4) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall identify features of paper charts to include:

   a. identifying features of a nautical chart;

   b. plotting latitude and longitude coordinates;

   c. describing the RCN's policy on the use of nautical charts, to include:

      (1) selecting best scale chart(s) for a specific area of operation and use them appropriately; and

      (2) explaining where to locate information regarding changes between editions (Notices to Mariners, NAVWARN (canhydropacs / lant), DND Notices).

   d. identifying the proprieties of chart symbols used by the Canadian hydrographic Service, to include;

      (1) topographic features;

      (2) hydrographic features;

      (3) aids to navigation; and

      (4) abbreviations.

e. correlating chart symbols to the visual navigation picture when in the role of OOW and 200W.

4. Teaching Points: See table below

5. Total EO Time: 4 x 50-minute periods.

6. References: A17, B43

7. Training and Learning Aids:

   a. Paper charts;

   b. Chart catalogue;

   c. Dividers;

   d. Parallel ruler; and or

   e. Computer with ECPINS installed and licenced.

8. Test Details:

   a. CEC Navigation Fundamentals; and

   b. Must achieve a minimum of 70% on the CEC.

9. Remarks:

   a. Instruction of this EO should be scheduled after EO 002.02 – Apply Nautical Measurements and EO 002.03 – Identify Aids to Navigation.

Period #	Description	Instructional	Period #	Description
1 and 2	Use a Nautical Chart 1. Explain features found on a chart: • Title • Chart number • Publication and edition notes • Surveys and survey authorities • Projection • Scale • Sounding units • Tidal information • Linear distance scale • Conversion table • Compass rose • Adjoining charts and insets • References to larger scale charts • Cautions and warnings 2. Read and plot latitude and longitude coordinates 3. Use the compass rose to measure bearings 4. Use the latitude scale to measure distances 5. Explain the RCNs policy on the use of charts to include: • Requirement to use charts distributed officially through HSOs • Requirement to use most recent edition of charts • Requirement to use charts that are correct and current 6. Discuss minimum carriage requirements for charts onboard HMC Ships 7. Using a chart catalogue, explain CHS coverage areas and numbering systems to include: • Small, large and best scale concepts:	**Method:** Lecture **Media:** Powerpoint**,** quizzes and assignments **Environment:** Classroom	2 x 50 mins	BWK Manual for the RCN, Chart 1

	• Selecting the best scale chart for a specific area of operation/passage.			
3 and 4	Chart Symbols 1. Identify and interpret topographical chart symbols from CHS Chart 1: • Natural features • Cultural features • Landmarks • Ports • Topographical terms • 2. Identify and interpret hydrographic chart symbols from CHS Chart 1: • tides and currents • depths • nature of the seabed • rocks • wrecks • obstructions • offshore installations • tracks • routes • calling-in points • pilot stations • areas • limits • hydrographic terms 3. Review the chart symbols for aids to navigation. Reference the distinction between fixed and floating aids. 4. Correlate navigation features visually with their charted symbol including • Land contours • Charted objects • Aids to navigation	**Method: Lecture** **Media: Powerpoint** **Environment: Classroom**	2 x 50 mins	BWK Manual for the RCN, Chart 1

EO 002.07 - Conduct Visual Fix and Dead Reckoning Techniques

1. Performance: Conduct visual fix and dead reckoning techniques.

2. Conditions:

   a. Given:

      (1) ship;

      (2) classroom;

      (3) paper charts;

      (4) parallel ruler;

      (5) dividers;

      (6) pencils;

      (7) art gum eraser;

      (8) speed-time-distance-ruler; and

      (9) OOW notebook.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall conduct fix and dead reckoning techniques to include:

   a. determining the ship's position with the aid of a chart by plotting the ship's position IAW CFCD 130) using a three-point fix;

   b. maintaining a dead reckon from the fix position to the next course alteration to include:

      (1) using ordered speed;

      (2) using ordered time; and

      (3) using course steered.

   c. updating the dead reckon every time the ship's course / speed changes;

d. determining if the ship’s track is safe;

e. adjusting the ship’s course when DR is determined unsafe;

f. determining navigational data using two consecutive fixes to include:

(1) course made good;

(2) speed made good;

(3) estimated position; and

(4) tidal set (direction and speed).

g. assessing the safety of the ship's actual track (EP) during tidal set;

h. making course adjustments as necessary if the EP is deemed unsafe;

i. adjusting the ship's course when the EP is considered unsafe;

j. solving basic speed-time-distance problems related to navigation, to include:

(1) ETD;

(2) ETA;

(3) SOA;

(4) speed required;

(5) DR time; and

(6) EP time.

i. using proper chart work symbols when plotting fixes, DRs, CMG, SMG, EPs, tidal set, to include:

(1) fixing:

(a) position lines with arrows pointing away from the objects;

(b) circle at the intersection of fix; and

(c) time of fix using four digits.

(2) dead reckoning (DR);

(3) Course Made Good (CMG) and Speed Made Good (SMG);

(4) Estimating Position (EP); and

(5) tidal set.

4. Teaching Points: See table below.

5. Total EO Time: 7 x 50-minute periods as follows.

6. References: A16, A17.

7. Training and Learning Aids:

a. Paper chart;

b. Navigation tools (parallel rule, dividers, speed-time –distance ruler); and

c. OOW notebook.

8. Test Details: Assessed in NWO II Mod 5 PTR Package.

9. Remarks: Practical portions of this EO are included in the NABS phase.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Basic Chartwork Skills 1. Introduce chartwork symbology on paper charts used in planning passage to include: • Waypoints, ETA, ETD • Tracks ○ Draw an open arrow in the direction of track; indicate the bearing label above the track at the arrow and parallel to the track. • Measure and draw 'distance to go' bubbles	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	CFCD 130, CFCD 131 Classroom exercise 'Basic Chartwork Skills' conducted concurrently.

	• Determine SOA and indicate it in a speed box below the first track in the vicinity of the start point. • Indicate calling-in points • Indicate chart changes using proper symbology.			
2 and 3	Fixing the Ship 1. Define lines of position and discuss methods for obtaining a fix, particularly by visual bearings and radar changes 2. Discuss how to choose the most optimal fixing points in a visual fix to include: • Objects must be fixed • Use edges of land as fixing points • Angle between objects should be large enough to provide an appropriate cut • Reciprocal objects should be avoided • Use objects relatively close to the ship for better accuracy. 3. Discuss how to pre-plan and shoot a visual fix: • Once the fixing points are determined, they are indicated by geographical name in the OOW • Notebook with room for the bearing observation. 4. Explain the order and speed in which the points are	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	2 x50 min	CFCD 130, CFCD 131

	taken matter when shooting a fix on paper charts. 5. Record the fix in the OOW notebook 6. Plot observed lines of bearing on a paper chart to generate a fix 7. Explain the following: • shoot-up • cocked hat • incorporating a transit in a fix and discuss accounting for two line of bearing			
4 and 5	Dead Reckoning 1. Define the dead reckon and how it is used on paper charts to establish the ship's track • explain how to maintain the DR on a paper chart using ship's course steered, ordered speed and fixing time interval • explain how to generate DR marks two fixing intervals ahead of the fix • Explain how to use chartwork symbols when plotting dead reckons. 2. Explain the importance of recording exact time in the OOW notebook and plotting any course or speed alteration on the chart IOT maintain the DR's integrity.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	2 x50 min	CFCD 130, CFCD 131

	3. Determine the DR time to a course alteration or to closest dangers 4. Discuss how to determine a safe course based on DR information 5. Discuss how to determine a safe course based on DR info. 6. Discuss how to calculate the speed required to the end of a passage from the current DR position.			
6 and 7	Estimating Position and Tidal Set 1. Define the estimated (EP) and how it is used on paper charts to establish the ship's actual track 2. Explain how to determine, using tow consecutive fixes, the course made good (CMG) speed made good (SMG) and estimated position (EP) of the ship. • CMG is indicated by two open arrowheads • SMG is indicated by speed box labelled SMG • EP position is determined along the CMG using the SMG and fixing time interval. It is indicated by a triangle and two-	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	2 x50 min	CFCD 130, CFCD 131

	digit time label (minutes only) 3. Explain how to use a tidal triangle to determine the direction and speed of the tidal set 4. Explain how to determine the EP time to a course alteration or to closest dangers; 5. Explain how to, when experimenting a tidal set, determine whether the ship's actual track (EP) is safe or if the course needs to be adjusted. Discuss how to adjust the ship's course when the ship's EP is determined unsafe. 6. Explain how to account for the tidal set when calculating a speed requirement.			

PO 003 - Apply Regulations for Preventing Collision at Sea

EO 003.01– Apply the Regulations for Preventing Collisions at Sea

1. Performance: Apply the regulations for preventing collisions at sea.

2. Conditions:

   a. Given:

      (1) ship or simulated bridge (NPTT or NABS);

      (2) assistance; and

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) day or night in all environmental conditions.

3. Standard: The Naval Officer Standard: The Naval Officer shall define key terminology and core concepts outlined in Rules 3 & 20-23 to include:

   a. interpreting and defining terms to include:

      (1) air cushion vessel;

      (2) composite unit;

      (3) exploration or exploitation vessel;

      (4) inconspicuous, partly submerged vessel or object;

      (4) inshore traffic zone;

      (5) seaplane;

      (6) separation zone;

      (7) traffic separation zone; and

      (8) trawling.

   b. Applying the principles from Rules 1-3 including:

      (1) the application of COLREGs to various vessels and circumstances;

(2) the responsibility for adhering to regulations; and

(3) general definitions critical to understanding and applying the rules.

4. Teaching Points: See table below.

5. Total PO Time: 18 x 50-minute periods as follows:

a. 2 periods – Intro on Day 1 of the course
nautical;

b. 2 periods x 6 – Quizzes (two periods/quiz; Quiz #1-6);

c. 1 period – Pre-CEC Review;

d. 2 periods – CEC; and

e. 1 periods – Post-CEC Review.

6. References:

7. Training and Learning Aids:

a. Flash cards – lights & shapes;

b. Flash cards – rules of the road;

c. Rules Master Pro software;

d. Simulator (NPTT and / or NABS); and

e. Ship at sea.

8. Test Details:

a. Formative: quizzes; and

b. Summative:

(1) CEC Collision Regulations;

(2) Must achieve a minimum of 70% on the CEC; and

(3) NWO II Mod 5 PTR Package.

9. Remarks:

a. The purpose of this EO is to introduce the students to the application of the rules with a specific focus on definitions, lights, shapes and sounds, so they can identify

b. different vessels while performing the duties of lookout during the NWO II sea phase;

c. Students are given homework assignments to learn the required rules prior to each classroom lesson so the time in class is spent confirming knowledge of the rules and providing additional explanations as required; and

d. Only selected International rules are covered in this EO as well as para 4 in the application section of the Canada Shipping Act, 2001 Collision Regulations, and specific definitions found in the Interpretation section.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1-2	<u>Introduction</u> Introduction to include explanation of the references to be used by the students. Location of electronic information (division's student folder, SharePoint and Rules Master Pro software program); 2.NWO II standard (as per this PO); 3. Learning format; 4. Milestones (homework assignments that need to be completed in advance of each class) for NWO II; and 5. Critical Enabling Check (CEC) Collision Regulations – pass mark, format, and method.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	2 x 50 min	
3-4	Explain and discuss the different aspects contained in the following sections of the Canada Shipping Act, 2001 – Collision Regulations		2x50 min	Ref: Collision Regulations (justice.gc.ca)

	<u>Interpretation, Application para 4, Rules 1-3</u> Discuss “Interpretation” and define the following: • Air cushion vessel; • Composite unit; • Exploration or exploitation vessel; • Give-way vessel; • Inconspicuous, partly submerged vessel or object; • Inshore traffic zone; • Seaplane; • Separation zone; • Traffic separation scheme; • Trawling. Discussion “Application” (para 4) Discuss Rules 1 – 3 • Application • Responsibility • General definitions	**Method:** Quizzes, exercise assignments, problem-based learning, demonstrations, peer learning and guided discussions **Media:** Powerpoint, Rules Master Pro, Videos, Audio recordings, NPTT activities **Environment:** Classroom		
5-6	<u>Rules 3 & 20-23</u> Review Rule 3 Explain application of Rules concerning lights and shapes – Rule 20. Define the following (Rule 21): • Sidelights • Masthead Light • Sternlight • Towing Light • All-round light • Flashing lights • Special flashing light • Blue flashing lights	Method: Quizzes, exercise assignments, problem-based learning, demonstrations, peer learning and guided discussions Media:	2 x 50 mins	

	Explain application of Rule 22 The Visibility of Lights – International Explain application of Rule 23 – Power –driven Vessels Underway.	Powerpoint, Rules Master Pro, Videos, Audio recordings, NPTT activities Environment: Classroom		

Optimal sequencing is as follows:

a. Quiz #1 should be at least two days after the Intro class (if the Intro is on Day 1 then Quiz #1 should be no earlier than Day 3)

b. There should be at least one day in between quizzes e.g. if Quiz #1 is on Day 3 then Quiz #2 should be no earlier than Day 5.

EO 003.02 – Apply Rules on Lights, Shapes, Sound Signals, and Restricted Visibility Under Rules 24-26

1. Performance: Apply rules on lights, shapes, sounds signals, and restricted visibility under Rules 24-26.

2. Conditions:

    a. Given:

        (1) ship at sea;

        (2) support, to include:

            (a) bridge team; and

            (b) on-watch personnel.

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (1) on board a ship at sea;

        (2) any time of day and night; and

        (3) any conditions of weather and visibility.

    b. Denied: Nil.

    c. Environment:

        (1) on board a ship at sea;

        (2) any time of day and night; and

        (3) any conditions of weather and visibility.

3 Standard: The Naval Officer shall apply rules on lights, shapes, sounds signals, and restricted visibility under Rules 24-26, to include:

    a. explaining application of Rules concerning lights and shapes (Rule 20);

    b. defining terms in Rule 21 to include:

        (1) sidelights;

        (2) masthead light;

(3) stern light;

(4) towing light;

(5) all-round light;

(6) flashing lights;

(7) special flashing light; and

(8) blue flashing lights.

c. explaining application of Rule 22: Visibility of Lights – International;

d. explaining application of Rule 23: Power- driven Vessels Underway;

e. explaining application of Rule 24: Towing and Pushing – International;

f. explaining application of Rule 25: Sailing Vessels Underway; and

g. explaining application of Rule 26: Fishing Vessels.

4. Teaching Points: See table below.

5. Total PO Time: 18 x 50-minute periods as follows:

a. 2 periods – Intro on Day 1 of the course nautical;

b. 2 periods x 6 – Quizzes (two periods/quiz; Quiz #1-6);

c. 1 period – Pre-CEC Review;

d. 2 periods – CEC; and

e. 1 periods – Post-CEC Review.

6. References:

7. Training and Learning Aids:

a. Flash cards – lights & shapes;

b. Flash cards – rules of the road;

c. Rules Master Pro software;

d. Simulator (NPTT and / or NABS); and

e. Ship at sea.

8. Test Details:

a. Formative: quizzes; and

b. Summative: CEC Collision Regulations.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1-2	<u>Rules 24-26</u> Explain application of Rule 24 – Towing and Pushing – International Explain application of Rule 25 – Sailing Vessels Underway Explain application of Rule 26 – Fishing Vessels	**Method:** Quizzes, exercise assignments, problem-based learning, demonstrations, peer learning and guided discussions **Media:** Powerpoint, Rules Master Pro, Videos, Audio recordings, NPTT activities **Environment:** Classroom	2 x 50 min	

EO 003.03 –Apply Regulations for Vessels with Restricted Maneuverability Under Rules 27-31:

1. Performance: Apply regulations for vessels with restricted maneuverability under Rules 27-31.

2. Conditions:

   a. Given:

      (1) ship at sea;

      (2) support, to include:

         (a) bridge team; and

         (b) on-watch personnel.

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) on board a ship at sea;

      (2) any time of day and night; and

      (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall apply regulations for vessels with restricted maneuverability under Rules 27-31 to include:

   a. explaining application of Rule 27: Vessels not under Command or Restricted in ability to Manoeuvre;

   b. explaining application of Rule 28: Vessels Constrained by their Draught – international;

c. explaining application of Rule 29: Pilot Vessels;

d. explaining application of Rule 30: Anchored Vessels and vessels aground – International; and

e. explaining application of Rule 31: Seaplanes.

4. Teaching Points: See table below.

5. Total PO Time: 18 x 50-minute periods as follows.

a. 2 periods – Intro on Day 1 of the course nautical;

b. 2 periods x 6 – Quizzes (two periods/quiz; Quiz #1-6);

c. 1 period – Pre-CEC Review;

d. 2 periods – CEC; and

e. 1 periods – Post-CEC Review.

6. References.

7. Training and Learning Aids:

a. Flash cards – lights and shapes;

b. Flash cards – rules of the road;

c. Rules Master Pro Software:

d. Simulator (NPTT and/or NABS); and

e. Ship at sea.

8. Test Details:

a. Formative: quizzes; and

b. Summative:

(1) CEC Collision Regulations.

Period #	Description	Instructional Strategies	Time	References/ Remarks

1-2	<u>Rules 27-31</u> Explain application of Rule 27 – Vessels not under Command or Restricted in their Ability to Manoeuvre. Explain application of Rule 28 – Vessels Constrained by their Draught – International Explain application of Rule 29 – Pilot Vessels Explain application of Rule 30 – Anchored Vessels and vessels aground- International. Explain application of Rule 31 - Seaplanes	**Method:** Quizzes, exercise assignments, problem-based learning, demonstrations, peer learning and guided discussions **Media:** Powerpoint, Rules Master Pro, Videos, Audio recordings, NPTT activities **Environment:** Classroom	2 x 50 Mins	

EO 003.04 – Identify and Interpret Signals Used by Vessels in Sight of One Another in Accordance with Rules 32-37:

1. Performance: Identify and interpret signals used by vessels in sight of one another in accordance with Rules 32-37.

2. Conditions:

   a. Given:

      (1) ship at sea;

      (2) support, to include:

         (a) bridge team; and

         (b) on-watch personnel.

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) on board a ship at sea;

      (2) any time of day and night; and

      (3) any conditions of weather and visibility.

3. Standard: The Naval Officer shall identify and interpret signals used by vessels in sight of one another in accordance with Rules 32-37 to include:

   a. defining terms to include:

      (1) whistle;

      (2) short blast; and

      (3) prolonged blast.

   b. explaining application of Rule 33: Equipment for sound signals;

   c. explaining application of Rule 34: Manoeuvring and Warning Signals - international;

   d. explaining application of Rule 35: Sound Signals in Restricted Visibility- International;

   e. explaining application of Rule 36: Signals to Attract Attention; and

f. explaining application of Rule 37: Distress Signals.

4. Teaching Points: See table below.

5. Total PO Time: 18 x 50-minute periods as follows:

a. 2 periods – Intro on Day 1 of the course nautical;

b. 2 periods x 6 – Quizzes (two periods/quiz; Quiz #1-6);

c. 1 period – Pre-CEC Review;

d. 2 periods – CEC; and

e. 1 periods – Post-CEC Review.

6. References:

7. Training and Learning Aids:

a. Flash cards – lights & shapes;

b. Flash cards – rules of the road;

c. Rules Master Pro software;

d. Simulator (NPTT and / or NABS); and

e. Ship at sea.

8. Test Details:

a. Formative: quizzes; and

b. Summative:

(1) CEC Collision Regulations.

Period #	**Description**	**Instructional Strategies**	**Time**	**References/ Remarks**
1-2	<u>Rules 32-37</u> Define the following: • Whistle • Short blast • Prolonged blast	**Method:** Quizzes, exercise	2 x 50 Mins	

	Explain application of Rule 33 – Equipment for sound signals Explain application of Rule 34 – Manoeuvring and Warning Signals – International Explain application of Rule 35 – Sound Signals in Restricted Visibility – International Explain application of Rule 36 – Signals to attract Attention Explain application of Rule 37 – Distress Signals	assignments, problem-based learning, demonstrations, peer learning and guided discussions **Media:** Powerpoint, Rules Master Pro, Videos, Audio recordings, NPTT activities **Environment:** Classroom		

EO 003.05 – Apply Technical Details for Lights, Shapes, and Distress Signals in Annex II & IV

1. Performance: Apply technical details for lights, shapes, and distress signals from Annex II & IV.

2. Conditions:

   a. Given:

      (1) ship at sea;

      (2) support, to include:

         (a) bridge team; and

         (b) on-watch personnel.

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) on board a ship at sea;

      (2) any time of day and night; and

      (3) any conditions of weather and visibility.

3. The Naval Officer shall apply technical details for lights, shapes, and distress signals from Annex II & IV to include:

   a. describing Annex II & IV: positioning and characteristics of lights, shapes and distress signals.

4. Teaching Points: See table below.

5. Total PO Time: 18 x 50-minute periods as follows:

   a. 2 periods – Intro on Day 1 of the course nautical;

   b. 2 periods x 6 – Quizzes (two periods/quiz; Quiz #1-6);

   c. 1 period – Pre-CEC Review;

   d. 2 periods – CEC; and

   e. 1 periods – Post-CEC Review.

6. References:

7. Training and Learning Aids:

a. Flash cards – lights & shapes;

b. Flash cards – rules of the road;

c. Rules Master Pro software;

d. Simulator (NPTT and / or NABS); and

e. Ship at sea.

8. Test Details:

a. Formative: quizzes; and

b. Summative:

(1) CEC Collision Regulations.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1-2	<u>Annex II &IV</u> Describe Annex II & IV	**Method:** Quizzes, exercise assignments, problem-based learning, demonstrations, peer learning and guided discussions **Media:** Powerpoint, Rules Master Pro, Videos, Audio recordings,	2 x 50 Mins	

		NPTT activities **Environment:** Classroom		

PO 004 - Conduct Shiphandling and Seamanship

EO 004.01 - Perform Basic Rope Work

1. Performance: Perform basic rope work.

2. Conditions:

   a. Given:

      (1) ship;

      (2) various types of rope and wire;

      (3) hawser laid rope;

      (4) whipping twine;

      (5) masking tape; and

      (6) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall perform basic rope work to include:

   a. describing rope and wires to include:

      (1) natural fibres;

      (2) synthetic fibres; and

      (3) metal.

   b. describing rope work, to include:

      (1) knots;

      (2) bends;

      (3) hitches; and

      (4) whippings.

c. describing tie knots to include:

(1) bends and hitches;

(2) figure eight;

(3) reef knot;

(4) bowline;

(5) double;

(6) sheet bend; and

(7) clove hitch.

d. describing tie whippings to include:

(1) common;

(2) west country; and

(3) sailmakers.

4. Teaching Points: See table below.

5. Total EO Time: 4 x 50-minute periods as follows:

a. 1 x 50-minute lecture; and

b. 3 x 50-minute practical.

6. References: A10, B24, B25, B26.

7. Training and Learning Aids:

a. aged sample lengths of each type of rope and wire;

b. Lengths of rope;

c. Stanchions, spar, eye, etc.;

d. Length of hawser laid rope and kevlar rope;

e. Heaving lines;

f. Whipping twine; and

g. Masking tape.

8. Test Details:

a. CEC General Naval Knowledge – Part I;

b. Must achieve a minimum of 70% on CEC; and

c. NWO II Mod 5 PTR Package.

9. Remarks:

a. This EO should be instructed prior to any further seamanship and boat work lectures; and

b. This EO should be instructed by the VENTURE CBM or delegated seamanship staff.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Perform Basic Rope Work 1. Describe the types of Ropes and their uses 2. Explain rope handling Precautions 3. Describe safety considerations. 4. Describe material utilized • Natural fibres • Synthetic fibres • Metal 5. Describe rope characteristics and terms • Explain the effects on rope of: ○ Insects ○ Humidity ○ Ultraviolet light 6. Explain the proper care and storage of rope.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	CFCD 105, Admiralty manual of seamanship vol(S) I, II, III

	7. Describe the difference between knots, bends and hitches. 8.Explain Knots, Bends and Hitches • Figure 8 • Reef Knot • Bowline • Double Sheet Bend • Clove Hitch 9. Describe whippings • Common • West Country • Sail Makers 10. Describe characteristics of coiling ropes • Coiling up a line • Coiling down a line • Mistakes in coiling down 11. Describe heaving lines			
2 - 4	Practical: 1. Students to tie: figure eight knot, reef knot, bowline, double sheet bend, and clove hitch; 2. Demonstrate that the slip reef knot is tied by tying the second overhand knot with a bight on one side; Explain that it is used when the knot will be underwater or likely to get wet; 3. Demonstrate: (1) How to prepare a coil for hanging – emphasize the importance of starting above the secured end to avoid kinks. (2) How to cheese and coil laid ropes in the same direction as their lay.	**Method: EDI/Practical** **Media: Instructor** **Environment Classroom**	3x50 min	

	4. Prepare and throw heaving lines. Discuss monkey's fist and other types of weighted knots. 5. Why the length of a whipping should be at least equal to the diameter of the rope, but not exceed twice the diameter; and 6.Apply the following whippings: Common, West Country, and Sailmaker's.			

EO 004.02 - Identify Upper Deck Fittings

1. Performance: Identify upper deck fittings.

2. Conditions:

   a. Given:

      (1) ship; and

      (2) References.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall identify upper deck fittings to include:

   a. locating upper deck fittings in a HMC Ships; and

   b. explaining the purpose of all upper deck fittings.

4. Teaching Points: See table below.

5. Total EO Time: 3 x 50-minute periods as follows.

6. References: A10, B24, B25, B26.

7. Training and Learning Aids: Nil.

8. Test Details:

   a. CEC General Naval Knowledge – Part I;

   b. Must achieve a minimum of 70% on CEC; and

   c. NWO II Mod 5 PTR Package.

9. Remarks:

Period #	Description	Instructional Strategies	Time	References/ Remarks

1	Identify Upper Deck Fittings
	
	1.Identify, locate, and explain the purpose of upper deck fittings to include:
	• Capstands, windlasses and cranes
	• Derricks, booms and davits
	• Eyebolts, ringbolts, eye plates and deck clenches
	• Fairleads, bullrings, angels and panama plates
	• Bits bollards, stag horns and cleats
	• Stanchions, guardrails and turnbuckles
	• Mushrooms, airvents, scuppers, guys, stays and halyards
	• Ladders
	  ○ Jacob
	  ○ Mediterranean
	  ○ Accommodation
	  ○ jumping
	
	2. Explain the purpose of upper deck gear and fittings
	
	3. Explain lifting equipment
	• Mode of operation
	• Maximum load rafting
		Method:
		Lecture
		
		Media:
		Powerpoint
		
		Environment:
		Classroom
			1x50 min	CFCD 105, Admiralty manual of seamanship vol(S) I, II, III

2-3	ORCA Tour	**Method:** Tour **Media:** **Environment:** ORCA Class Vessel	2x50 min	

EO 004.03 - Con the Ship

1. Performance: Con the Ship.

2. Conditions:

   a. Given:

      (1) ship;

      (2) helmsman;

      (3) rudder angle indicator; and

      (4) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall con the ship to include:

   a. maneuvering the ship ISE under normal circumstance, to include:

      (1) delivering conning orders IAW the SSO's (class specific);

      (2) conning the ship consistently to within five degrees of the intended course; and

      (3) explaining ship handling theory to adjust conning.

   b. maneuvering the ship in company with other ships under normal circumstances.

4. Teaching Points: See table below.

5. Total EO Time: 17x 50-minute periods as follows:

   a. 1 x 50-minute lecture;

   b. 2 x 50-minute practical; and

   c. 14 x 50 min periods NABS phase.

6. References: A73, A61.

7. Training and Learning Aids:

   a. NABS; and

   b. NPTT.

8. Test Details:

   a. CEC Navigation Fundamentals;

   b. Must achieve a minimum of 70% on CEC; and

   c. NWO II Mod 5 PTR Package.

9. Remarks: See Annex P for NABS Phase.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Con the Ship 1. Explain the importance of proper conning 2. Deliver conning orders 3. Describe speed orders 4. Describe compass course 5. Describe steering orders 6. Describe helm orders 5. Explain action conning 6. Turning over 7. Adjust conning using ship handling theory. 8. Explain the protocols for altering course • Orders for <5 degree • Orders for 5-10 degrees • Orders for >11	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1 x 50min	SSOs, Orca Class Standing Orders

2	Explain the steering arrangement in CPF and ORCA class vessels to include: • port and stbd • primary • jog • secondary • emergency	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	SSOs, Orca Class Standing Orders
3	CTO practice exercise: -changing helm and speed orders (ORCA) -Requesting permission to turnover the conn -Turning over the conn	**Method:** NABS **Media:** NABS **Environment:** Simulator	1x50 min	
	NABS Exercise	**Method:** NABS **Media:** NABS **Environment:** Simulator	14x50 min	As detailed in Annex P - NABS Handbook

EO 004.04 - Act as Part Ship Officer in Seamanship Evolutions

1. Performance: Act as part ship officer in seamanship evolutions.

2. Conditions:

    a. Given:

        (1) ship; and

        (2) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall describe the duties of part ship officer in seamanship evolutions to include:

    a. explaining duties of part ship officer to include:

        (1) during RAS at sea;

        (2) during towing evolutions;

        (3) when coming and slipping a buoy;

        (4) during anchoring; and

        (5) during line handling evolutions.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute periods as follows.

6. References: A10, A17, B16.

7. Training and Learning Aids: Nil.

8. Test Details:

    a. CEC General Naval Knowledge – Part II; and

    b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	1. Discuss part ship officer safety considerations; 2. Describe the safety considerations throughout RAS evolutions: • liquid, • light jackstay, • heavy jackstay (explanation should include line handling) 3. Describe the safety considerations at different locations while conducting towing: • supply forward, • Supply aft, • receive forward, and • receive aft; 4. Explain part ship officer considerations during mooring for the following: • control of boat • buoy jumpers 5. Explain part ship officer responsibilities during anchoring to include: • safety, • reporting the cable, • and cable terminology. 6. Explain the duties of part ship officer during line handling evolutions	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom **:**	1x50 min	**References:** BWK manual, CFCD 105, ATP 16, CFCD130

EO 004.05 - Act as OOW/OOD for Berthing and Slipping

1. Performance: Act as OOW/OOD for berthing and slipping.

2. Conditions:

   a. Given:

      (1) ship;

      (2) bridge team; and

      (3) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall act as OOW/OOD for berthing and slipping IAW refs, to include:

   a. identifying the personnel required and their duties;

   b. identifying the lines used and their purpose;

   c. identifying types of fenders and how they are used;

   d. explaining all communications and signals used during slipping / berthing;

   e. explaining procedures for securing / slipping tugs;

   f. explaining procedures for berthing / slipping;

   h. describing orders given and their meaning;

   i. describing ceremonial procedures;

   j. describing standard berthing / slipping arrangements;

   k. explaining all safety considerations, gear and procedures;

   l. performing the duties of part ship officer during line handling evolutions;

   m. directing preparations for seamanship evolutions; and

n. initiating special sea duty and special parties' organization.

4. Teaching Points: See table below.

5. Total EO Time: 2 x 50-minute periods.

6. References: A10, A16, B29, B30, B31.

7. Training and Learning Aids:

a. Reading assignment;

b. Line / bollard mock-up;

c. NABS; and

d. Ship.

8. Test Details:

a. CEC General Naval Knowledge – Part II; and

b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period	Description	Instructional Strategies	Time	References/ Remarks

1	Act as OOW for Berthing and Slipping
	
	1.Identify personnel required and their duties and their roles:
	• Special Sea Dutymen
	  ○ Bridge
	  ○ MCR
	  ○ Ops
	  ○ Steering Gear Comp
	  ○ Part ship comms, Aft lookout
	
	• Cable Party
	  ○ Required as per SSOs
		Method:
		Lecture
		
		Media:
		Powerpoint
		
		Environment:
		Classroom
			1x50 min	- CFCD 105: chapter 4 article 4.2.4, and chapter 6 articles 6.1-6.7.2,
				- Admiralty Manual of Seamanship Vol I, II, III,
				- CFCD130

	○ Deck O • Part ship hands • supplemental staff ○ boat crews and lowerers 2. Explain line orders • turn Up • hold • double Up 3. Describe line handling terminology 4. Explain fenders when berthing and slipping • types of fenders • personal safety when handling fenders • marrying fenders • securing fenders 5. Describe berthing lines when coming alongside • four working lines ○ breast lines ○ hurricane hawsers • head rope, fore spring, aft spring, stern rope, breast, after breast, for and aft hurricane hawsers. 6. Describe ship's typical berthing arrangement. 7. Describe the safety precautions for line handling. 8. Explain the arrangements for securing and slipping a tug • use of tug lines • locations of securing lines • safety considerations • line tenders			

	9. Describe part ship communication • SHINCOM/ICS • sound powered telephones • PRCs • Ships internal telephone • General broadcast • Deck weapons net(MCDV) 10. Describe tugs and their role 11. Describe procedure for securing for sea • Lines • Hawsers and fenders • Boats • Brow equipment • Ensign staff • Jack staff 12. Describe procedure for securing alongside in harbour • Rigging hurricane hawsers • Making shore connections • Placing rat guards • Securing gear • Disposing of gash 13. Explain ceremonial considerations • Falling in • Eyes on ship 14. Describe part ship officer considerations • safety • line handling terminology • sequence of events • layout of gear 15. Explain how to initialize special sea duty and special parties' organization • controlling ship's boats • towing • coming to/leaving a buoy			

	• anchoring • entering and leaving harbour. 16. Explain OOW Sequence of Events • preparations the day prior • bridge setup • OOD/OOW turnover • initial pipe • navigation and seamanship brief • OOW checklist and reports • relationship with the Ops room • station in control • roles of 2OOW, XO 17. Describe orders heard from XO Chief Boatswain mate during line handling.			
2	Shiphandling: 1. Discuss heavy weather to include: heavy seas, and strong winds; 2. Discuss berthing/slipping to include: standard approach, standard departure, on/off jetty sets, mental math techniques, propulsion system and rudder effects, and hydrodynamic effects; 3. Discuss shallow water to include: shallow water effect, smelling the ground, speed implications, and manoeuvring implications; and 4. Review canals, locks, and rivers to include: canal effect, bank effect, sheer correcting, ship interaction, and current effects.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1 x 50 min	

EO 004.06 - Perform the Duties of a Small Boat Coxswain

1. Performance: Perform the duties of a small boat coxswain.

2. Conditions:

   a. Given:

      (1) ship;

      (2) Rigid Inflatable Boat (RIB)

      (2) outboard motorboat; and

      (3) references.

   b. Denied: Nil.

   c. Environment:

      1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall perform the duties of a small boat coxswain, to include:

   a. conducting boat communications to include:

      (1) using proper voice procedures;

      (2) interpreting flag signals; and

      (3) interpreting visual signals.

   b. operating a rigid inflatable boat (RIB) to include:

      (1) launching a RIB;

      (2) recovering a RIB;

      (3) handling a RIB under power; and

      (4) conducting troubleshooting.

   c. responding to emergencies to include:

      (1) conducting a rescue and recovery of a casualty;

(2) conducting a tow; and

(3) conducting response to environmental emergency.

d. enforcing all safety requirements for personnel embarked; and

e. overseeing equipment checks.

4. Teaching Points: See table below.

5. Total EO Time: 7 x 50-minute periods as follows:

a. 4 x 50-minute Lecture; and

b. 3 x 50-minute Practical.

6. References: A10.

7. Training and Learning Aids.

8. Test Details: Tested during practical portions of training.

9. Remarks: Teaching points taken from Boat Coxswain Course Modules 2 and 3.

Period #	Description	Instructional Strategies	Time	References/ Remarks

1	Conduct Boat Communications
	
	1. Use communication equipment
	• VHR Radio
	• Loud hailers
	
	2. Use proper voice procedures
	• Operating rules
	• Circuit discipline
	• Prowords
	• Rules for spelling
	• Rules for figures
	• Radiocheck
	  ○ Signal strength.
	  ○ Readability
	
	3. Describe the following
	• Sound signals
	• Light signals
	• Hand signals
	• Boat Hails
	
	4. Explain the use of Pyrotechnics
	• List types
	• Explain operating procedures
	• Describe Safety considerations
	  ○ Safe handling
	  ○ Storage
	
	6. Explain Emergency radio procedures
	• Responding/initiating:
	  ○ PAN PAN
	  ○ Securite
	  ○ MAYDAY
	• RCC/QHM/VTMS
	  ○ Function
	  ○ Responsibility
		Method:
		Lecture
		
		Media:
		Powerpoint
		
		Environment:
		Classroom
			1 x 50 mins	CFCD 105

2 - 4	Conduct Rasic RIB Handling 1. Characteristics of a RIB • Parts • Fuel capacity • Propulsion • Payload • Limitations 2. Enforce all safety requirements for personal embarked 3. Oversee equipment checks. 4. Launch RIB • Crane5. Recover RIB • Crane 5. Operate RIB under power • Engine start/stop procedure • Departing berth • Coming alongside berth ○ Jetty ○ Other boats ○ Ladders 6. Conduct Troubleshooting • Responding to engineering emergences ○ Engine will not start ○ EOPs ○ Problems underway 7. Respond to Emergency • Conduct rescue and recovery of a casualty • Conduct a row • Conduct response to environmental emergency.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	3 x50 min		

5-7	Practical	**Method:** EDI **Media:** Instructor **Environment:** Naval	3 x 50 mins		

PO 005 - Perform Watch Administration

EO 005.01 – Describe the duties and responsibilities of the Watch on Deck

1. Performance: Describe the duties and responsibilities of the Watch on Deck.

2. Conditions:

    a. Given:

        (1) ship;

        (2) CSO;

        (3) SSO, and

        (4) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall describe the duties and responsibilities of the Watch on Deck, to include:

    a. stating the members of the watch on deck;

    b. explaining turnover reports;

    c. stating the members of the engineering watch and their duties and responsibilities;

    d. explaining the duties of the duty supply tech;

    e. listing the members of the home port duty watch;

    f. identifying the difference between a home port duty watch and an away from home port duty watch;

    g. explaining the duties and responsibilities of the members of the home port duty watch IAW SSOs;

h. explaining the process of completing the ship's log;

i. explaining the types of rounds conducted by the duty watch;

j. explaining the purpose and use of the night rounds book;

k. standing as member of the duty watch; and

l. carrying out proper execution of the duties and responsibilities of the OOW/OOD as they relate to watch administration IAW refs.

4. Teaching Points.

5. Total EO Time: 2 x 50-minute periods.

6. References: A7, A17, A73.

7. Training and Learning Aids:

a. NABS;

b. Ship's log;

c. Night rounds book;

d. OOW notebook; and

e. Boatswain's call.

8. Test Details:

a. CEC General Naval Knowledge – Part II; and

b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #1	Description	Instructional Strategies	Time	References/Remarks
1	Direct Watch on Deck 1. Explain composition and responsibilities of the Watch on Deck • OOW • POOW • Lookouts • Boatswain's Mate • Helmsman • Quartermaster 2. Describe duty locations 3. Describe steering console 4. Explain the duties of the duty supply tech. 5. Report to and from the OOW: • lookout report • helm reports • conning (brief discussion, detailed explanation of conning is in EO 004.03) 6. Explaining the need for additional members of the away from home port duty watch; 7. List other personnel and resources that the OOW has available to them • 2OOW • NavComms • The Captain	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	CSO's, SSOs, CFCD 131 - BWK manual for the RCN

	8.List the members of the engineering watch • MCCOP • ORO • Track supervisor • TOW • Engineering roundsperson 9. Explain the degrees of readiness 10. Explain the difference between a home port duty watch and an away from home port duty watch; Direct Watch on Ship			
2	1. Discuss the members of the Home Port Duty Watch and their roles • OOD • OOD Rounds • Duty Petty Officer • Brow watchkeeper • MCR Watchkeeper • Duty cook • Boatswain's mate 2. Discuss members of the Away from Home Port and their roles 3. List the record-keeping responsibilities of the OOD. • Ship's log • Night rounds book • DC exercise log • DC Zone tour log • OOD Turnover sheet	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1 x 50 mins	SSOs, CFCD 131 - BWK Manual for the RCN

EO 005.02 - Implement Ship's Routines

1. Performance: Implement Ship's routines.

2. Conditions:

   a. Given:

      (1) ship;

      (2) CSOs;

      (3) SSOs; and

      (4) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall implement ship's routines to include:

   a. stating the six harbour routines;

   b. explaining when each of the above routines would be used;

   c. explaining the differences in the different daily sea routines;

   d. explaining the meaning of the following:

      (1) out pipes;

      (2) stand easy;

      (3) clear up decks, return gear;

      (4) make and mend; and

      (5) pipe down.

   e. describing night clothing;

   f. explaining which pipes are normally made in the second degree of readiness;

   g. Maintaining the OOW notebook and the ship's log;

h making routine and emergency pipes;

i identifying the parts of a boatswain's call; and

j identifying the pipes made using a boatswain's call.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: A7, A17, A73.

7. Training and Learning Aids:

   a. OOW notebook; and

   b. Boatswain's call.

8. Test Details:

   a. CEC General Naval Knowledge – Part II; and

   b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

TP #	Description	Instructional Strategies	Time	References/ Remarks
1	Implement ship's routines 1. Describe ship's routine at sea throughout the 1st through 4th degree of readiness and the various states of preparedness; 2. Describe ship's daily at sea routines, to include; • daily at sea; • Friday at sea; • sunday at sea. 3. Explain the following: • Out pipes • Stand Easy • Clear up Decks, Return Gear • Make and Mend • Pipe Down 4. Explain the routine pipes 5. Explain piping procedures for second degree of readiness 6. Identify all pipes IAW ship's Daily Routines 7. Identify the parts of a boatswain's call; 8. Identify the pipes made using a boatswain's call. 9. Describe night clothing.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	CSO's, SSO's, CFCD 131 – BWK Manual for the RCN

PO 006 - Direct the Operation of Onboard Engineering Systems

EO 006.01 - Describe the Key Components of Marine and Combat Systems Engineering

1. Performance: Describe the key components of marine and combat systems engineering.

2. Conditions:

   a. Given:

      (1) ship; and

      (2) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall describe the key components of marine and combat systems to include:

   a. the five components of ship's engineering;

   b. basic propulsion;

   c. steering configurations;

   d. electrical configurations; and

   e. auxiliary and combat systems onboard HALIFAX class and ORCA class vessels.

4. Teaching Points as per table below.

5. Total EO Time: 2 x 50-minute periods.

6. References: A51, A67.

7. Training and Learning Aids: Nil.

8. Test Details:

   a. CEC General Naval Knowledge – Part I; and

   b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Describe elements of Ship's engineering 1.Describe the five components of engineering • Propulsion • Steering • Electrical power • Auxiliaries • Navigation/combat systems 2. Describe basic propulsion • Prime movers • Gearboxes • Shaft line and CRPP 3.Explain basic steering engineering • Steering configurations of a Halifax class frigate • Number of rudders • 3 steering positions 4. Explain basic electrical engineering • Power generation and distribution on Halifax class ships • Power generation and distribution on ORCA class vessels.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	Naval Engineering Manual, Safety and Environmental Management System (Halifax Class)

	5. Explain Auxiliaries engineering • production, storage and disposal of black water and grey water. • Storage of fresh water • AC, Ventilation and cooling systems on ORCA and Halifax class frigate. • Electric, diesel pumps used to support a ship 6. Describe components of navigation and combat systems engineering • GPS • SATCOM • Radar • Compass • Speed log • AIS 7. Describe communications and radar systems • 3D surface radar • EW sensors • Fire control radars • SEOSS • HMS • Air search raders • HF/UHF/CHF communications • SATCOM 8. Describe weapon systems • 57mm Bofors II • MASS • Harpoon surface to surface missile/ ESSM surface to air missiles • CIWS • Torpedoes • Towed array systems • Nixie towed acoustic decoy 9. Propulsion configurations of • HALIFAX • KINGSTON • ORCA			

	• CODOG • CODOD • HDW class • JSS 10. Explain planned vs routine maintenance • Procedure to facilitate repair to engineering equipment ○ Assessment, information, repair			
2	Explain engineering systems of RCN ships 1. Engineering systems of RCN ships 2. Maintenance of engineering systems 3. Shore facility support to maintenance.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	Naval Engineering Manual, Safety and Environmental Management System (Halifax Class)

PO 007 - Use Naval Communications

EO 007.01 - Use Flags and Pennants

1. Performance: Use flags and pennants.

2. Conditions:

    a. Given:

        (1) ship;

        (2) flags, pennants flashcards;

        (3) assistance; and

        (4) all meteorological conditions.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall use flags and pennants to include:

    a. recognizing any naval flag or pennant within 5 seconds of observation; and

    b. interpreting the military and civilian meanings.

4. Teaching Points as per the table below.

5. Total EO Time: 1x 50-minute period.

6. References: B14, C23.

7. Training and Learning Aids: Flash cards.

8. Test Details:

    a. CEC General Naval Knowledge – Part I;

    b. Must achieve a minimum of 70% on CEC; and

    c. NWO II Mod 5 PTR Package.

9. Remarks:

a. This lecture should be scheduled before boat coxswain training.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Use Flags and Pennants 1. Explain why flags and pennants are used 2. List equipment associated with flag hoisting 3. Describe all flags and pennants, with emphasis on those commonly used 4. Discuss flags that hold different meanings when at sea or alongside, and the various positions on the mast such as at the dip, or close-up. 5. List noteworthy flag combinations and their meanings. 6. Recognize flags and understand their meanings to include: • Alpha, • Bravo, • Echo, • Gulf, • Hotel, • India (including procedures), • Kilo, • Lima, • Mike, • Oscar, • Papa, • Quebec, • Romeo (including procedures for RAS), • Uniform (including procedures), Zulu, • Flag 5, • Flag 8, • 3rd sub, • Code, • Corpen, • Desig,	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	International Code of Signals, ATP 1 Vol II - Allied Maritime Tactical Signal and Manoeuvring Book

	• Port, • • Prep, and • Starboard.			

PO 008 - Direct Action and Emergency Procedures

EO 008.01 - Describe the Action and Emergency Organization.

1. Performance: Describe the action and emergency organization.

2. Conditions:

    a. Given:

        (1) ship; and

        (2) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall describe the action and emergency organization to include:

    a. explaining the damage control organization;

    b. defining the Emergency Response Team;

    c. outlining the Casualty Clearing Organization;

    d. integrating into the fighting organization;

    e. integrating into the functional organization;

    f. employing the emergency organization; and

    g. identifying when each of the above are to be closed-up.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50 minute period.

6. References: A73.

7. Training and Learning Aids:

    a. HALIFAX class. Know your ship book.

8. Test Details:

a. CEC General Naval Knowledge – Part II; and

b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Describe Action and Emergency Organization 1. Explain the role of Command 2. Explain the damage Control Organization (DC) • Composition • Role • Location of close up 3. Describe the emergency Response Team (ERT) • Occasions for ERT close up • Location 4. Describe Action/Emergency Medical Organization • Casualty Clearing Team ○ Occasions for close up ○ Difference between casualty clearing and medical rapid response team.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	SSOs

EO 008.02 - Act as OOW in Response to Emergencies

1. Performance: Act as OOW in response to emergencies.

2. Conditions:

   a. Given:

      (1) ship; and

      (2) references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall act as the OOW/OOD in response to emergencies to include:

   a. stating the occasions that the general, DCS, Man overboard, and crash on deck alarms are used;

   b. stating the locations from which the general, DCS, man overboard and crash on deck alarms may be initiated and where there are sounds;

   c. explaining the use of action up / down ladders;

   d. identifying the characteristics of ship's pipes;

   e. explain procedures for responding to ship's pipes;

   e. describing security procedures at the brow;

   f. explaining the purpose and duties of jetty / shore patrols; and

   g. defining emergency destruction and its use.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: A73.

7. Training and Learning Aids:

   a Ship;

b. Ship's diagram;

c. SSO; and

d. Sample bridge pack.

8. Test Details:

a. CEC General Naval Knowledge – Part II; and

b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #	Description	Instructional	Period #	Description
1	Act as OOW for Response to Emergencies 1. Explain Emergencies-Basic Principles 2. Describe ship alarm systems • DCS Alarm • General alarm • Lifebuoy release alarm • Crash on deck alarm 3. Identify characteristics of ship's pipes to include pipes for: • Emergency stations • Emergency flying stations • man overboard • abandon ship • verification muster • survival muster • survival stations • abandon ship • bacteriological or chemical attack • shelter station • bomb threat	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	SSOs

	• explain reactions to pipes listed above 4. Describe battle Damage Control System 5. Describe DCS Alarm vs Confirmed incident 6.Explain how to brace for shock 7. Describe Emergency stations • Emergency Flying stations • Action stations • Rescue stations • Shelter stations • Survival stations 8. Locate the Ship's ladders 9. Explain Emergency destruction			

EO 008.03 - Explain the Ship's Fighting Organization

1. Performance: Explain the ship's fighting organization.

2. Conditions:

   a. Given:

      (1) watch and station bill; and

      (2) special parties and RAS board.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall explain the ship's fighting organization to include:

   a. describing the watch system;

   b. describing the watch and station bill;

   c. describing the stations of the watch;

   d listing the different special parties and RAS teams; and

   e. describing the degrees of readiness and states or preparedness.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: A73.

7. Training and Learning Aids:

   a. Watch and station bill;

   b. Special parties / RAS board; and

   c. Know your ship book.

8. Test Details:

   a. CEC General Naval Knowledge – Part I; and

b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Explain the Ship's Fighting Organization 1. Explain the difference between the functional and fighting organizations; 2. Explain how the fighting organization is governed by the watch and station bill and special parties and RAS boards; 3. Explain the Cox'n's role in maintaining the watch and station bill and special parties board. 4. Explain watch systems • two, • three • four 5. Explain each department's watch and station bill • describe how positions are determined using SSOs; 6. Explain the positions found on each of the documents and what they are/when they would be closed up; 7. Explain the abbreviations used • FX, • AX • SSD • PSH, etc...); 8. Explain the difference between SSD and PSH and when they would be closed up; 9. Explain the importance of the Watch and Stations Bill and the	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1 x 50min	SSOs

	responsibility each sailor has to know their positions. 10. Explain degrees of readiness 1 – 5. 11. Explain the states of preparedness and the difference between them.			

PO 009 - Support Operations other than Warfare

EO 009.01 - Describe Key Events in the History of the RCN

1. Performance: Describe key events in the history of the RCN.

2. Conditions:

    a. Given: References.

    b. Denied: Nil.

    c. Environment: N/A.

3. Standard: The Naval Officer shall describe key events in RCN history from its inception to the present to include:

    a. Events leading to the formation of the RCN;

    b. Canada's contribution to the WWI effort;

    c. Post-WWI naval reduction;

    e. Creation of the naval reserves and their role between WWI / WWII;

    g. Canada's contribution to WWII;

    h. Post-WWII demobilization;

    i. Formation of NATO and Canadian involvement;

    h. The Mainguy report;

    j. Fleet acquisitions; and

    j. Current operations.

4. Teaching Points: See table below.

5. Total EO Time: 8 x 50-minute periods total:

    a. 6 x 50 minute lecture; and

    b. 2 x 50 minute museum tour.

6. References: The Official History of the RCN.

7. Training and Learning Aids:

   a. Lecture;

   b. Guided discussion; and

   c. Museum tour.

8. Test Details: Not tested.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	1. Discuss Canada's geographic and geostrategic circumstances • Militia Act of 1862 • Fenian Raids of 1866 2. Explain the implications of the following Maritime affairs: • The fishery disputes, • The Treaty of Washinton 1871 • The disbandment of the Marine Police. 3. Discuss the RCN's withdrawal of fleet to home waters in 1903. 4. Describe the BattleShip Race (Dreadnought) 5.Describe the 1909 Canadian Naval debate 6. Discuss the Austrialian approach to imperial defence. 7. Discuss the arrival of NIOBE and RAINBOW. 8. Explain the difficulties with the RN and establishment of RCN Operating Areas. 9. Explain the consequences of the 1911 Election.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	The Official History of the RCN

2	1.Discuss the Post-WWI naval reduction; 2. Discuss the outcome of the 1917 conference on Post –War Imperial Naval Defence 3.Explain the significance of the Jellicoe Report its recommendations. 4. Discuss the First Crisis (Budget cuts) 5. Discuss the founding of the naval reserves by Walter Hose, and discuss their their role between WWI / WWII; 6. Discuss the first warships purpose built for Canada. • Saguenay • Skeena 7. Discuss the second crisis (1933) 8. Describe the Naval Policy on the eve of WWII.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1 x 50 mins	The Official History of the RCN <u>Quick research project</u>: Divide students into small groups to discuss the potential long-term impacts of the Jellicoe Report's recommendations on the RCN. Each group shares their thoughts.

3	1.Explain the challenges that came with the rapid expansion of the RCN. 2. Explain the -Role of Women in the RCN • -The Women's royal Canadian naval service • -Western approach tactical unit • Jobs • -Integration into the RCN 3. Discuss the German western offensives of 1940. 4. Discuss the RCN's first wartime losses • Fraser • Margaree • Saquenay 5. Explain the Uboat Peril the battle of the Atlantic 6. Describe the U-boat numbers throughout the war. 7. Discuss technological and tactical advances to include: • Sonar • Hedge • Hog • Radar • HFDF • Acoustic Torpedoes • Canadian Anti Torpedo device (CAT) • Western Approaches Tactical Unit 8. Explain the Lende Lease 9. Discuss the 'Corvette' • Life in a Corvette – LS Franks Curry Audio	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	Life in a Corvette – LS Franks Curry Debby Piers audio on Winter Weather of 42/43

	10. Discuss the significance of ‘Engima’and ‘Ultra’. 11. Discuss the impact of the Winter Weather of 42/43 • Near loss of Resstigouche 12. Discuss the presence of the U Boats in the Gulf of St Lawrence. 13. Describe the RCN Operational Pause of winter 1943, including Churchill's letter and Mackenzie King's response." 14. Discuss the Escort Groups • The famous 5th. 15. Describe the Tribal Class Destroyer 16. Discuss Canada’s first carriers • NABOB • PUNCHER 17. Discuss the involvement of the RCN in the Normandy Landings: • Air, Land, and Sea – Joint Operations • ASW Screens • Force J • Mine sweeping Operations • 14th, 31st, and 32nd Flotillas • The Shore Bombardment and NGF support • Troop transports, LCI(L) Flotillas, and LCA Flotillas • RCN Beach Commando "W"			

	18. Discuss the involvement of the RCN in the mediterranean. 19. Discuss the actions of LT(N) Robert Hampton Grey – Canada's last Victoria Cross recipient. 20. Explain HMCS Ontario and HMCS Uganda 21. Discuss the actions of LT(N) Robert Hampton Grey – Canada's last Victoria Cross recipient. 22. Discuss Women at War – the Women's Royal Canadian Naval Service. 23. Discuss the conclusion of WWII.			
4-6	1. Explain the rapid demobilization of the RCN 2. Discuss key events from 1947 to 1949. • Ontario • Magnificent • Cayuga • Crescent 3. Discuss the implications of the Mainguy report 4. Explain the formation of NATO. 5. Explain Train busting in the Korean war. 6. Describe the St. Laurent Class 7. Explain Canada's focus on ASW. 8. Describe the following: • Variable depth Sonar towed Arrays 9. Describe the operation of helicopters off small ships 10. Discuss the Cuban Missile Crisis. 11. Describe Canadian Carriers post WWII. 12. Describe HMCS Labrador.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom/ museum	3x 50 min	

	13. Discuss the role of Canada's O boats 14. Describe the Unification crisis. 15. Describe Canada's AORs 16. Discuss Canada's participation in the NATO mission to re-enforce Norway on the left Flanks. 17. Discuss Canada's inclusion of women in the sea going Navy. 18. Discuss the 'Sisters of the Space Age' - new tribal class. 19.Discuss the end of the Cold war and the new missions (Maritime Interdiction). 20. Discuss the first Gulf War of 1991. 21. Discuss the arrival of the Halifax and Victoria classes. 22. Discuss the implications of 911 and the RCN's role in the OP Apollo. 23. Discuss RCN HADR missions. 24. Describe the RCN Operations off Libya. 25. Discuss recent INDOPAC Operations 26. Discuss the return to a new cold war and the resurgence of NATO.			

7-8	Students will have a museum visit for 2 periods	**Method:** Museum visit **Media:** Powerpoint **Environment:** Class/museum	2 x 50 mins	

EO 009.02 - Practice Customs and Traditions of the RCN

1. Performance: Practice customs and traditions of the RCN.

2. Conditions:

   a. Given:

      (1) PPT;

      (2) classroom;

      (3) PP&E;

      (4) computer;

      (5) projector;

      (6) references; and

      (7) self-study materials.

   b. Denied: Nil.

   c. Environment: Classroom.

3. Standard: The Naval officer shall practice customs and traditions of the RCN to include but not limited to:

   a. participating in naval traditions;

   b. abiding by naval customs;

   c. displaying proper wardroom etiquette;

   d. attending mess events; and

   e. abiding by dress regulations.

4. Teaching Points: See table below.

5. Total EO Time: 2 x 50-minute periods.

6. References:

7. Training and Learning Aids:

8. Test Details: Not tested.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1 and 2	Practice Customs and Traditions of the RCN Explain the following: 1. Saluting the brow 2. Christmas 3. New Year's 4. Baptisms 5. Crossing the Line 6. Barber Pole Group 7. Up Spirits 8. Splicing the Main Brace Explain the following components of Mess Events: 1. Mess Structure 2. Wardroom Etiquette 3. Cocktail Parties 4. Mess dinners 5. Dress regulations	**Method:** Self-study with presentation **Media:** Peer to peer. **Environment:** Classroom	2x50 min	Student will be divided into pairs ahead of this lesson. Each pair will research one custom/tradition of the RCN and present their findings to the course during this lesson. Instructor is to ensure that any information missed during the research is discussed at the end of each presentation.

EO 009.03 - Apply Naval Terminology

1. Performance: Apply naval terminology.

2. Conditions:

    a. Given:

        (1) ship; and

        (2) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall apply naval terminology to include:

    a. defining naval terms to include:

        (1) naval rank structure;

        (2) origin of naval ranks;

        (3) origin of common terms; and

        (4) definitions.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: A10, A16.

7. Training and Learning Aids: Nil.

8. Test Details:

    a. CEC General Naval Knowledge – Part I;

    b. Must achieve a minimum of 70% on CEC; and

    c. NWO II Mod 5 PTR Package.

9. Remarks: This EO should be instructed at an early stage, to enable subsequent practical application.

Period#	Description	Instructional Strategies	Time	References/ Remarks

1	Apply Naval Terminology
	
	1.Explain traditions and fundamentals
	• Salty Dips Project
	• Hearts of Oak
	
	2. Describe naval Ranks and Appointments
	• Origin of naval ranks
	• Naval ranks
	  ○ Flaq officers
	  ○ Senior officers
	  ○ Senior NCOs
	  ○ Junior NCMs
	• Origin of Naval Appointments
	
	3. Define naval terminology to include:
	• General: hull, fore part, midships part, after part, the stem, the stern, fore and aft, starboard, port; and the house;
		Method:
		Lecture
		
		Media:
		Powerpoint
		
		Environment:
		Classroom
			1x50 min	CFCD 105, CFCD 130

2	• The hull surfaces: keel, bow, transom, and top part; • Terms applied to the hull: freeboard, draught, beam, length overall, camber, bilge keel, masthead height, shaft, rudder, “A” bracket, sonar dome, discharges, intakes, decks, weather, flats, forecastle (FX), quarterdeck/aftercastle (AX), and deck numbering; • Naval slang: BZ, cart, defaulters, dhobey, duff, flats, Irish pennant, killick, molly, oggie, pick, pusser, rack, scran, scran locker, scull, skate, skive, splice the mainbrace, stand easy, stoker, tiddly, tiffy, and zoomie; • Movement about the ship: going, forward, lay aft, below, aloft; • Movement of the ship: underway, way on, headway, sternway, gathers way, and steerage way; • Sideways moving vessel: broadside to, leeway, weather side, lee side, and drift rate; • Descriptions relative to ship: on the quarter, on the bow, dead ahead, dead astern, fine on the quarter/bow, abaft the beam, abeam, and objects relative bearings (i.e. object bears green six zero) and how it translates to true bearings; and	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 Min	CFCD 105, CFCD 130

	• Nautical measurements: nautical mile, cable, shackle, fathom. 7. Naval terms and origins 8. Naval punishments 9. Abbreviations			

EO 009.04 - Conduct Ceremonial Procedures

1. Performance: Conduct ceremonial procedures.

2. Conditions:

    a. Given:

        (1) ship; and

        (2) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall perform ceremonial procedure, IAW references to include:

    a. executing ceremonial procedures, to include:

        (1) colours;

        (2) sunset; and

        (3) half-masting.

    b. explaining ceremonial procedures to include:

        (1) occasions for dressing ship;

        (2) dress ship procedures;

        (3) terminology relating to flags and ceremonial; and

        (4) passing honours between ships.

4. Teaching Points: See table below.

5. Total EO Time: 2 x 50-minute periods.

6. References: A39.

7. Training and Learning Aids: Nil.

8. Test Details:

a. CEC General Naval Knowledge – Part I;

b. Must achieve a minimum of 70% on CEC; and

c. NWO II Mod 5 PTR Package.

9. Remarks:

a. This EO should be instructed early in the course to enable trainees to participate / practice the ceremonies of Colours and Sunset as a member of the HMCS VENTURE duty watch.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	Conduct Ceremonial Procedures 1. Define: General naval terms: • Ship's ensign • Naval jack • Out of routine • Dress overall • Tracking 2. Describe procedures: • Bent on • In hand • Close up • Breaking 2. Describe flags and pennants • Out of routine • Starboard Pennant • Preparative pennant 3. Explain proceedings for colours and sunset • Use of PREP • Bell • Orders • Pipes • Times • Sunday ensign/jack • Personnel required • Half masting procedures	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	Manual of Ceremony for HMC ships

	4. Explain dressing ship • Purpose of strongback • use of signals • use of masthead flag 5. Explain use of masthead flags and breaking masthead flags 6. Explain location and precedence of masts and halyards 7. Explain dipping the Ensign 8. Explain procedure for mark of respect passed between warships in harbour 9. Demonstrate method of folding an ending/jack for storage. 10. Explain Flag handling 11. Describe Flags and Pennants	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2	CTO should use VENTURE halyards (Nixon Gym) to demonstrate ceremonial procedures including several slow time practices.	**Method:** Practical **Media:** **Environment:** Gym	1x50 min	Manual of Ceremony for HMC ships

EO 009.05 - Employ Personal Safety Procedures

1. Performance: Employ personal safety procedures.

2. Conditions:

   a. Given:

      (1) ship;

      (2) references;

      (3) checklists;

      (4) safety equipment; and

      (5) supervision.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall employ personal safety procedures IAW references to include:

   a. following occupational safety requirements;

   b. describing precautions to be taken in times of heavy weather;

   c. explaining safety procedures for personnel aloft;

   d. explaining safety procedures for personnel over the side;

   e. explaining safety procedures for working in or around hazardous areas;

   d. identifying the safety equipment located at the brow and how it operates; and

   e. maintaining awareness of the weather deck and its safety precautions.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: A67.

7. Training and Learning Aids:

   a. Sample burning and welding chit;

   b. Sample DND 2145 man aloft chit;

   c. Cane;

   d. Rubber gloves;

   e. Flash gear (gloves);

   f. Diagram of emitters; and

   g. NABS.

8. Test Details:

   a. CEC General Naval Knowledge – Part I;

   b. Must achieve a minimum of 70% on CEC; and

   c. NWO II Mod 5 PTR Package.

9. Remarks:

   a. This EO is applied practically during the Mod 5 OJT.

1	Employ Personal Safety Procedures 1. Follow safety procedures in void spaces • identify hazards • actions to take • encountering a casualty 2. Follow safety procedures when operating in a cold room • define cold room • identify hazards • explain alarm system • handling casualties when at risk of freon leaks 3. Follow safety procedures when operating near electrical hazards • cane and rubber gloves • handling casualties around electrical hazards 4. Follow safety procedures when burning and welding • process to follow when burning or welding • duties of a fire sentry and their dress 5 Operate safely when aloft • define aloft • identify hazards • permissions needed to go aloft	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	Safety and Environmental Management System
	6. Discuss Brow safety • identify safety equipment located at Brow • how to use brow safety equipment 7. Operate in rough weather • explain out of bounds			

	• explain life lines • securing heavy equipment • close opening on KINGSTON class vessels • explain POOW rounds			

EO 009.06 - Explain Naval Doctrine

1. Performance: Explain naval doctrine.

2. Conditions:

   a. Given: references.

   b. Denied: Nil.

   c. Environment:

      (1) at sea or alongside; and

      (2) all meteorological conditions.

3. Standard: The Naval Officer shall explain the principal elements of naval doctrine, to include:

   a. key doctrinal terms as defined in naval doctrine publications; and

   b. the roles of the RCN in domestic and international operations.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References:

7. Training and Learning Aids: Nil.

8. Test Details:

   a. CEC General Naval Knowledge Part I; and

   b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	1. Define “policy”, “doctrine”, and “strategy”. - Canada’s Defence Policy (2017 to 2024) - RCN’s Strategy (Leadmark 2020 to Trident) 2. Discuss the relevance of naval power to include: - Canada and the international framework of maritime commerce - Sea lines of communication 3. Discuss the unique attributes of naval forces: flexibility/high readiness, diplomacy/demonstration of national interests, freedom to pre-position/forward deploy. 4. Discuss Booth’s triangle: - Military - Diplomatic - Constabulary 5. Discuss examples of RCN activities and operations that support each role.	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	Canada’s Defence Policy 2017 – Strong, Secure, Engaged Canada’s Defence Policy 2024 – Our North Strong and Free Leadmark: The RCN Strategy for 2020 Trident: Royal Canadian Navy Strategic Doctrine CFJP 01 – Canadian Military Doctrine (2009)

EO 009.07 - Identify General Ship Types

1. Performance: Identify general ship types.

2. Conditions:

    a. Given: general ship images.

    b. Denied:

        (1) references; and

        (2) assistance.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The naval officer shall identify general ship types to include but not limited to:

    a. working vessels;

    b. and pleasure craft;

    c. Canadian Government vessels to include:

        (1) coast guard;

        (2) fisheries and oceans; and

        (3) RCMP.

    d. other vessels to include:

        (1) trawlers;

        (2) trolling vessels;

        (3) tugs;

        (4) ferries; and

        (5) hovercraft.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: C29.

7. Training and Learning Aids:

   a. Janes Fighting Ships.

8. Test Details:

   a. CEC General Naval Knowledge – Part I; and

   b. Must achieve a minimum of 70% on the CEC.

9. Remarks:

   a. Focus on describing distinguishing features of each vessel.

TP #	Description	Instructional Strategies	Time	References/ Remarks
1	1. Describe types of small vessels and passenger vessels: • Pleasure craft • Fishing vessels • Tugs • Barges, Booms and other marine rigs • Ferries 2. Describe government vessels • Coast guard • Fisheries and oceans • RCMP 3.Describe and state the general characteristics of each of the major types of merchant vessels and tankers: • break bulk, • bulk, • auto-carrier, • ro-ro, • tugs, • barges (different types) etc.; 4. Describe other vessels encountered including: • trawlers, • trolling vessels,	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	Jane's Fighting Ships

	• tugs, • ferries, • hovercraft, and • pleasure craft.			

EO 009.08 - Identify RCN Ships and Supporting Aircraft

1. Performance: Identify RCN ships and supporting aircraft.

2. Conditions:

    a. Given: various images of RCN supporting aircraft.

    b. Denied:

        (1) references; and

        (2) assistance.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall identify RCN ships and supporting aircraft to include:

    a. recognizing RCN ships to include:

        (1) CPF;

        (2) AOPS;

        (3) AOR;

        (4) MCDV;

        (5) auxiliary and tender vessels; and

        (6) VICTORIA class.

    b. stating the capabilities of RCN ships to include:

        (1) Aurora;

        (2) Cyclone;

        (3) Cormorant; and

        (4) CF 18 Hornet.

    c. stating the capabilities of RCN aircraft.

4. Teaching Points: See table below.

5. Total EO Time: 1 x 50-minute period.

6. References: Nil.

7. Training and Learning Aids: Nil.

8. Test Details:

   a. CEC General Naval Knowledge – Part I; and

   b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Focus on describing distinguishing features of each aircraft.

Period #	Description	Instructional Strategies	Time	References/ Remarks
1	1.Describe the capabilities and characteristics of RCN ships to include: • HALIFAX class (FFH), • VICTORIA class (SSK), • PROTECTEUR class (AOR), • KINGSTON class (MCDV), • HARRY DE WOLF class, • ORCA class (PCT), • ketch sail training (HMCS ORIOLE); 2. Describe the capabilities and characteristics of RCN supporting aircraft of the maritime air group to include: • Aurora, • Sea King, • Cormorant, • CF 18 Hornet. 3. Describe the capabilities and characteristics of Canadian Government assets • CH-149 Cormorant • Bell 429 GlobalRanger • Bell412EPI • Department of Fisheries and Oceans Aircraft • RCMP Aircraft	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	

PO 0010 - Support Air Operations

EO 010.01 - Describe Basic Safety Procedures for Shipbourne Air Operations

1. Performance: Describe basic safety procedures for shipbourne air operations.

2. Conditions:

    a. Given:

        (1) helicopter;

        (2) ship; and

        (3) references.

    b. Denied: Nil.

    c. Environment:

        (1) at sea or alongside; and

        (2) all meteorological conditions.

3. Standard: The Naval Officer shall describe basic safety procedures during helicopter operations, to include:

    a. conducting FOD rounds;

    b. securing head dress and other gear;

    c. identifying and avoiding out of bounds areas;

    d. identifying safety procedures to be followed on the flight deck when embarking / disembarking a helicopter or when working on the flight deck with an on deck helicopter;

    e. identifying safety procedures to be followed when working on the flight deck with an on-deck helicopter

    f. explaining safety procedures when operating in high-risk areas to include:

        (1) JP5;

        (2) exhaust areas;

        (3) rotor safety zones;

        (4) embarking a helicopter alongside;

g. explaining procedures for smoking safety; and

h. explaining procedures for operating in out of bounds areas.

4. Teaching Points: See table below.

5. Total EO Time:1 x 50-minute period.

6. References: A72, A73.

7. Training and Learning Aids: Nil.

8. Test Details:

a. CEC General Naval Knowledge – Part I; and

b. Must achieve a minimum of 70% on the CEC.

9. Remarks: Nil.

Period#	Description	Instructional Strategies	Time	References/ Remarks
1	Describe Basic Safety Procedures for Shipbourne Air Operations 1. Describe air departments on a ship 2. Describe characteristics of a helicopter 3. Explain components of the Air Department 4. Define Foreign Object Damage (FOD) 5. Explain coming to flying stations 6. Define FOD rounds 7. Explain OOB areas during flying stations. 8. Define Crash on deck 9. Explain Crash on deck safety precautions 10. Explain safety around the helicopter	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom	1x50 min	SSOs, Shipborne Helicopter Operating Procedures (SHOPs)

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		Official CAF Publications and Directives that relate to this Qualification
A1	A-AD-265-000/AG-001	Canadian Armed Forces Dress Manual
A2	C-09-153-003/TS-000	Canadian Armed Forces Explosives Manual
A3	A-PD-201-000/PT-000	Canadian Armed Forces Manual of Drill and Ceremonial
A4	B-GJ-005-501/FP-000	Canadian Armed Forces Use of Force Manual
A5		Canadian Forces Administrative Orders (CFAOs)
A6		Canadian Forces General Messages (CANFORGEN)
A7		Captain's Standing Orders
A8		CFCD 102 - Maritime Command Combat Readiness Requirements
A9		CFCD 103 – Canadian Submarine Combat Readiness Requirement
A10		CFCD 105 - Seamanship Rigging and Procedures Manual
A11		CFCD 106 – Maritime Tactical Instructions
A12		CFCD 107 - Maritime Command Communications
A13		CFCD 108 - Naval Boarding and Force Protection Operations Manual
A14		CFCD 109 - Naval Signature Management Manual
A15		CFCD 114 - Maritime Command Electronic Warfare Orders
A16		CFCD 130 - Canadian Navigation Manual
A17		CFCD 131 - BWK Manual for the Royal Canadian Navy
A18		CHP 1 - Hydrographic Supplies Manual
A19		CHP 5 – Regulations Affecting HMC Ships in Navigable Waters
A20		CHP 7 - Nautical Chart Symbols, Abbreviations and Related Material (DND Item 839)
A21		CHP 9 - Ship's Log (DND Item 910)
A22	B-GS-316-011/AG-000	Concepts and Weapons Effects
A23		Defence Administrative Orders and Directives (DAODs)
A24		Defence Team Writing Guide
A25		DIP 100
A26	C-02-040-009/AG-000	DND General Safety Standards
A27		Environmental Management System (EMS)
A28	A-PA-005-000/AP-139	Fighting Spirit: The Profession of Arms in Canada
A29	B-GS-316-014/FP-001	Formation and Unit Procedures
A30	A-AE-000-001/AG-E00	Guide to the Divisional System
A31		HMCS VENTURE Routine Orders
A32	B-GS-316-013/FP-001	Individual Procedures
A33		JTF(P) CONPLAN SUBSMASH
A34		KHM – Esquimalt Harbour Practices and Procedures
A35		King's Regulations and Orders (KR&Os)
A36	B-GJ-005-014/FP-021	Law of Armed Conflict

A37	C-22-502-000/MF-000	Life Preserve Yoke Martime Pouch
A38	C-22-552-000/MF-001	Life Preserver Yoke Hazardous Duty
A39		Manual of Ceremony for HMC Ships, Submarines and Naval Reserve Divisions
A40		MAOP 331 – Master Air Operations Plan
A41		Maritime Atlantic Orders (MARLANTORDS)
A42		Maritime Pacific Orders (MARPACORDS)
A43		MARPAC Guide to Administrative Practices and Writing Procedures
A44		MCCR
A45		Military Justice at the Unit Level
A46		MILOC
A47		Mk 46 Mod 5 Torpedo Technical Manual
A48		National Defence Act
A49		National Defence Security Instructions
A50		National Defence Security Policy
A51	C-03-005-033/AA-001	Naval Engineering Manual Vol II
A52		Naval General Messages (NAVGEN)
A53		Naval Orders (NAVORDS)
A54		Navigation Data Book (Class Specific)
A55		Navigation Data Book (NABs Bridge Pack)
A56		Navigator's Information Sharing Project (Website)
A57		NCPM 1000 – Above Water Warfare Procedures Manual
A58		NCPM 1001 – Underwater Warfare Procedures Manual
A59		NCPM 704 – 50 Calibre Heavy Machine Gun Drills & Procedures Manual
A60		ORCA Class Bridge Pack
A61		ORCA Class Standing Orders
A62		Our North, Strong and Free: A renewed Vision for Canada's Defence
A63		PaCE References and Quick Reference Guides
A64		Pacific Coast Catalogue of Nautical Charts and Related Publications (HSO Item #836)
A65		Pronotes – The RCN Shiphandler's Journal
A66		RelVel Handbook (ODIP 201)
A67		Safety and Environmental Management System (SEMS)
A68		Ship's Stability and Notes for Ship's Officers
A69	C-23-045-000/AG-001	Shipboard Damage Control
A70	C-23-045-000/AG-002	Shipboard Damage Control Vol II
A71	C-23-045-000/AG-002	Shipboard Nuclear, Biological and Chemical Defence
A72		Shipborne Helicopter Operating Procedures (SHOPs)
A73		Ships Standing Orders (SSOs)
A74	B-GL-382-002/FP-001	Shoot to Live and Training in Battle
A75	AN/AY 503	Sonobuoy Processor Operating Procedures
A76		Strong, Secure, Engaged – Canada's Defence Policy

A77		Survival at Sea
A78	A-PA-005-000/AP-138	The CAF Ethos: Trusted to Serve
A79		Tug Assistance Manual
B		Non-CAF Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification
B1		ACP 125 -Communication Instructions – Radiotelephone Procedures
B2		ACP 130 - Communications Instructions: Signaling Procedures in the Visual Medium
B3		ACP 167 – Glossary of Communications Electronic Terms
B4		ACP 176 – Allied Naval and Maritime Air Communications Instructions
B5		ADATP 11 Vol 1&2 – Allied Data Processing Publication 11 SOP for NATO Link 11 & Link 11b C/W Chg 1
B6		ADMIRALTY List of Lights and Fog Signals
B7		ADMIRALTY Total Tides Software
B8		ANP 1 - NATO Master Navigation Supplement
B9		APP 1 Vol 1 - Allied Maritime Voice Reporting Procedures
B10		APP 11 – NATO Message Catalogue
B11		APP 2 – Helicopter Operations from Ships Other Than Aircraft Carriers (HOSTAC)
B12		APP 7 – Joint Brevity Words Publication
B13		ATP 1 Vol I – Allied Maritime Tactical Instructions and Procedures
B14		ATP 1 Vol II - Allied Maritime Tactical Signal and Manoeuvring Book
B15		ATP 10 – Search and Rescue
B16		ATP 16 - Replenishment at Sea (RAS)
B17		ATP 24 Vol 1 – Naval Mine Countermeasures Tactics and Execution
B18		ATP 28 – Allied Anti-Submarine Warfare Manual (NU)
B19		ATP 31 – NATO Above Water Warfare Manual
B20		ATP 43 – Ship-to-Ship Towing -
B21		ATP 57 – Submarine Search and Rescue Manual
B22		AXP 5 – NATO Experimental Tactics and Amplifying Tactical Instructions
B23		BR 134 – Collisions and Groundings and Other Accidents
B24		BR 45 - Admiralty Manual of Navigation Vol I
B25		BR 45 - Admiralty Manual of Navigation Vol II
B26		BR 45 - Admiralty Manual of Navigation Vol III
B27		BR 45 - Admiralty Manual of Navigation Vol IV
B28		BR 45 - Admiralty Manual of Navigation Vol VI
B29		BR 67 - Admiralty Manual of Seamanship Vol I
B30		BR 67 - Admiralty Manual of Seamanship Vol II
B31		BR 67 - Admiralty Manual of Seamanship Vol III
B32		BR 67 - Admiralty Manual of Seamanship Vol IV
B33		BR 67 - Admiralty Manual of Seamanship Vol V
B34		BR 67 - Admiralty Manual of Seamanship Vol VI

B35		Canada Shipping Act, 2001
B36		CCG - Annual Notice to Mariners
B37		CCG – List of Lights, Buoys and Fog Signals (4 volumes)
B38		CCG - Navigation Warnings (NAVWARN)
B39		CCG - Radio Aids to Marine Navigation
B40		CHS – Canadian Atlas of Tidal Currents
B41		CHS – Canadian Sailing Directions
B42		CHS - Canadian Tide and Current Tables
B43		CHS – Chart 1 – Symbols, Abbreviations, and Terms
B44		Fisheries Act
B45		Fundamentals of Naval Leadership (US Naval Academy)
B46		Manual of Marine Weather Observations (MANMAR)
B47		MC 362 - NATO Rules of Engagement
B48		MXP 1 – Multinational Submarine and Anti-Submarine Exercise Manual
B49		MXP 2 – Multinational Maritime Above-Water Warfare Exercises Manual
B50		NP 134a - Admiralty Manual of Hydrographic Surveying (AMHS) Vol I
B51		NP 134b - Admiralty Manual of Hydrographic Surveying (AMHS) Vol II
B52		NP 135 – Admiralty General Instructions for Hydrographic Surveyors
B53		Pub No.9 – American Practical Navigator
B54		Quick Reference Guide to Canada's Navigation Aids
B55		TP968 – Canadian Aids to Navigation System
B56		Transportation of Dangerous Goods Act
B57		Transportation Safety Board Marine Investigation and Reports
C		Civilian Publications and Manuals that relate to this Qualification
C1		57MM Technical Manual
C2		A Guide to the Collision Avoidance Rules (Cockcroft & Lameijer)
C3		A Seaman's Guide to the Rule of the Road (Ford)
C4		AIS Operating Manual
C5		AN/SQS-510 Operator's Manual
C6		Behaviour In Organizations (Cdn Edition - Prentice Hall Canada, Inc.)
C7		Bridge Master 180/250 Radar Series Ship's Manual
C8		Bridge Procedure Guide
C9		Bridge Team Management Guide
C10		CANTASS User Manual
C11	B-GS-316-012/FP-001	CBRN Operational Equipment
C12	C-77-260-000/MB-001	Chemical Agent Monitor (CAM) Manual
C13		Collisions and Their Causes (Cahill)
C14		Communications Set – SONAR AN/WQC 501(V) UWT
C15		Echo Sounder User's Manual
C16		ECPINS-M Operators Manual

C17		Farewell's Rules of the Nautical Road (Allen)
C18		Flash Cards – Lights and Shapes
C19		Flash Cards – Rules of the Road
C20		Fundamentals of Naval Leadership (US Naval Academy)
C21		Harpoon Users Guide Vol I
C22		Industry Canada RIC-23 – Issue 3, dated 3 March, 1997
C23		International Code of Signals
C24		International Hydrographic Organization (IHO) Special Publication 52 – Specifications for Chart Content and Display Aspects of ECDIS
C25		International Hydrographic Organization (IHO) Special Publication 57 – Transfer Standard for Digital Data Edition 3.0
C26		International Maritime Organization (IMO) Performance Standards for ECDIS
C27		International Maritime Organization, Model Course 1.27 – The Operational Use of ECDIS
C28		International Regulations for Preventing Collisions at Sea
C29		Jane's Fighting Ships (Jane's Defence Publishing)
C30		Marine Accident Investigation Branch (MAIB) Safety Digests
C31		Mariner's Handbook
C32		Maritime New Zealand Incident & Investigation Reports and Insights
C33		Nautical Almanac
C34		Naval Shiphandling (Crenshaw)
C35		Navigation Accidents and Their Causes (The Nautical Institute)
C36		Nuclear Weapons and their Effects (Brassey)
C37		Personal Time Management 3rd Edition (Marion E. Haynes)
C38		Raytheon Instruction Manual Vols I-III Pathfinder / ST Radar and ARPA System
C39		Rules Master Pro Software – A guide to the Nautical Rules of the Road
C40		Safety of Life at Sea (SOLAS) Consolidated
C41		Sailing Directions / US Coast Pilots
C42		SEOSS Handbook
C43		SHINCOM Operating Manual
C44		SM2 Firing Guidance Manual
C45		The Navigator Magazine (The Nautical Institute)
C46		Workplace Hazardous Material Information System (WHMIS)
D		Other Publications and Manuals that relate to this Qualification
D1		Sail Skills – Collision Regulations (website)
D2		
D3		
D4		
D5		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

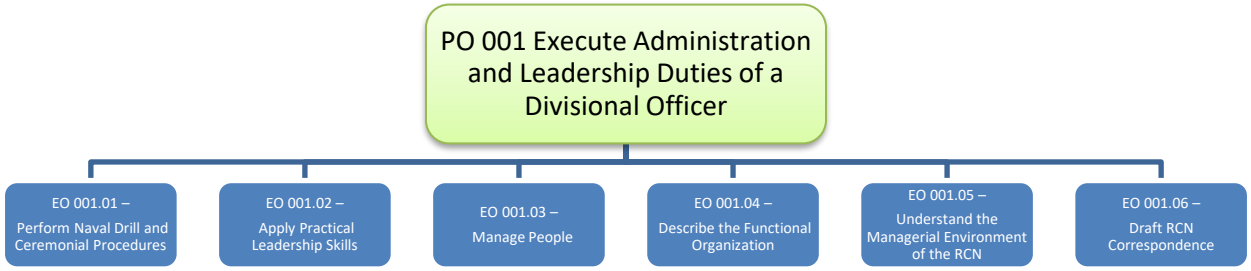
CRITERIA	DEFINITION
A	Previously qualified. Tasks and knowledge already possessed by the student by CAF and / or RCN courses, training and / or experience.
B	Will receive on future courses. Tasks and knowledge will be covered as part of other CAF and / or RCN courses and / or training.
C	Difficulty. Task is not difficult to learn; formal training is not required.
D	Importance. Task is not important; it has no significant impact on mission success.
E	Frequency. Task is rarely performed; no formal training is required.
F	Lack of equipment. Currently there is no equipment in the CAF inventory to train this task.

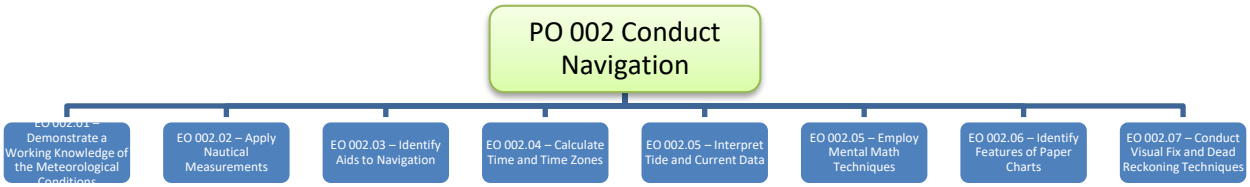
TASK #	Task Statement	Train / No Train	PO #	Substantiation/Comments
	Common Work Requirements - OGS			
A-1, A-2, A-3, A-4, B-1	All tasks at SLt level	NT		A – Tasks covered in BMOQ
B-2, B-3, C-1, C-2, C-3	All tasks at SLt level	NT		B – Tasks to be covered in future courses
C0320	Administer a remedial measure	NT		B – Task to be covered in future courses
C0330	Advise personnel on benefits and entitlements	NT		B – Task to be covered in future courses
C0340	Inform personnel on policies, procedures, orders, and regulations	T	PO 001	
C0370	Enable interpersonal conflict resolution	NT		B – Task to be covered in future courses
C0380	Produce an administrative product	T	PO 001	
C0390	Update administrative records of personnel	NT		B – Task to be covered in future courses
	SER Master Task List - OGS			

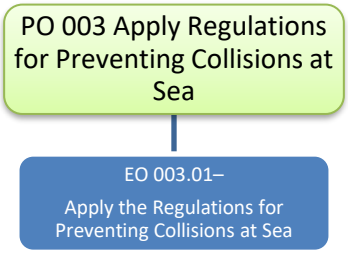
A0010	Conduct naval ceremonial requirements	T	PO 009	
A0020	Initiate damage control (DC) procedures in naval environment	T	NETP Mod 3	
A0030	Execute damage control (DC) duties in Naval environment	T	NETP Mod 3	
A0060	Conduct responsibilities of a Naval Duty Watch	T	PO 005	
A0070	Initiate firefighting procedures in Naval environment	T	NETP Mod 3	
A0080	Execute firefighting duties in Naval environment	T	NETP Mod 3	
A0110	Direct Naval Duty Watch	T	PO 005	
A0140	Supervise seamanship and shipboard evolutions at sea	T	PO 004	
A0160	Supervise seamanship and shipboard evolutions alongside in harbour	T	PO 004	
A0180	Enforce shipboard safety regulations at sea	T	PO 009	
A0190	Enforce shipboard safety regulations alongside in harbour	T	PO 009	
A0200	Enforce MARCOM Environmental Protection policies (e.g. Formation Safety and Environmental Management System (SEMS), Ship Class EMS, and Marine Mammal Mitigation)	T	PO 009	
A0210	Initiate shipboard CBRN protective measures	T	NETP Mod 3	
A0240	Conduct sea survival procedures	T	NETP Mod 4	
A0250	Supervise ship's husbandry procedures	T	Mod 5 OJT	
A0270	Secure for sea	T	Mod 5 OJT	
A0280	Perform pre-sail procedures	T	Mod 5 OJT	
A0310	Perform Naval Sword Drill	T	PO 001	

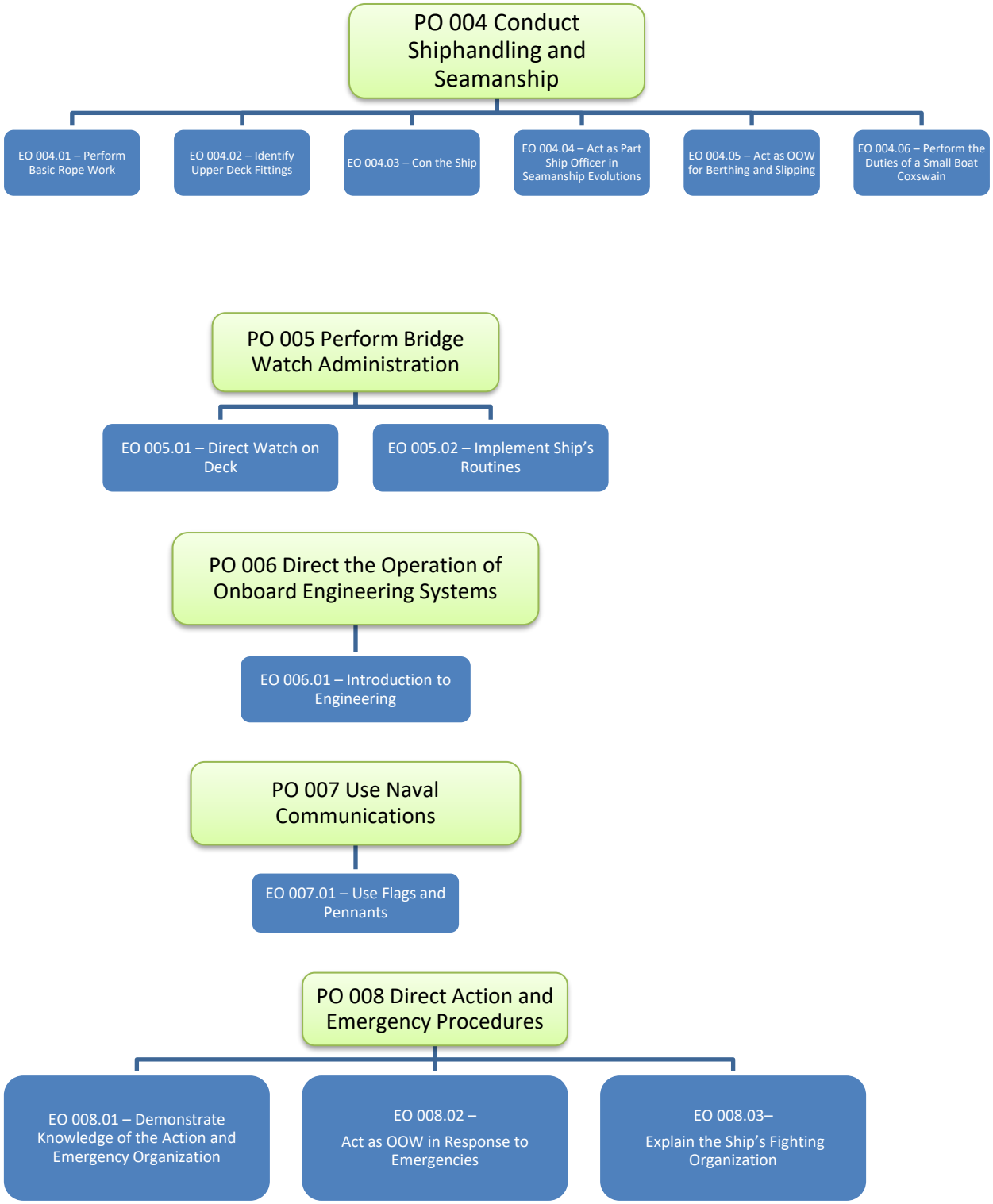
A0320	Maintain Naval Readiness Requirements	T	PO 009	
A0340	Enforce Naval Readiness Requirements	T	PO 009	
A0360	Execute Maritime emergency response plans	T	PO 008	
A0380	Execute Maritime Force Protection plans	T	NETP Mod 2	
C0010	Manage personnel IAW The Divisional System	T	PO 001	
C0020	Execute charge alongside	NT		B – Covered on OOD OJT
	Master Task List - NWO			
AT0010	Navigate the ship in coastal / ocean waters	T	PO 002	Introductory knowledge only
AT0060	Prevent Collisions at Sea	T	PO 003	
AT0080	Maintain ship's logs and records	T	PO 005	
AT0100	Direct personnel during seamanship evolutions and routines	T	PO 004	
AT0110	Coordinate blind pilotage / anti-collision information	NT		E – task not performed in the fleet
AT0120	Conduct charge of a ship at sea	T	PO 004	Introductory knowledge only
BT0240	Contribute to the Recognized Maritime Picture (RMP)	T	PO 009	

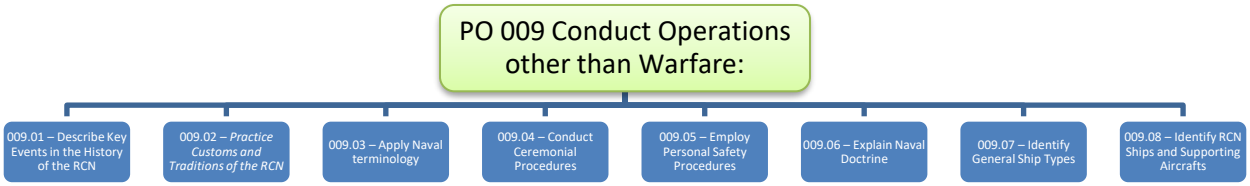
ANNEX C – QS SCALAR

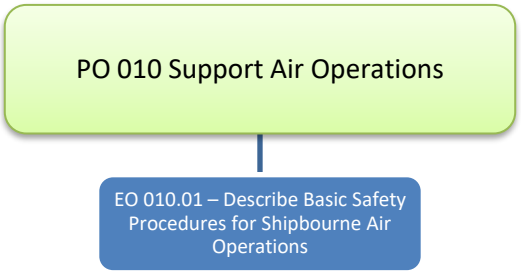







PO 004 Conduct Shiphandling and Seamanship
EO 004.01 – Perform Basic Rope Work
EO 004.02 – Identify Upper Deck Fittings
EO 004.03 – Con the Ship
EO 004.04 – Act as Part Ship Officer in Seamanship Evolutions
EO 004.05 – Act as OOW for Berthing and Slipping
EO 004.06 – Perform the Duties of a Small Boat Coxswain
PO 005 Perform Bridge Watch Administration
EO 005.01 – Direct Watch on Deck
EO 005.02 – Implement Ship's Routines
PO 006 Direct the Operation of Onboard Engineering Systems
EO 006.01 – Introduction to Engineering
PO 007 Use Naval Communications
EO 007.01 – Use Flags and Pennants
PO 008 Direct Action and Emergency Procedures
EO 008.01 – Demonstrate Knowledge of the Action and Emergency Organization
EO 008.02 – Act as OOW in Response to Emergencies
EO 008.03– Explain the Ship's Fighting Organization


PO 009 Conduct Operations other than Warfare:
009.01 – Describe Key Events in the History of the RCN
009.02 – *Practice Customs and Traditions of the RCN*
009.03 – Apply Naval terminology
009.04 – Conduct Ceremonial Procedures
009.05 – Employ Personal Safety Procedures
009.06 – Explain Naval Doctrine
009.07 – Identify General Ship Types
009.08 – Identify RCN Ships and Supporting Aircrafts


PO 010 Support Air Operations
EO 010.01 – Describe Basic Safety Procedures for Shipbourne Air Operations

ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions, and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title		NWO Training Review		
Location of Board		HMCS VENTURE		
Dates		11-14 December 2023		
Board Composition				
	Initials	Surname	Rank	Unit
Co-Chair (content)				
Co-Chair (process)	D	Gray	LCdr	VEN
TM	M	Parent	Lt(N)	NTG HQ
SME	T	Kennedy	Civ	VEN
SME	K	Greenwood	Civ	VEN
SME	R	Macquarrie	Civ	VEN
SME	S	Kucher	Lt(N)	VEN
SME	M	Couturier	Lt(N)	VEN
Secretary	D	Hoare	A/SLt	VEN
Advisor	R	Whitman	SLt	VEN

	Details	Action
Pre-Board discussion points regarding modularization of NWO II between Occupation Managers and HMCS VENTURE NWO Division Commander; LCdr C. Erickson		
1	Increased recruiting and higher demand on TE mean longer waits for training after completion of BMOQ.	Modularize NWO II to enable occupation BTL managers to maximize throughput by sending trainees on NETP modules separately from NWO II training serials. Compress NWO II academic material into single module.
2	Log and Int occupations do not require extra navigation training for junior officers, however, require exposure to life at sea prior to A/HOD posting or deployment.	Log and Int O training will take NETP modules 1, 2, 3, and 4, and then complete the NWO II at sea OJPR. Log and Int BLT managers to coordinate sea time directly with PCTU during quarterly ORCA Crewing Working Group.
3	NTO occupation does not require NWO specific training and has requested a 2-week abbreviated course to cover general naval knowledge and safety IOT increase throughput.	PO 002 and 003, and portions of 004 have been removed from the training for NTO students decreases the course length to 10 days of training from 25. Specific details for PO 004 are described in the remarks section. New MITE created to differentiate between NWO and NTO training: NWO II mod 1 NTO (123935)
4	NTO students will complete their at-sea practical requirements during the NEI course.	No OJT package required to complete NWO II requirements.
5	The time between training for Weapons Directing Officer procedures and .50 cal weapons drills and the point at which the trainee is expected to use these skills is too long. There is a significant amount of skill fade.	Move WDO performance objective to NWO IV.
6	Trainees who transfer to NWO after completing the Log O / Int O or NTO IET plan with not have the baseline foundational knowledge to continue to NWO III.	Pre-requisite for NWO III will be: NWO II mod 1 (123898), NETP mod 2 (118532), NETP mod 3 (118533), NETP mod 4 (118534), and NWO II mod 5 (123899).
Discussion points during December 2023 board:		

7	The board's overall objective was to discuss threading NWO training from NWO II through to NWOPQ and best practices for NWOPQ training to increase throughput of qualified NWOs into the fleets.	
8	The board looked at all the existing POs and EOs for NWO training and agreed upon a set of common POs for all levels of training. The board moved existing EOs under the new PO structure. No content was removed; however, some training was moved to a level that was more appropriate for the learner. NWO II will be the first course formalized under this PO structure.	PO 001 - Execute Administration and Leadership Duties of a Divisional Officer PO 002 - Conduct Navigation PO 003 - Apply Regulations for Preventing Collisions at Sea PO 004 - Conduct Shiphandling and Seamanship PO 005 - Perform Bridge Watch Administration PO 006 - Direct the Operation of Onboard Engineering Systems PO 007 - Use Naval Communications PO 008 - Direct Action and Emergency Procedures PO 009 - Support Operations other than Warfare PO 010 - Support Air Operations PO 011 - Support Above Water Warfare Operations PO 012 - Support Under Water Warfare Operations PO 013 - Implement Weapon Safety Precautions PO 014 - Operate Bridge Equipment PO 015 - Apply Bridge Resource Management Best Practices PO 016 - Direct Formation Manoeuvres and Screening as OOW
9	PO 011 – PO 016 will not be trained at the NWO II level and are not included in this QS.	
10	Two elements of training were added to the performance objectives; Apply bridge resource management and ship construction and stability.	Concepts introduced in NWO II, and further amplified through NWO III, NWO IV, and NWOPQ.

11	All Performance Objectives in this QS are designed and written in a building block format that continues through NWO III, NWO IV, and ending with NWOPQ. The conditions are based on conditions an NWO will be expected to perform under when qualified, though the elements of the standard are designed for the specific level of training being conducted.	
12	Changed PO 005 name from ‘Perform Bridge Watch Administration’ to ‘Perform Watch Administration’.	

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification for the amendment section, the member imputing the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	
Date:	Who Initiated Change:
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	
Training Manager:	

TRAINING PLAN ANNEXES

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities / Trainers

SER	DESCRIPTION	QTY	REMARKS
a.	Navigation and Bridge Simulator (NABS)		
b.	Naval Part Task Trainer (NPTT)		
c.	Boatshed with qualified staff		Boat coxswain (IRB) training
NOTES			

3. Specialized / Technical Equipment

SER	DESCRIPTION	QTY	REMARKS
a.	Laptop	1/student	Includes case, power supply, mouse, and ethernet cable.
b.	Admiralty Total Tides license	1/student	
c.	ECPINS license	1/student	
NOTES			

4. Transport / Vehicles

SER	DESCRIPTION	QTY	REMARKS
a.	Bus transportation	As req.	Off campus tours and training
b.			
c.			
NOTES			

7. Communication Equipment / Frequencies

SER	DESCRIPTION	QTY	REMARKS
a.			
b.			
c.			
NOTES			

8. Training Aids

SER	DESCRIPTION	QTY	REMARKS

a.	Rules Master Pro 2		
b. NSN: 1R6910-00-5124-2033-TD	International Flags and Pennants Flash Cards	1/student	
c.			
NOTES			

10. General Purpose Stores

SER	DESCRIPTION	QTY	REMARKS
a.	Rolling/Parallel ruler	1/student	30cm or greater in length
b.	Dividers	1/student	Must be sturdy and able to hold shape when set
c.	Compass	1/student	Must be sturdy and able to hold shape when set
d. 6605-99-523-6999	Blundell Harling 0655.37 Portland Speed-Time-Distance Slide Ruler/Calculator	1/student	30 cm
e.	IALA Bravo Buoyage Flash Cards	1/student	
f.	Rules of the Road Lights and Shapes Flash Cards	1/student	

g. NSN: 7530-21-872-2301	Officer of the Watch/Navigator's Notebook (CF 481)	1/bridge	
h.			
NOTES			

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.	Printer Paper & Toner		1 box	Exams, quizzes, and handouts.
b.	Pens, Pencils, Erasers		2 of each /student	Chartwork, Exams, Quizzes.
c.				
NOTES				

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5-year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course:		Serial:				
TD	No. of Students / Instructors	Cost per each	# Days (If applicable)	Initial TE Set-up	Each Successive Course	5 Year Total
Airfare (Return)	24/4		1			
R&Q	24/4		35			
Incidentals	24/4		35			
Subtotal						
Material	Quantity	Cost per each	# Days (Enter 1 if not applicable)			
PP&S (Estimate)						
Classroom (Square Meters)	1					
IT Equipment (Projector, Smart board, etc.)	1	$7000.00	1			
DWAN connection (Set up)	25					
Link Toolkit (on DVD)						
Instructional Materials (Printed / Handouts, etc.)	24					
Subtotal						
PYs (only if course includes incremental on SWE, or course to be delivered off site)	Quantity	Cost per day (SWE if necessary)	# Days (If applicable)			
Instructor (PO1 / WO) (Incremental)						
Incremental Civilian Instructor	2		6			
		Subtotal				

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as “Notes”” following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

	NWO Mod 1		NTO Mod 1	
Allocation of ATT	Periods	Days	Periods	Days
Actual Training Time	124	17.7	62	8.9
ATT Subtotal				
Allocation of BAT, TAT, TTT, PT				
Base Administrative Time (BAT)	6	0.9	1	0.1
Training Administrative Time (TAT)	22	3.1	7	1
Training Travel Time (TTT)	-	-	-	-
Physical Fitness (PT)	16	2.3	0	0
Subtotal				
Course Duration Calculation (ATT + BAT + TTT + PT)				
Total Training Days	168	24	70	10

OPTIMUM TRAINING SCHEDULE – NWO II Mod 1

Week 1 Time	Day 1	Day 2	Day 3	Day 4	Day 5
Period 1 0800 - 0850	Course Introduction	009.04 – Conduct Ceremonial Procedures	001.01 – Perform Naval Drill and Ceremonial Procedures	009.02 – Practice Customs and Traditions of the RCN	Physical Training
Period 2 0900 - 0950	TAT – CTO/Student Interview	009.04 – Conduct Ceremonial Procedures	009.06 – Explain Naval Doctrine	009.02 – Practice Customs and Traditions of the RCN	Physical Training
Period 3 1000 - 1050	TAT – CTO/Student Interview	009.03 – Apply Naval Terminology	009.05 – Employ Personal Safety Precautions	001.02 – Apply Practical Leadership Skills (OLQ)	002.01 – Explain the Effect of Meteorological Conditions on Shiphandling
Period 4 1100 - 1150	TAT – Stores Issue	007.01 – Use Flags and Pennants	010.01 – Describe Basic Safety Procedures for Shipbourne Air Operations	001.02 – Apply Practical Leadership Skills (Ldrshp Discussion)	002.01 – Explain the Effect of Meteorological Conditions on Shiphandling
Period 5 1300 - 1350	003.01 – Apply the Regulations for Preventing Collisions at Sea	009.07 – Identify General Ship Types	003.01 – Apply the Regulations for Preventing Collisions at Sea	006.01 – Describe Key Components of Marine and Combat Systems	003.01 – Apply the Regulations for Preventing Collisions at Sea
Period 6 1400 - 1450	003.01 – Apply the Regulations for Preventing Collisions at Sea	009.08 – Identify RCN Ships and Supporting Aircraft	003.01 – Apply the Regulations for Preventing Collisions at Sea	006.01 – Describe Key Components of Marine and Combat Systems	003.01 – Apply the Regulations for Preventing Collisions at Sea
Period 7 1500 - 1550	001.02 – Apply Practical Leadership Skills (PLQ)	001.04 – Describe the Functional Organization	Physical Training	008.03 – Explain the Ship's Fighting Organization	BAT – Personal Admin

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

Week 2 Time	Day 6	Day 7	Day 8	Day 9	Day 10
Period 1 0800 - 0850	004.01 – Perform Basic Rope Work	004.06 – Perform Duties of a Small Boat Coxswain	001.01 – Perform Naval Drill and Ceremonial Procedures	**CEC - General Naval Knowledge – Part 1**	Physical Training
Period 2 0900 - 0950	004.01 – Perform Basic Rope Work	004.06 – Perform Duties of a Small Boat Coxswain	004.05 – Act as OOW for Berthing and Slipping	**CEC - General Naval Knowledge – Part 1**	Physical Training
Period 3 1000 - 1050	004.01 – Perform Basic Rope Work	004.06 – Perform Duties of a Small Boat Coxswain	004.05 – Act as OOW for Berthing and Slipping	Physical Training	008.01 – Describe Action and Emergency Procedures
Period 4 1100 - 1150	004.01 – Perform Basic Rope Work	004.06 – Perform Duties of a Small Boat Coxswain	004.04 – Act as Part Ship Officer in Seamanship Exercises	001.03 – Manage People	088.02 – Act as OOW in Response to Emergencies
Period 5 1300 - 1350	004.02 – Identify Upper Deck Fittings	004.06 – Perform Duties of a Small Boat Coxswain	003.01 – Apply the Regulations for Preventing Collisions at Sea	001.05 – Describe the Managerial Environment of the RCN	003.01 – Apply the Regulations for Preventing Collisions at Sea
Period 6 1400 - 1450	004.01/004.02 – ORCA Tour	004.06 – Perform Duties of a Small Boat Coxswain	003.01 – Apply the Regulations for Preventing Collisions at Sea	001.05 – Describe the Managerial Environment of the RCN	003.01 – Apply the Regulations for Preventing Collisions at Sea
Period 7 1500 - 1550	004.01/004.02 – ORCA Tour	004.06 – Perform Duties of a Small Boat Coxswain	TAT – General Naval Knowledge – Part 1 – Pre-Exam Review	BAT – Personal Admin	002.05 – Employ Mental Math techniques

Week 3 Time	Day 11	Day 12	Day 13	Day 14	Day 15
Period 1 0800 - 0850	001.02 – Apply Practical Leadership Skills (Ldrshp in CAF)	002.02 – Apply Nautical Measurements	001.01 – Perform Naval Drill and Ceremonial Procedures	002.04 – Calculate Time and Time Zones	Physical Training
Period 2 0900 - 0950	001.02 – Apply Practical Leadership Skills (Ldrshp in CAF)	002.03 – Identify Aids to Navigation	004.03 – Con the Ship	002.04 – Calculate Time and Time Zones	Physical Training
Period 3 1000 - 1050	005.01 – Direct Watch on Deck	002.03 – Identify Aids to Navigation	004.03 – Con the Ship (NPTT)	002.03 – Identify Aids to Navigation (NPTT)	009.01 – Describe Key Events in the History of the RCN
Period 4 1100 - 1150	005.01 – Direct Watch on Deck	002.05 – Employ Mental Math Techniques	004.03 – Con the Ship (NPTT)	002.03 – Identify Aids to Navigation (NPTT)	009.01 – Describe Key Events in the History of the RCN
Period 5 1300 - 1350	005.02 – Implement Ship's Routines	003.01 – Apply the Regulations for Preventing Collisions at Sea	**CEC - General Naval Knowledge – Part 2**	003.01 – Apply the Regulations for Preventing Collisions at Sea	002.06 – Identify Features of Paper Charts (Nautical Chart)
Period 6 1400 - 1450	001.06 – Draft RCN Correspondence	003.01 – Apply the Regulations for Preventing Collisions at Sea	**CEC - General Naval Knowledge – Part 2**	003.01 – Apply the Regulations for Preventing Collisions at Sea	002.06 – Identify Features of Paper Charts (Nautical Chart)
Period 7 1500 - 1550	001.06 – Draft RCN Correspondence	Physical Training	**CEC - General Naval Knowledge – Part 2**	002.05 – Employ Mental Math Techniques	TAT – Collision Regulations Pre-Exam Review

Week 4 Time	Day 16	Day 17	Day 18	Day 19	Day 20
Period 1 0800 - 0850	**CEC – Collision Regulations**	Physical Training	001.01 – Perform Naval Drill and Ceremonial Procedures	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	Physical Training
Period 2 0900 - 0950	**CEC – Collision Regulations**	Physical Training	002.07 – Conduct Visual Fix and Dead Reckoning Techniques (EP & Tidal Set)	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	Physical Training
Period 3 1000 - 1050	002.06 – Identify Features of Paper Charts (Symbols)	002.05 – Employ Mental Math Techniques	002.07 – Conduct Visual Fix and Dead Reckoning Techniques (EP & Tidal Set)	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	**CEC – Navigation Fundamentals**
Period 4 1100 - 1150	002.06 – Identify Features of Paper Charts (Symbols)	002.05 – Employ Mental Math Techniques	BAT – Personal Admin	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	**CEC – Navigation Fundamentals**
Period 5 1300 - 1350	002.07 – Conduct Visual Fix and Dead Reckoning Techniques (Chartwork skills)	002.07 – Conduct Visual Fix and Dead Reckoning Techniques (Dead Reckoning)	009.01 – Describe Key Events in the History of the RCN	002.05 – Employ Mental Math Techniques	001.02 – Apply Practical Leadership Skills
Period 6 1400 - 1450	002.07 – Conduct Visual Fix and Dead Reckoning Techniques (Fix the Ship)	002.07 – Conduct Visual Fix and Dead Reckoning Techniques (Dead Reckoning)	009.01 – Describe Key Events in the History of the RCN	002.05 – Employ Mental Math Techniques	001.02 – Apply Practical Leadership Skills
Period 7 1500 - 1550	002.07 – Conduct Visual Fix and Dead Reckoning Techniques (Fix the Ship)	BAT – Personal Admin	TAT – Collision Regulations Post-Exam Review	TAT – Navigation Fundamentals Pre-Exam Review	BAT – Personal Admin

Week 5 Time	Day 21	Day 22	Day 23	Day 24	Day 25
Period 1 0800 - 0850	Navigation & Bridge Simulator – Practical Exercise	Navigation & Bridge Simulator – Practical Exercise	001.01 – Perform Naval Drill and Ceremonial Procedures	Physical Training	
Period 2 0900 - 0950	Navigation & Bridge Simulator – Practical Exercise	Navigation & Bridge Simulator – Practical Exercise	BAT – Personal Admin	Physical Training	
Period 3 1000 - 1050	Navigation & Bridge Simulator – Practical Exercise	Navigation & Bridge Simulator – Practical Exercise	009.01 – Describe Key Events in the History of the RCN	TAT – Hot Wash Up	
Period 4 1100 - 1150	Navigation & Bridge Simulator – Practical Exercise	Navigation & Bridge Simulator – Practical Exercise	009.01 – Describe Key Events in the History of the RCN	TAT – Return Stores/Out routine	
Period 5 1300 - 1350	Navigation & Bridge Simulator – Practical Exercise	Navigation & Bridge Simulator – Practical Exercise	009.01 – Museum Tour	TAT – Return Stores/Out routine	
Period 6 1400 - 1450	Navigation & Bridge Simulator – Practical Exercise	Navigation & Bridge Simulator – Practical Exercise	009.01 – Museum Tour	TAT – Course Reports & Certificates	
Period 7 1500 - 1550	Navigation & Bridge Simulator – Practical Exercise	Navigation & Bridge Simulator – Practical Exercise	TAT – Course Critique	TAT – Course Reports & Certificates	

OPTIMUM TRAINING SCHEDULE – NWO II Mod 1 - NTO

Week 1 Time	Day 1	Day 2	Day 3	Day 4	Day 5
Period 1 0800 - 0850	TAT – Course Intro	009.07 – Identify General Ship types	004.01 – Perform Basic Rope Work	008.01 – Describe the Action and Emergency Organization	Physical Training
Period 2 0900 - 0950	TAT – Stores Issue	009.08 – Identify RCN Ships and Supporting Aircraft	004.01 – Perform Basic Rope Work	008.02 – Act as OOW in Response to Emergencies	Physical Training
Period 3 1000 - 1050	001.02 – Apply Practical Leadership Skills (PLQ)	101.01 – Describe Basic Safety Procedures for Shipbourne Air Operations	004.01 – Perform Basic Rope Work	008.03 – Explain the Ship's Fighting Organization	009.01 – Describe Key Events in the History of the RCN
Period 4 1100 - 1150	009.03 – Apply Naval Terminology	009.05 – Employ Personal Safety Precautions	004.01 – Perform Basic Rope Work	006.01 – Describe Key Components of Marine and Combat Systems Engineering	009.01 – Describe Key Events in the History of the RCN
Period 5 1300 - 1350	007.01 – Use Flags and Pennants	009.02 – Practice Customs and Traditions of the RCN	004.02 – Identify Upper Deck Fittings	004.04 – Act as Part Ship officer in Seamanship Evolutions	005.02 – Implement Ship's Routines
Period 6 1400 - 1450	009.04 – Conduct Ceremonial Procedures	009.02 – Practice Customs and Traditions of the RCN	001.02 – Apply Practical Leadership Skills	004.05 – Act as OOW for Berthing and Slipping	TAT – General Naval Knowledge – Part 1 – Pre-Exam Review
Period 7 1500 - 1550	009.04 – Conduct Ceremonial Procedures	009.06 – Explain Naval Doctrine	001.02 – Apply Practical Leadership Skills	004.05 – Act as OOW for Berthing and Slipping	BAT – Personal Admin

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

Week 2 Time	Day 6	Day 7	Day 8	Day 9	Day 10
Period 1 0800 - 0850	**CEC - General Naval Knowledge – Part 1**	009.01 – Describe Key Events in the History of the RCN	001.06 – Draft RCN Correspondence	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	Physical Training
Period 2 0900 - 0950	**CEC - General Naval Knowledge – Part 1**	009.01 – Describe Key Events in the History of the RCN	001.06 – Draft RCN Correspondence	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	Physical Training
Period 3 1000 - 1050	001.03 – Manage People	001.05 – Describe the Managerial Environment of the RCN	009.01 – Describe Key Events in the History of the RCN	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	TAT – Hot Wash Up
Period 4 1100 - 1150	001.04 – Describe the Functional Organization of the RCN	001.05 – Describe the Managerial Environment of the RCN	009.01 – Describe Key Events in the History of the RCN	001.02 – Apply Practical Leadership Skills (Ldrshp Panel)	TAT – Return Stores/Out routine
Period 5 1300 - 1350	005.01 – Direct Watch on Deck	001.02 – Apply Practical Leadership Skills	**CEC - General Naval Knowledge – Part 2**	009.01 – Museum Tour	TAT – Course Reports & Certificates
Period 6 1400 - 1450	005.01 – Direct Watch on Deck	001.02 – Apply Practical Leadership Skills	**CEC - General Naval Knowledge – Part 2**	009.01 – Museum Tour	TAT – Course Reports & Certificates
Period 7 1500 - 1550	001.02 – Apply Practical Leadership Skills	001.02 – Apply Practical Leadership Skills	**CEC - General Naval Knowledge – Part 2**	TAT – Course Critique	TAT – Course Reports & Certificates

ANNEX H – ASSESSMENT PLAN

EC / CEC	Scope	Title	How	Time	Resources	Limitations	Pass / Fail Criteria	Impact
Module 1:								
CEC	PO 002	Navigation Fundamentals	Written	2 periods	Computer and internet access	Require access to DLN	70%	First unsuccessful attempt will result in remedial training and a second attempt at the CEC. A second unsuccessful attempt will result in a TRB.
CEC	PO 003	Collision regulations	Written	2 periods			70%	First unsuccessful attempt will result in remedial training and a second attempt at the CEC. A second unsuccessful attempt will result in a TRB.
CEC	PO 004 PO 005 PO 006 PO 007 PO 008 PO 009 PO 010	General Naval Knowledge – Part 1	Written	2 periods	Computer and internet access	Require access to DLN	70%	First unsuccessful attempt will result in remedial training and a second attempt at the CEC. A second unsuccessful attempt will result in a TRB.
CEC	PO 001 PO 004 PO 005 PO 008	General Naval Knowledge – Part 2	Written	3 periods	References, DWAN computer access	Require DWAN account	70%	First unsuccessful attempt will result in remedial training and a second attempt at the CEC. A second unsuccessful attempt will result in a TRB.

Module 5:								
CEC		OJT Package	Practical	5 training days	Ship at sea	Must be completed in a sea environment.	Complete	Unsuccessful completion of OJT package will result in TRB.

ANNEX I – ON JOB PERFORMANCE RECORD (OJPR)

NWO II

PRACTICAL TRAINING REQUIREMENTS

AFLOAT TRAINING

AIM OF THE AFLOAT TRAINING PHASE

1. The aim of Afloat Training Phase is to introduce new officers to the bridge, basic ship organization, routines as well as mariner and seamanship skills required for operating as a ship's officer and to ensure that necessary training takes place.

2. The PTRs listed below are to be completed as pre-requisites for successful completion of NWO II. Instructions for assessment of the trainee completing the PTRs are contained in the NWO QSP and the NWO II Underway Handbook.

PTR INSTRUCTIONS

3. As the trainee completes each PTR to the required standard the Assessing Officer (AO) shall sign and date the applicable item in the sign off sheet.

4. Should the trainee be unable to complete any part of an enabling objective practically due to the training limitations of the platform, then they shall complete the requirement either verbally or by completion of an assignment / project assigned by the CTO or Training Supervisor. Once the individual has demonstrated a sound understanding of the material then the requirement may be signed off.

5. When an individual has completed all of the training requirements for the NWO II sea phase, then the requirements will be certified complete by signing the certification at the end of the PTR package.

6. Students should be given a copy of the completion certification, a copy is to remain on the Officer's training file, and a copy is to be emailed to +ESQ NPTG HMCS Venture Scheduling & Training Coordination Center@NPTG HMCS Venture@Esquimalt.

PTR SIGNING AUTHORITIES

7. The CTO, seamanship instructor (or chief boatswain's mate) or ship's executive officer will normally sign the PTRs. The following PTR tables will indicate the specific signing authorities. The CTO may authorize additional authorities depending on the staff carried in the ORCA. (Note that when XO is indicated, the XO must be NOPQ qualified). When training is conducted on a HMC ship other than an ORCA the closest equivalent to the signing authorities indicated in the PTR table are to act as signing authority.

8. At sea the CTO is the final authority regarding interpretation of NWO training standards and the completion of PTRs.

2-1 GENERAL				
No.	**Description**	**Authority**	**Date**	**Signature**
001 (2-1-1)	Discuss line handling issues for alongsides and departures	SWK / CBM Or equivalent		

002 (2-1-2)	Complete Know Your Ship Book	CBM / CTO / XO Or equivalent		
003 (2-1-3)	Act as member of ship's emergency organisation	CBM / CTO / XO Or equivalent		
004 (2-1-4)	Identify DC, FF and first aid systems	CBM / OCE / CTO / XO Or equivalent		
2-2 WATCH ON DECK				
005 (2-2-1)	Act as competent helm	OOW / CTO / CBM Or equivalent		**(not req'd for Log / Int O)**
006 (2-2-2)	Act as competent lookout	OOW / CTO / CBM Or equivalent		**(not req'd for Log / Int O)**
007 (2-2-3)	Participate in rescue stations	OOW / CTO / CBM Or equivalent		
2-3 OOW SKILLS (Not req'd for Log O / Int O)				
008 (2-3-1)	Stand as 2OOW	OOW / CTO		**(not req'd for Log / Int O)**
009 (2-3-2)	Con the ship	OOW / CTO		**(not req'd for Log / Int O)**
010 (2-3-3)	Use a pelorus effectively	OOW / CTO		**(not req'd for Log / Int O)**
011 (2-3-4)	Plot a visual fix and DR on a paper chart	OOW / CTO		**(not req'd for Log / Int O)**
012 (2-3-5)	Identify Naval Flags and Pennants	OOW / CTO		**(not req'd for Log / Int O)**
013 (2-3-6)	Prepare ship for night steaming	CTO / XO / OIC / POOW		**(not req'd for Log / Int O)**
014 (2-3-7)	Maintain basic bridge records	CTO / XO / OIC		

				(not req'd for Log / Int O)
015 (2-3-8)	Identify Canadian Aids to Navigation	OOW / CTO / OIC		**(not req'd for Log / Int O)**
2-4 SMALL BOAT OPERATIONS				
016 (2-4-1)	Participate in small boat lowering and recovery	CBM / Deck IC Or equivalent		
017 (2-4-2)	Operate the ship's boat (zodiac)	CBM		**(not req'd for Log / Int O)**
2-5 SEAMANSHIP				
018 (2-5-1)	Locate and explain the use of upper deck fittings	CBM Or equivalent		
019 (2-5-2)	Act as part ship hand	CBM / PartShipIC Or equivalent		
020 (2-5-3)	Competently tie knots, bends and hitches, and identify their uses.	CBM Or equivalent		
2-6 GENERAL SAFETY				
021 (2-6-1)	Describe confined space entry	OCE / DCPO Or equivalent		
022 (2-6-2)	Describe burning and welding procedures	OCE / DCPO Or equivalent		
023 (2-6-3)	Describe going aloft procedures	OCE / RadHazO Or equivalent		

PTR REQUIREMENTS

2-1 GENERAL

PTR 001 (2-1-1)

1. Performance: Discuss line handling issues for alongsides and departures
2. Conditions:
    a. Given:
        (1) a ship entering or leaving harbour;
        (2) Part Ship Hands closed up; and
        (3) supervision;
    b. Denied: references; and
    c. Environmental:
        (1) daylight; and
        (2) good visibility.
3. Standard: The officer shall in accordance with CFCD 105 Fleet Seamanship Rigging and Procedures Manual, the Admiralty Manual of Seamanship, BR 67
    a. Explain the effect on the ship's movement that each line has;
    b. Under supervision give appropriate orders for line handling;
    c. Ensure the safety of the ship and the personnel throughout the evolution; and
    d. Complete a diagram showing arrangement of berthing lines / fenders.

PTR 002 (2-1-2)

1. Performance: Complete the Know Your Ship Book
2. Conditions:
    a. Given: a ship's diagram (Know-Your-Ship Booklet);
        (1) blank diagrams of the engineering compartments (Know-Your-Ship Booklet); and
        (2) references;
    b. Denied: assistance;
    c. Environmental: n/a.
3. Standard: The officer shall:
    a. Identify the compartments in the ship;
    b. State the main components / purpose of each compartment;
    c. Locate and label all Lifesaving and First Aid equipment; and
    d. Explain the operation and use of all such equipment.

e. Draw the main components of the main propulsion plant; and

f. Describe briefly the function and operation of each major component within the lube system, fuel system and cooling system.

PTR 003 (2-1-3)

1. Performance: Act as a Member of the Ship's Emergency Organization.
2. Conditions:
    a. Given:
        (1) A ship at sea or alongside;
        (2) Watch and Station Billboard;
        (3) Ship’s Emergency Team; and
        (4) supervision;
    b. Denied:
        (1) references; and
        (2) assistance;
    c. Environmental: n/a.
3. Standard: The officer shall participate in a simulated fire exercise.

PTR 004 (2-1-4)

1. Performance: Identify Damage Control, Firefighting and First Aid Systems in a Ship.
2. Conditions:
    a. Given:
        (1) blank diagrams of the engineering compartments (Know-Your-Ship Booklet); and
        (2) references;
    b. Denied: n/a; and
    c. Environmental: n/a.
3. Standard: The officer shall:
    a. locate by diagram the risk and control markings;
    b. locate by diagram and identify all first aid firefighting equipment;
    c. locate by diagram and identify and trace the components of the fixed firefighting system;
    d. explain the use and limitations of all firefighting equipment;
    e. draw the ship’s fire main system;
    f. identify the normal fire main pressure range;

g. locate and describe shoring and leak stopping equipment; and

h. sketch the principle components of the power supplies.

2-2 WATCH ON DECK

PTR 005 (2-2-1)

1. Performance: Act as Competent Helmsman.

2. Conditions:

   a. Given:

      (1) a ship at sea; and

      (2) supervision;

   b. Denied: assistance; and

   c. Environmental: in all conditions.

3. Standard: The officer shall:

   a. stand at least 4 rotations as Helmsman;

   b. correctly respond to helm and engine orders;

   c. maintain the ordered course within 5 degrees;

   d. give the proper report and react in accordance with vessel Standard Operating Procedures for a steering gear breakdown; and

   e. without error provide the OOW with a Helmsman turnover report.

PTR 006 (2-2-2)

1. Performance: Act as Competent Lookout.

2. Conditions:

   a. Given:

      (1) a ship at sea; and

      (2) OOW supervision;

   b. Denied: assistance; and

   c. Environmental: in all conditions.

3. Standard: The officer shall:

   a. Stand at least 4 rotations as Lookout:

   b. Without error provide the OOW with Lookout reports for vessels and other contacts (i.e. aircraft, aids to navigation, flotsam and other dangers) to include:

      (1) identifying vessels / aircraft encountered by type or class; and

      (2) identifying aids to navigation by type and colour (i.e. red buoy, white light flashing 5 times every 20 seconds etc.);

(3) without error provide the OOW with a Lookout turnover report for both night and day;

(4) properly use and care for binoculars and other equipment entrusted to the Lookout; and

(5) react to a man overboard as a Lookout.

PTR 007 (2-2-3)

1. Performance: Participate in Rescue Stations.
2. Conditions:
   a. Given:
      (1) a ship at sea;
      (2) supervision; and
      (3) a simulated man overboard;
   b. Denied: assistance; and
   c. Environmental:
      (1) sea state 0-2;
      (2) day or night; and
      (3) good visibility.
3. Standard: The officer shall:
   a. Participate in Rescue Stations as the:
      (1) 2OOW
      (2) Lookout; and
      (3) Lifebuoy Sentry;
   b. As 2OOW, under the supervision of the OOW, con the vessel to recover a simulated man overboard; and
   c. As the Lookout and Lifebuoy Sentry, raise the alarm, launch lifesaving devices*, point to the man in the water, and report to the OOW as required.

*Note: the launching of Marine Markers (Comets) is to be simulated unless otherwise directed by ship's staff.

2-3 OOW SKILLS

PTR 008 (2-3-1)

1. Performance: Stand as 2OOW.
2. Conditions:
   a. Given: a vessel at sea and underway;

b. Denied: nil

c. Environmental:

(1) sea state 0-2;

(2) day and night;

(3) safe water; and

(4) good visibility.

3. Standard: The officer shall:

a. Stand at least 4 rotations as 2OOW:

b. Assist the OOW with:

(1) Conducting rounds prior to a watch, where practical;

(2) Fixing the ship's position, plotting the position on a paper chart and applying a DR to the fix;

(3) Following the navigation plan as promulgated by the vessel OIC;

(4) Order and supervise the launching and recovery of the ship's boat

(5) Administering the ship's routine;

(6) Responding to shipboard emergency exercises and;

(7) Maintaining the OOW notebook and Ship's log IAW CFCD 130.

PTR 009 (2-3-2)

1. Performance: Con a Vessel.

2. Conditions:

a. Given: a vessel at sea and underway;

b. Denied: assistance; and

c. Environmental:

(1) sea state 0-2;

(2) daylight;

(3) safe water; and

(4) good visibility.

3. Standard: The officer shall in accordance with Maritime Command Ship's Standing Orders and class specific Standing Orders con the vessel in a series of at least three turns by:

a. Using correct helm orders and replies;

b. Using correct counter helm; and

c. Steadying the vessel within 5 degrees of the intended course

d. If deployed in Orca Class vessels:

(1) Explain troll and:

(2) Steer using JOG.

PTR 010 (2-3-3)

1. Performance: Use the Pelorus
2. Conditions:
    a. Given: a vessel at sea and underway
        (1) a functioning pelorus;
        (2) denied: assistance; and
    b. Environmental:
        (1) sea state 0-2;
        (2) day or night;
        (3) safe water; and
        (4) good visibility.
3. Standard: The officer shall:
    a. take visual bearings (true and relative) of objects for navigation and contact avoidance within ½ ° of the actual bearing.
    b. Determine bearing rate change of vessel traffic and bearing trend as either drawing left, drawing right or closing on a steady bearing.

PTR 011 (2-3-4)

1. Performance: Fix the Position of the Ship.
2. Conditions:
    a. Given:
        1. references;
        2. assistance;
        3. paper charts;
        4. a parallel ruler;
        5. a speed times distance ruler; and
        6. a set of dividers;
    b. Denied: n/a; and
    c. Environmental: n/a.
3. Standard: The officer shall:
    a. Demonstrate an understanding of chart symbology by identifying fixing points and relating them to charted objects;
    b. Take and plot 10 visual fixes; and

c. Demonstrate an understanding of units of measurement by assisting the OOW in plotting a fix including a dead reckoning (DR) and / or determining distance to go and ranges to objects.

PTR 012 (2-3-5)

1. Performance: Identify Naval Flags and Pennants and Their Meanings.

2. Conditions:

a. Given: naval flag and pennant flashcards;

b. Denied: assistance; and

c. Environmental: n/a.

3. Standard: The officer shall:

a. Recognize any naval flag or pennant within five seconds of observation; and

b. State the correct military and INTERCO meaning for all flags and pennants:

c. State the correct procedure for using flag India for vessels coming alongside other vessels;

d. State the correct procedure for utilizing flag Uniform for coming to anchor;

e. State the correct procedure for using flag Romeo during a RAS; and

f. Identify ORCA class vessels by international call signs.

PTR 013 (2-3-6)

1. Performance: Prepare ship for night steaming.

2. Conditions:

a. Given:

1. ship at sea

b. Denied: nil

c. Environmental: dusk

3. Standard: The officer shall assist the CBM with the darken ship routine and assist the OOW with dimming the bridge equipment and switching on navigation lights.

PTR 014 (2-3-7)

1. Performance: Maintain basic bridge records.

2. Conditions:

a. Given: a vessel at sea and underway

b. Denied: assistance; and

c. Environmental:

1. day and night;

2. safe water; and

3. good visibility.

3. Standard: The officer shall maintain the OOW notebook and the Ship's Log IAW the requirements outlined in CFCD 130.

PTR 015 (2-3-8)

1. Performance: Identify Canadian Aids to Navigation.
2. Conditions:

a. Given:

1. a ship at sea;
2. OOW supervision;
3. Chart or chartlet; and
4. Flash cards

b. Denied: assistance

c. Environmental:

1. sea state 0-2;
2. day and night;
3. safe water; and
4. good visibility

3. Standard:

a. The officer shall by day:

1. Identify the names, colour, top mark and significance of a variety of buoys, day marks and any other aid to navigation. This may be accomplished by actually sighting the object or by flash card and / or quiz.

b. The officer shall by night:

1. Identify the names, light colour, flashing characteristics and significance of a variety of buoys or any other aid to navigation. This may be accomplished by actually sighting the object or by flash card and / or quiz.

2-4 SMALL BOAT OPERATIONS

PTR 016 (2-4-1)

1. Performance: Participate in small boat lowering and recovery.
2. Conditions:

a. Given:

1. a ship's boat at sea or alongside;
2. crane; and
3. supervision.

b. Denied:

1. references; and

2. assistance

c. Environmental: in all conditions.

3. Standard: the officer shall act as a member of the boat's lowering party to launch and recover the ship's boat.

PTR 017 (2-4-2)

1. Performance: Operate the ship's boat.

2. Conditions:

a. Given: a RHIB or Zodiac;

b. Denied:

1. references; and

2. assistance;

c. Environmental:

1. sea state 0-1;

2. daylight; and

3. good visibility.

3. Standard: The officer shall:

a. Act as the Cox'n for the ship's boat to include:

1. receive a departure brief from OOW / OOD and conduct all communications checks;

2. recognize and rectify or report any discrepancies with the ship’s boat;

3. enforce personnel safety and discipline; and

4. observe all marks of respect and ceremonial procedures;

b. Operate the motor and steer the boat to accomplish the following in a safe manner:

1. leave from alongside the vessel;

2. execute a series of tight turns at speed;

3. come alongside a float in a smooth and controlled approach, stopping the boat within half a boat length of a designated target area;

4. come alongside the vessel at the scramble net in a position to easily pass the bow and stern lines; and

5. shut down the motor and secure the boat so as to leave it clean, tidy and ready for use;

c. Apply rules of the road as they pertain to a vessel of that size;

d. Follow a pre-planned route as briefed by the OOW / OOD and identify the boat's position with the aid of a chart / chartlet;

e. Identify and utilize navigational aids;

f. Adhere to all safety and navigational rules and regulations;

g. Act as a Bowsman for the ship's boat; and

h. Describe basic boat maintenance procedures to include inflation, cleaning and patching repairs.

2-5 SEAMANSHIP

PTR 018 (2-5-1)

1. Performance: Locate and Explain the Use of Upper Deck Fittings in a Ship.

2. Conditions:

a. Given: ship diagrams (Know-Your-Ship Booklet)

b. Denied:

1. references; and

2. assistance

c. Environmental: N/A

3. Standard: The officer shall in accordance with CFCD 105 Fleet Seamanship Rigging and Procedures Manual and the Admiralty Manual of Seamanship BR 67:

a. Identify (if fitted) upper deck gear and fittings to include all fairleads, bollards, cleats, eye plates, deck clenches, winches, windlasses and capstans;

b. State the purpose / operation of all upper deck gear and fittings; and

c. Indicate, using the Know-Your-Ship Booklet, the location of all upper deck gear and fittings.

PTR 019 (2-5-2)

1. Performance: Act as a part ship hand.

2. Conditions:

a. Given:

1. a ship; and

2. supervision

b. Denied: references

c. Environmental: in all conditions

3. Standard: The officer shall in accordance with CFCD 105 Fleet Seamanship Rigging and Procedures Manual and the Admiralty Manual of Seamanship BR 67:

a. Perform the duties of a line handler for entering and leaving harbour to include:

1. coiling, cheesing and faking a line;

2. responding correctly to line handling orders;

3. performing ceremonial procedures for entering / leaving harbour; and

4. following all safety practices;

b. Perform the duties of a fender handler for entering / leaving harbour to include:

1. responding correctly to fender handling orders; and
2. following all safety practices;

c. Participate as a Foc'sle Part Ship Hand for:

1. anchoring; and
2. weighing anchor;

d. Describe the standard procedures followed in berthing, slipping, anchoring and weighing anchor to include:

1. naming / numbering conventions for ship's lines;
2. sequence of events;
3. orders and actions taken when they are given;
4. safety precautions;
5. use of flags and shapes; and
6. reports to command.

PTR 020 (2-5-3)

1. Performance: Competently tie knots, bends and hitches, and identify their uses.

2. Conditions:

a. Given: ropes

b. Denied:

1. references; and
2. assistance

c. Environmental: N/A

3. Standard: The officer shall in accordance with CFCD 105 Seamanship Rigging and Procedures Manual:

a. Tie the following knots, bends and hitches:

1. Figure Eight knot;
2. Reef knot;
3. Bowline;
4. Double Sheet Bend;
5. Clove Hitch; and
6. Round Turn and Two Half Hitches

b. State the use of each of the above knots, bends and hitches.

2-6 GENERAL SAFETY (Log O / Int O only)

PTR 021

1. Performance: Describe safety precautions for working in or around confined / void spaces.

2. Conditions:

a. Given:

1. References;

2. Checklists / Forms;

3. Safety equipment; and

4. Supervision.

b. Denied: Nil.

c. Environmental: Ship or alongside

3. Standard: The officer shall describe and follow occupational safety requirements IAW with references when working in or around confined / void spaces:

a. Describe the following with assistance:

1. Definition of void / confined space;

2. Hazards associated with void / confined space;

3. Actions taken prior to entering a void / confined space; and

4. Safety procedures when encountering a casualty in a void / confined space.

PTR 022

1. Performance: Describe safety precautions for burning and welding (hot work).

2. Conditions:

a. Given:

1. References;

2. Checklists / Forms;

3. Safety equipment; and

4. Supervision.

b. Denied: Nil.

c. Environmental: Ship or alongside

3. Standard: The officer shall describe and follow occupational safety requirements IAW with references when managing burning and welding (hot work):

a. Describe the following with assistance:

1. Hazards associated with burning and welding;

2. Fill out the Safe to Work Certificate;

3. Procedures to be followed prior to starting hot work;

4. Duties of a fire sentry; and

5. Equipment and dress required to perform sentry duties.

PTR 023

1. Performance: Describe safety precautions for working aloft.

2. Conditions:

a. Given:

1. References;

2. Checklists / Forms;

3. Safety equipment; and

4. Supervision.

b. Denied: Nil.

c. Environmental: Ship or alongside

3. Standard: The officer shall describe and follow occupational safety requirements IAW with references when working aloft and managing personnel aloft:

a. Describe the following with assistance:

1. Definition of aloft;

2. Hazards associated with work aloft or over the side;

3. Fill out the DND 2145 (Person Aloft Radiate / Antenna Rotate Control Chit);

4. Procedures to be followed prior to starting work aloft or over the side;

a. RADHAZ / MEL

b. HERP

c. Exhaust

d. Boat wake

5. Role of the OOD / OOW in aloft procedures; and

6. Safety Equipment required to perform work aloft.

AFLOAT TRAINING PHASE REQUIREMENTS

COMPLETION OF ALL REQUIREMENTS

Ship: _____

At Sea from: _____ **to:** _____

Student: (Print)

_____	_____	_____
Rank	**Name**	**SN**

I hereby certify that the above mentioned officer has fully met the standard of NWO II practical training and demonstrated the necessary watch on deck and part-ship skills.

Course Training Officer or Training Supervisor: (Print)

_____	_____	_____
Rank	**Name**	**Position**
	_____	
	Email	

Once complete, a scanned copy of this certificate is to be emailed to:
+ESQNPTGHMCSVENTURESchedulingTrainingCoordinationCenter@intern.mil.ca
(+ESQ NPTG HMCS Venture Scheduling & Training Coordination Center@NPTG HMCS Venture@Esquimalt)

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

ANNEX J – TT PLAN

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented. - Realized benefits - e/w Coast Delta
002.03	NPTT/NABS		
002.07	NPTT/NABS		
004.03	NABS/NPTT	NPTT if NABS not available	Realism lost, can be recouped during sea phase OJT
003.01	Rules Master Pro		

ANNEX K – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions, and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Board Composition: SLt Wydysh, Lt(N) Kucher	
Topic	Outcome
Moved teaching point table.	Restructured layout of EO to have the teaching point table below the EO. to improve readability and encourage instructors' adoption of the training plan EO point 4 will state: reference teaching table.
Moved reference table to end of QS	There will be one reference table in the annexes, as opposed to a reference table after every PO. This will improve readability.
EO 001.05	EO 001.05 - Changed name from Understand the Managerial Environment of the RCN to Describe the Managerial Environment of the RCN to make EO a performance. Describe is a performance; understand is not.
EO 002.01 Name change	EO 002.01: Demonstrate a Working Knowledge of the Meteorological Conditions – change title to EO 002.01: Describe the Effects of Meteorological Conditions on Shiphandling.
EO 006.01	Title changed to Describe the Key Components of Marine and Combat Systems Engineering to make EO a performance.
EO 008.01	Title changed to Describe the Action and Emergency Organization.
Amendment to History EO	History EO added a museum visit and a ship tour to improve student engagement with subject matter.
Teaching table format	Teaching point table will be divided by period to encourage instructor use of QSP.
Drill EO	There will be no marching drill in NWO II, with this change, the schedule will make room for instruction on sword drill. There will be 2 periods of refresher drill.
Mental Math	MSI recommends having 6 periods for MM. Three periods of instruction and three periods of application/consolidation to be woven into material. Furthermore, MSI recommends spacing three math periods one day apart to aid student retention. Nautical measurements should be taught before the first MM lesson.
Tides and Currents	New policy and practice have eliminated the need for tabular tide and current calculations. Content originally instructed in NWO II has been moved to NWO III to provide training at the relevant level.
Placement of material	MSI recommends having more core material (ex mental math) in Phase II reduces failure rate in NWO III.
Name change	EO 004.05 – Act as OOW/OOD for Berthing and Slipping – name amended to be inclusive of NTOs.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	X.X
Date:	Who Initiated Change:
Requirement for the Amendment	
Description of Changes:	
Change Tracking Number:	
Trg Mgr:	

ANNEX L – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

Acronyms / Abbreviations	Term / Definitions
2OOW	Second Officer of the Watch
AAW	Anti-Air Warfare
AAWC	Anti-Air Warfare Controller
ACP	Allied Communications Publication
ADC	Air Defence Controller
ADEX	Air Defence Exercise
AIS	Automatic Identification System
AMN	Admiralty Manual of Navigation
AMS	Admiralty Manual of Seamanship
AOPV	Actic and Offshore Patrol Vessel (*Harry DeWolf Class*)
AOR	Auxiliary Oiler Replenishment (*Protecteur Class*)
APP	Allied Procedural Publication
ARPA	Automatic Radar Plotting Aid
ART	Article
ASMD	Anti-Ship Missile Defence
ASuW	Anti-Surface Warfare
ASuWC	Anti-Surface Warfare Controller
ASWC	Anti-Submarine Warfare Commander
ATP	Allied Tactical Publication
ATR	Any Trade Requirement
AWW	Above Water Warfare
AXP	Allied Exercise Publication
BCP	Bridge Control Panel
BLA	Below Limit Approach
BP	Boarding Party
C4ISR	Command, Control, Computers, Communications, Intelligence, Surveillance and Reconnaissance
CANEWS	Canadian Naval Electronic Warfare System
CANTASS	Canadian Towed Array Sonar
CBRN	Chemical Biological Radiation Nuclear
CCA	Curriculum Control Authority
CCG	Canadian Coast Guard
CCS	Command and Control System

CEC	Critical Enabling Check
CFAO	Canadian Forces Administrative Orders
CFCD	Canadian Forces Command Document
CFITES	Canadian Forces Individual Training and Education System
CFMWC	Canadian Forces Maritime Warfare Centre
CGS 100	Communications Gateway System 100
CHP	Canadian Hydrographic Publication
CIO TEAM	Communications Intercept Operations Team
CIPs	Calling In Points
CIWS	Close-in Weapons System
CMG	Course Made Good
CNTG	Commander Naval Training Group
CNS	Canadian Navigation Supplement
CO	Commanding Officer
COLREGS	Collision Regulations
COMEX	Commence Exercise
COMPLAN	Communication Plan
CP	Cable Party
CPA	Closest Point of Approach
CPF	Canadian Patrol Frigate
CRCN	Commander Royal Canadian Navy
CRT	Cathode Ray Tube
CSOs	Captain’s Standing Orders
CTO	Course Training Officer
C-TRAINS	Combat Training
CWC	Composite Warfare Commander
DAMCAT	Damage Category
DC	Damage Control
DNC	Distance to New Course
DNCS	Data Net Control Station
DOTAH	Daily Optask Air Helo
DR	Dead Reckoning
DRP	Davit Rescue Position
DST	Daylight Savings Time
ECM	Electronic Counter Measures
ECPINS	Electronic Chart Precise Integrated Navigation System
EDI	Explain, Demonstrate, Imitate
EMCON	Emission Control
EO	Enabling Objective

EP	Estimated Position
EPM	Electronic Magnetic Pulse Modulation
ESM	Electronic Sensor Measures
ESUS	Electronic Sound Underwater Signal
ETA	Estimated Time of Arrival
ETD	Estimated Time of Departure
EWS	Electronic Warfare Supervisor
EXTORP	Exercise Torpedo
FAF	Final Approach Fix
FBA	Fighter Bomber Attack
FF	Fire Fighting
FFH	Multi-role Patrol Frigate – Helicopter (*Halifax Class*)
FIAC	Fast Insert Attack Craft
FINEX	Finish Exercise
FIXO	Fixing Officer
FLYCO	Flying Controller
FN	Feedback Note
FOC	Furthest On Circle
FOD	Foreign Object Debris
FREQPLANS	Frequency Plans
FTC-SUBSURFACE	Force Track Coordinator Subsurface
FTC-SURFACE	Force Track Coordinator Surface
FTC-AIR	Force Track Coordinator Air
FXO	Fos'cle Officer
GCCS	Global Command and Control System
GDS	Global Distress Signal
GMDSS	Global Maritime Distress and Safety System
GMT	Greenwich Mean Time
GMVLS	Guided Missile Vertical Launch System
GOP	General Operations Plot
GPS	Global Positioning System
GRU	Grid Reporting Unit
HAZMAT	Hazardous Materials
HERF	Hazards of Electro-Magnetic Radiation Fuel
HERO	Hazards of Electro-Magnetic Radiation Ordinance
HERP	Hazards of Electro-Magnetic Radiation Personnel
HF	High Frequency
HIFR	Helicopter In-Flight Re-Fueling
HMCS	His Majesty Canadian Ship

HMS	Hull Mounted Sonar
HOTORP	Hot Torpedo
HVU	High Value Unit
IALA	International Association of Maritime Aids to Navigation and Lighthouse Authority
IAW	In Accordance With
IFR	Instrument Flight Rules
IMC	Instrument Meteorological Conditions
IMD	Information Management Director
INMARSAT	International Marine Satellite
INS	Inertial Navigation System
IRB	Inflatable Rubber Boat
ISE	Independent Ship Exercise
IWD	Information Warfare Director
JBOS	Job Based Officer Specification
JRCC	Joint Rescue Coordination Center
LCP	Local Control Panel
LDL	Limiting Danger Line
LMT	Local Mean Time
MANMAR	Meteorology For Mariners and Marine Weather Observing
MAP	Member Aspiration Profile
MCDV	Maritime Coastal Defence Vessel
MCR	Machinery Control Room
MCTS	Maritime Communication Transport System
MED TECH	Medical Technician
MEL DISTANCES	Maximum Exposure Levels Distances
MET TECH	Meteorological Technician
METOC	Meteorological Operations Centre
MH	Maritime Helicopter
MHz	Mega Hertz
MIO	Maritime Interdiction Operations
MLP	Master Lesson Plan
MM	Mechanical Minesweeper (*Kingston Class*)
MM	Monitor Mass
MMTT	Monitor Mass Training Tracker
MOB	Man Overboard
MOBEX	Man Overboard Exercise
MPRR	Military Personnel Resume Report
N/A	Not Applicable
NABS	Navigation and Bridge Simulator

NAEB	Naval Anti-Escort Beacon
NAVO	Navigating Officer
NAVORDS	Naval Orders
NCM	Non-Commissioned Member
NCPM	Naval Combat Procedural Manual
NOTAM	Notice to Mariners
NTG	Naval Training Group
NPTT	Naval Part Task Trainer
NTO	Naval Technical Officer
NTS	Naval Training System
NUC	Not Under Command
NUCO TABLES	Numerical Code Tables
NWO	Naval Warfare Officer
OCS	Officer Conducting Serial
ODS	Ozone Depleting Substances
OGS	Officer General Specifications
OOW	Officer of the Watch
OPGEN	Operations General
OPORDS	Operational Orders
OPS	Operations
OPS ROOM	Operations Room
OPSTAT	Operational Status
OPTASKS	Operation Tasking
ORO	Operations Room Officer
ORS	Operations Room Supervisor
OSC	On-Scene Commander
OTC	Officer In Tactical Command
PaCE	Performance and Competency Evaluation
PACE	Pocket Accelerated Crypto Equipment
PAR	Performance Appraisal Report
PC	Performance Check
PLQ	Professional Leadership Qualities
PO	Performance Objective
POOW	Petty Officer of the Watch
PPI	Planned Position Indicator
PRC	Portable Radio Communications
PRF	Pulse Repetition Frequency
PSH	Part Ship Hands
PSO	Part Ship Officer

PSR	Predicted Sonar Range
PT	Part Task
PTL	Part Task Lesson
PTR	Practical Training Requirements
QR	Quick Reaction
QRS	Quick Reaction Setting
QSP	Qualification Standard and Plan
RADAR EPM	Radar Electronic Protection Measures
RADHAZ	Radiation Hazards
RAMSES	Reprogrammable Advanced Multimode Shipboard ECM System
RAS	Replenishment At Sea
RCC	Rescue Coordination Centre
RCN	Royal Canadian Navy
RCP	Remote Control Panel
RCS	Radar Cross Section
RelVel	Relative Velocity
RHIB / RIB	Rigid Hull Inflatable Boat
RIC	Radiocommunication Information Circulars
RMP	Recognized Maritime Picture
ROs	Routine Orders
ROE	Rules of Engagement
RoR	Rules of the Road
RSI	Radiation Status Indicator
RSP	Recognized Surface Picture
RV	Rendezvous
SAC	Shipborne Air Controller
SAG	Surface Action Group
SAR	Search And Rescue
SATCOM	Satellite Communications
SAU	Search And Attack Unit
SCS	Sonar Control Supervisor
SDM	Shinnads Dual Mode
SEOSS	Shipboard Electro-Optical Surveillance System
SHINCOM	Shipboard Integrated Interior Communication System
SHINNADS DUAL M	Shipboard Integrated Navigation and Display System Dual Mode
SMG	Speed Made Good
SOA	Speed Of Advance
SR	Speed Required
SRO	Speed Rung On

SSBN	Submarine Ballistic Nuclear
SSC	Coastal Submarine
SSD	Special Sea Duty
SSGN	Nuclear Powered Guided Missile Submarine
SSK	Diesel Attack Submarine (*Victoria Class*)
SSN	Nuclear Attack Submarine
SSOs	Ship Standing Orders
STD	Speed / Time / Distance
SUS	Submarine Underwater Signal
SWC	Sensor & Weapons Controller
TACCO	Tactical Air Coordinator
TCM	Torpedo Counter Measures
TCPA	Time To Closest Point of Approach
TDA	Torpedo Danger Area
TDZ	Torpedo Danger Zone
TG	Task Group
TP	Teaching Point
TRACK SUP	Track Supervisor
TRB	Training Review Board
TRU	Target Reporting Unit
TSR	Tactical Sonar Range
UTC	Coordinated Universal Time
USCG	United States Coast Guard
USN	United States Navy
UWT	Underwater Telephone
UWW	Under Water Warfare
VERTREP	Vertical Replenishment
VFR	Visual Flight Rules
VHF	Very High Frequency
VLS	Vertical Launch System
VMC	Visual Meteorological Conditions
VOI	Vessel Of Interest
VTMS	Vessel Traffic Management System
VTS	Vessel Traffic Services
W/O	Wheelover
WDO	Weapons Directing Officer
WRT	With Respect To

ANNEX M – ACADEMIC CAUTION

(To be retained in Training File)

Issued To:		
Rank:	Name:	SN:
Course:	Division:	
Issued By:		
Rank:	Name:	SN:

Level One Academic Caution:

IAW the NWO II QSP, chapter 3 para 12, you are hereby issued a Level One Academic Caution for failure of CEC _____.

You will be given the opportunity to re-write this CEC. If you pass the re-write no further administrative action will be required.

If you fail the re-write, a Training Review Board IAW the NWO II QSP, chapter 3 para(s) 10, 16-17 will be convened to determine your suitability for further training and employment in the CAF.

Issuer's Signature:_____Date: _____

I HAVE BEEN DEBRIEFED WITH RESPECT TO THE REASONS FOR FAILURE AND UNDERSTAND THAT I WILL BE GIVEN AN EXAM REVIEW AND RE-WRITE. I HAVE READ AND UNDERSTAND THE CONSEQUENCES OF A SUBSEQUENT FAILURE.

Trainee's Signature:_____Date: _____

Level Two Academic Caution:

IAW the NWO II QSP, chapter 3 para 12, you are hereby issued a Level Two Academic Caution for failure of a re-write of CEC_____.

Pursuant to your Level One Academic Caution issued on_____, a Training Review Board IAW the NWO II QSP, chapter 3 para(s) 10, 16-17 will be convened to determine your suitability for further training and employment in the CAF.

Issuer's Signature:_____Date: _____

I UNDERSTAND THAT I AM RECEIVING A LEVEL TWO ACADEMIC CAUTION DUE TO _____. I HAVE BEEN DEBRIEFED WITH RESPECT TO THE REASONS FOR FAILURE AND UNDERSTAND THAT A TRAINING REVIEW BOARD IAW THE NWO II QSP CHAPTER 3 PARA 10 WILL BE CONVENED TO DETERMINE MY SUITABILITY FOR FURTHER TRAINING AND EMPLOYMENT IN THE CAF.

Trainee's Signature:_____Date: _____

ANNEX N – PERFORMANCE REGRESSION

(To be retained in Training File)

Issued To:		
Rank:	Name:	SN:
Course:	Division:	

IAW the NWO II QSP, chapter 3 para 13, you are hereby issued a (check one of the following):

□ Initial Performance Regression □ 2nd Performance Regression □ 3rd Performance Regression

Reason for Issue: (narrative in detail below)

Failure to rectify the performance deficiency outlined above will result in either a:

- Second Performance Regression; or
- Third Performance Regression and referral to a Training Review Board IAW the NWO II QSP, chapter 3 para 10 to determine your suitability for further training and employment in the CAF.

_____ Date:

Course Training Officer

_____ Date:

Trainee

ANNEX O – PROFESSIONAL LEADERSHIP QUALITIES (PLQ) CHIT

(To be retained in Training File)

Part A – Reason for Assessment		
Issued To:		
Rank:	Name:	SN:
Course:	Division:	
Negative Assessment	Positive Assessment	
Issued By:		
Rank:	Name:	SN:
Reason for Issue: (narrative in detail, including the specific qualities displayed or transgressed)		
Issuer's Signature:_____Date: _____ Witness's Signature:_____Date: _____		
Note: Part B (on reverse side of form) is to be completed for negative PLQ Assessments		

- As a minimum, trainees are evaluated on the following PLQs:
  - Honesty, Integrity, Loyalty, Obedience, Courage, Responsibility, Fairness, Dress, Conduct (Deportment and Dedication (Diligence)

Part B – Record of Counselling
IAW the NWO II QSP, chapter 3 para 14: • one negative PLQ Assessment automatically constitutes a verbal warning; • a second negative PLQ Assessment automatically constitutes a formal caution (note a second negative PLQ Assessment for the same deficiency may result in an IC; and • a third negative PLQ Assessment will result in remedial measures (initial counselling).
The issuance of this assessment constitutes a: □ 1st Negative Assessment □ 2nd Negative Assessment □ 3rd Negative Assessment
I have been informed of the reason for my negative PLQ Assessment and counselled with respect to areas for improvement and understand the consequences of further assessments as outlines above. Trainee's Signature: _____ Date: _____

Note: The following table illustrates the PLQs that are to be observed and assessed. Trainees at VENTURE are expected to develop and display, as a minimum, the following qualities:

Honesty	Implies the refusal to lie, steal, defraud or deceive under any circumstances. An honest person recognizes and adheres to virtues such as truthfulness, candor, respect for the possessions of others, sincerity, and fairness. There are no degrees of honesty, either a person is honest or is not.
Courage	We face challenges, whether physical or moral, with determination and strength of character.
Integrity	Integrity could be more accurately described as moral courage. This means simply having the courage of your convictions, the fortitude to call things as you see them (inside the proper bounds of respect for seniors and subordinates) and the readiness toadmit a mistake.
Responsibility	The person accepts responsibility and consequences for his/her own actions.Disinclination to assume responsibility is an indication of lack of integrity.
Loyalty	The "quality, state, or instance of being faithful or faithful adherence to a person, government, cause or duty". Servicemen and women dedicate themselves to Canada. The loyal officer supports the chain of command; this means wholeheartedly serving the chain of command efficiently and well, and in no way undermining the superior's authority or prestige by any word or action. Lastly, the officer should always give the best possible service to the ship and motivates and supports the team.
Obedience	Getting on with the job when given clear direction. Outright refusal is clearly not acceptable. Consider also that reluctance, the "oh, all right I'll do it if I have to" attitude is inappropriate for the leader in the Royal Canadian Navy or Canadian Armed Forces
Fairness	In addition, the good officer is always supportive of subordinates; having consideration for the welfare of juniors and a willingness to look out for their legitimate interests. A leader cannot afford to play favourites, but must be eminently scrupulous and fair in dealing with subordinates, in making awards and assigning punishments and in grantingfavourable assignments and dispensing onerous ones.
Diligence	Leaders undertake all tasks with dedication and perseverance and recognize their dutyto perform with competence and to strive for excellence.

(Dedication)	
Dress	Uniform kept up in a smart fashion to reflect the professional status of officers and bring credit to the Royal Canadian Navy and Canadian Armed Forces.
Deportment (Conduct)	Smart and alert in bearing, discrete in his or her actions and words. Appropriate behaviour inside the mess. Appropriate behaviour with superiors and subordinates onboard and ashore; knows the difference between friendliness and familiarity. A leader develops good "word of command" and has the drive and enthusiasm to take charge, control, direct and inspire their team. A leader who cannot develop a commandpresence will soon find that the members of their team will look for someone else who can.

NWO II

NABS HANDBOOK

MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

TABLE OF CONTENTS

1. NWO II NABS Phase Overview
2. NWO II NABS Gridlock Brief
3. Outline of Buoy Slalom

NWO II – NABS Overview

Day 1 Forenoon

Ship handling

1. DEMO –Turn at rest
   - 4 x BRIDGES independent of each other
2. Buoy slalom
   - 4 x BRIDGES independent of each other

Afternoon

Navigation Basics - Beaver Pt to Patey Rk

1. CTO DEMO –Take a bearing of a light, edge of land and vessel and taking/plotting a fix
   - 1 X BRIDGE (preferably ALPHA)
2. Beaver Pt to Patey Rk
   - 4 X BRIDGES independent of each other. No interacting contacts.

Day 2 Forenoon

Ship handling

1. CTO DEMO Smoke approach
   - 1 X BRIDGE (preferably ALPHA)
2. Smoke approaches
   - 4 x BRIDGES 1nm separation from each other

Conga Lines

1. Cubicles to be linked, students to follow guide during turns/corpens
   - 4 x BRIDGES in same exercise

Afternoon

Navigation Basics - Howe Sound - E Ekins Pt (RamilliesChannel)SB to Lookout Pt by Horseshoe Bay

1. E Ekins Pt to Lookout Pt
   - 4 x BRIDGES independent of each other

NWO II NABS CTO / MENTOR GRIDLOCK BRIEF

Day 1 – Forenoon

Division:	Lead CTO:	Number of Students:
Objectives	To practice manoeuvring ships with main engines To develop student confidence in conning	
Database	CPAC A1	
Exercise	NWO II Day 1 Buoy Slalom templates	
Route	Course 000 Spd 0	
Bridges	4 cubicles - separated	
Manning	1 NABS operators	
Ship Type	ORCA	
Environmentals	Day, good visibility	
Contacts	None	
Errors	None induced	
Intent	**Scenario 1: Turn at Rest (2hrs)** **Start Position: South of Buoy Slalom** • Demonstrate manoeuvring at rest to student in cubicle 1 • Students commence 10 minute runs manoeuvring ships with engines **Scenario 2: Buoy Slalom (2hrs)** **Start Position:** (Buoy Slalom Scenario 1-5) • Ships to navigate buoy slalom based on types of buoys • Students conduct turns at North end of run • Run ends when students exit buoy slalom • Each student to have one 20 minute run; operator manually move to start of next run	
Notes		

NWO II NABS CTO / MENTOR GRIDLOCK BRIEF

Day 1 – Afternoon

Division:	Lead CTO:	Number of Students:
Objectives	To practice taking bearings of contacts, navigation markers, edges of land To develop basic understanding of fixing	
Database	CPAC A1	
Area/Exercise	Satellite Channel/NWO II Satellite Channel Template	
Route	0.5nm SE Beaver Pt (Stationary at start) to Patey Rock	
Bridges	4 cubicles – separated	
Manning	1 NABS operators	
Ship Type	ORCA	
Environmentals	Day, good visibility	
Contacts	No contacts requiring manoeuvring, but contacts available to take bearings of	
Errors	None induced	
Intent	**Scenario 1: Basic Navigation Start Position: Beaver Pt** • to demonstrate taking and plotting a fix • Students practice fixing while ship transits from Beaver Pt to Patey Rk • Each run to be NMT 20 minutes	
Notes		

NWO II NABS CTO / MENTOR GRIDLOCK BRIEF

Day 2 – Forenoon

Division:	Lead CTO:	Number of Students:
Objectives	To practice manoeuvring ships with main engines and rudder To develop student confidence in conning	
Database	CPAC A1	
Area/Exercise	Strait of Georgia/NWO II Day 2 Conga line template	
Route	Course 000 Spd 0	
Bridges	4 cubicles - not separated	
Manning	1 NABS operators	
Ship Type	ORCA	
Environmentals	Day, good visibility	
Contacts	None	
Errors	None induced	
Intent	**Scenario 1: Smoke Approaches (2hrs)** **Start Position: Georgia Strait - 1 nm btw ships to conduct approaches** • Demonstrate smoke approaches • Students commence 10 minute runs **Scenario 2: Conga Lines (2hrs) Start Position: Georgia Strait** • Guide/signals controlled by control. Ships formed in a column astern of guide at 500x spacing. Guide crse 310 sp 9 • Students to follow guide conducting corpens/turns as able during 20 minute runs	
Notes		

NWO II NABS CTO / MENTOR GRIDLOCK BRIEF

Day 2 – Afternoon

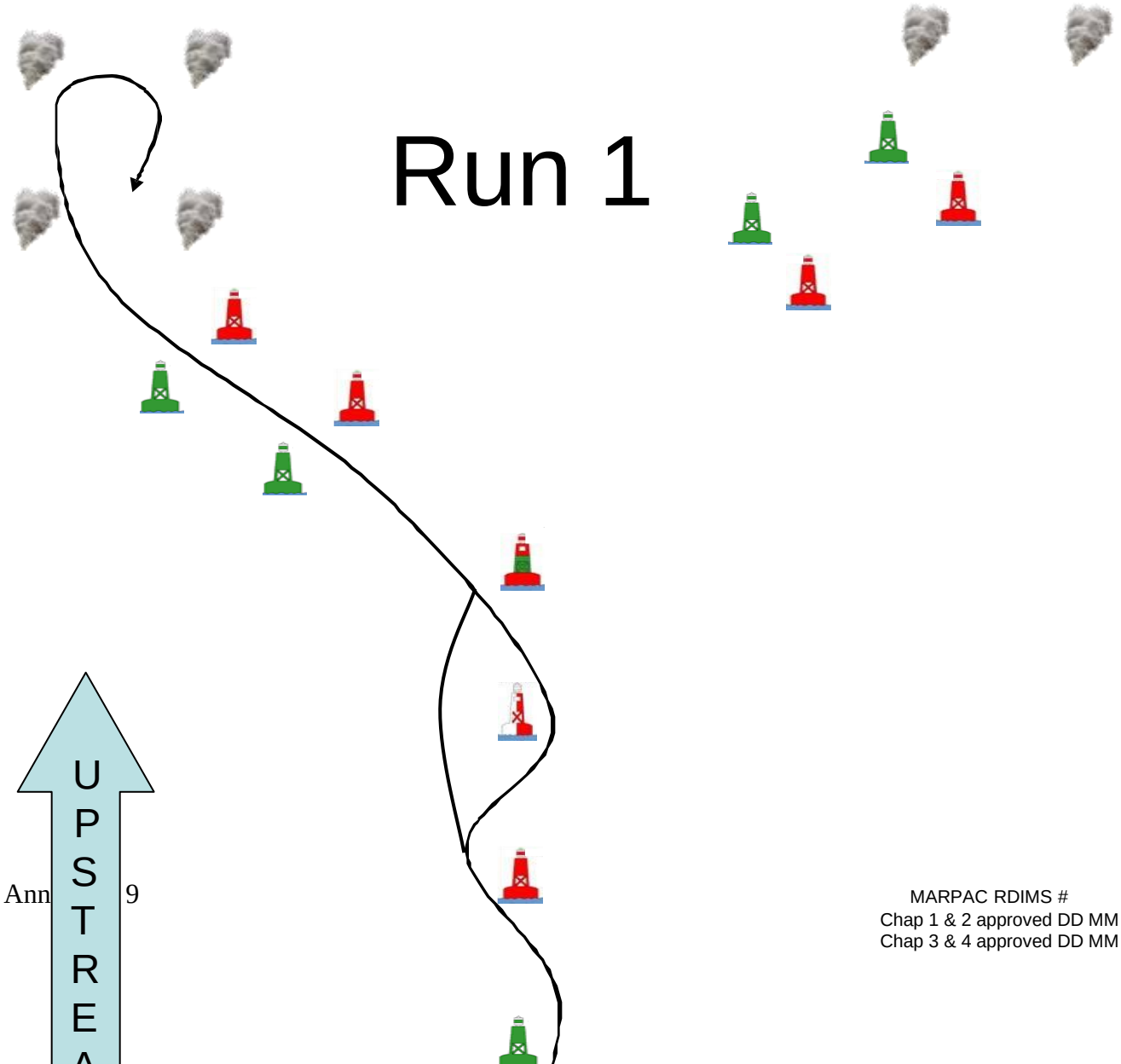
Division:	Lead CTO:	Number of Students:
Objectives	To practice taking bearings of contacts, navigation markers, edges of land To develop basic understanding of fixing	
Database	CPAC A1	
Area/Exercise	Howe Sound/ Exercise NWO II Day 2 Howe sound Template	
Route	E Ekins Pt (Ramillies Channel) SB to Lookout Pt by Horseshoe Bay	
Bridges	4 cubicles – separated	
Manning	1 NABS operators	
Ship Type	ORCA	
Environmentals	Day, good visibility	
Contacts	No contacts requiring manoeuvring, but contacts available to take bearings of	
Errors	None induced	
Intent	**Scenario 1: Basic Navigation** • Students to practice fixing while ship transits pre-planned route • Each run to be NMT 30 minutes	
Notes		

Nabs Buoy Slalom

Day 1

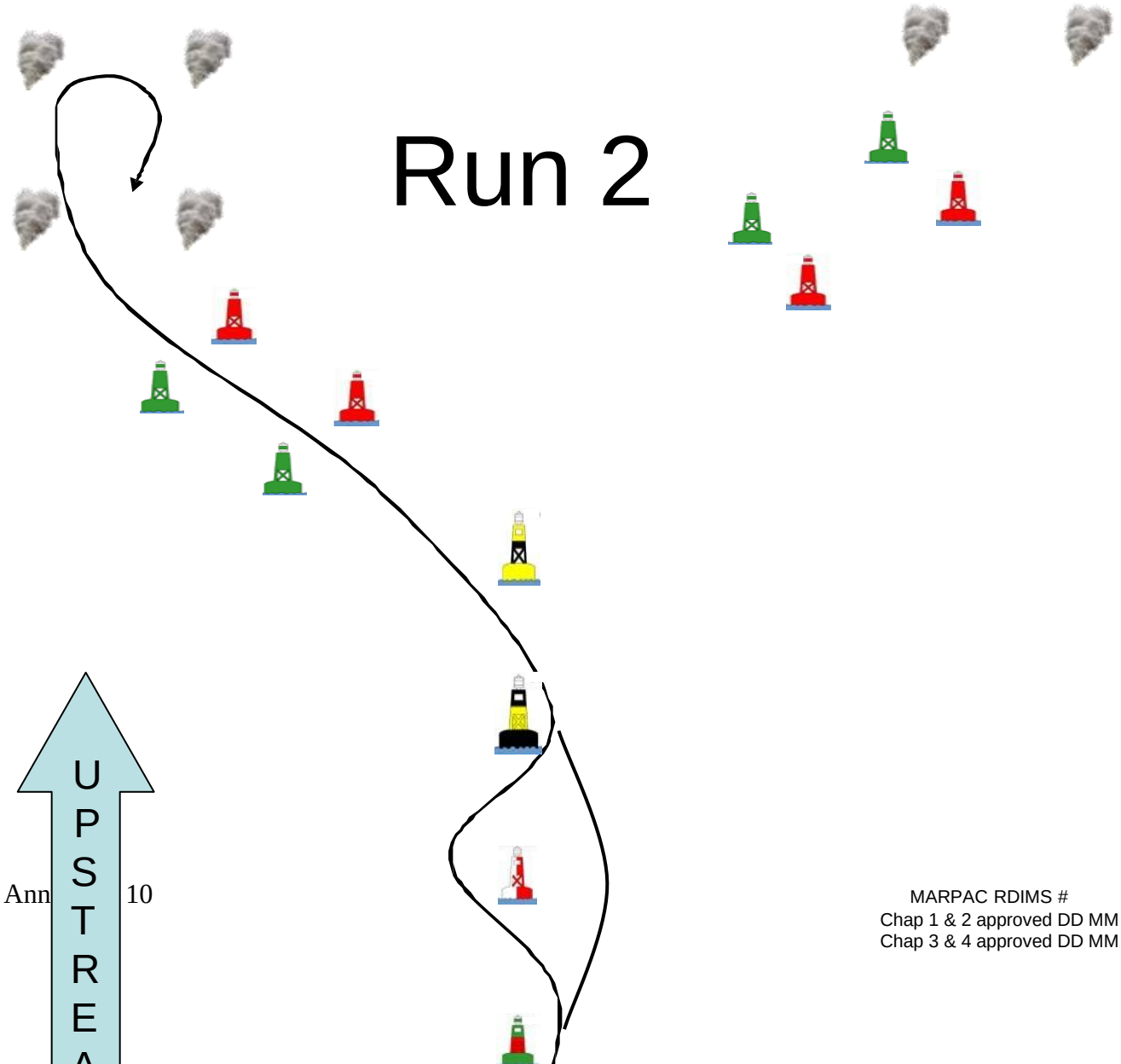
MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

Run 1



MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

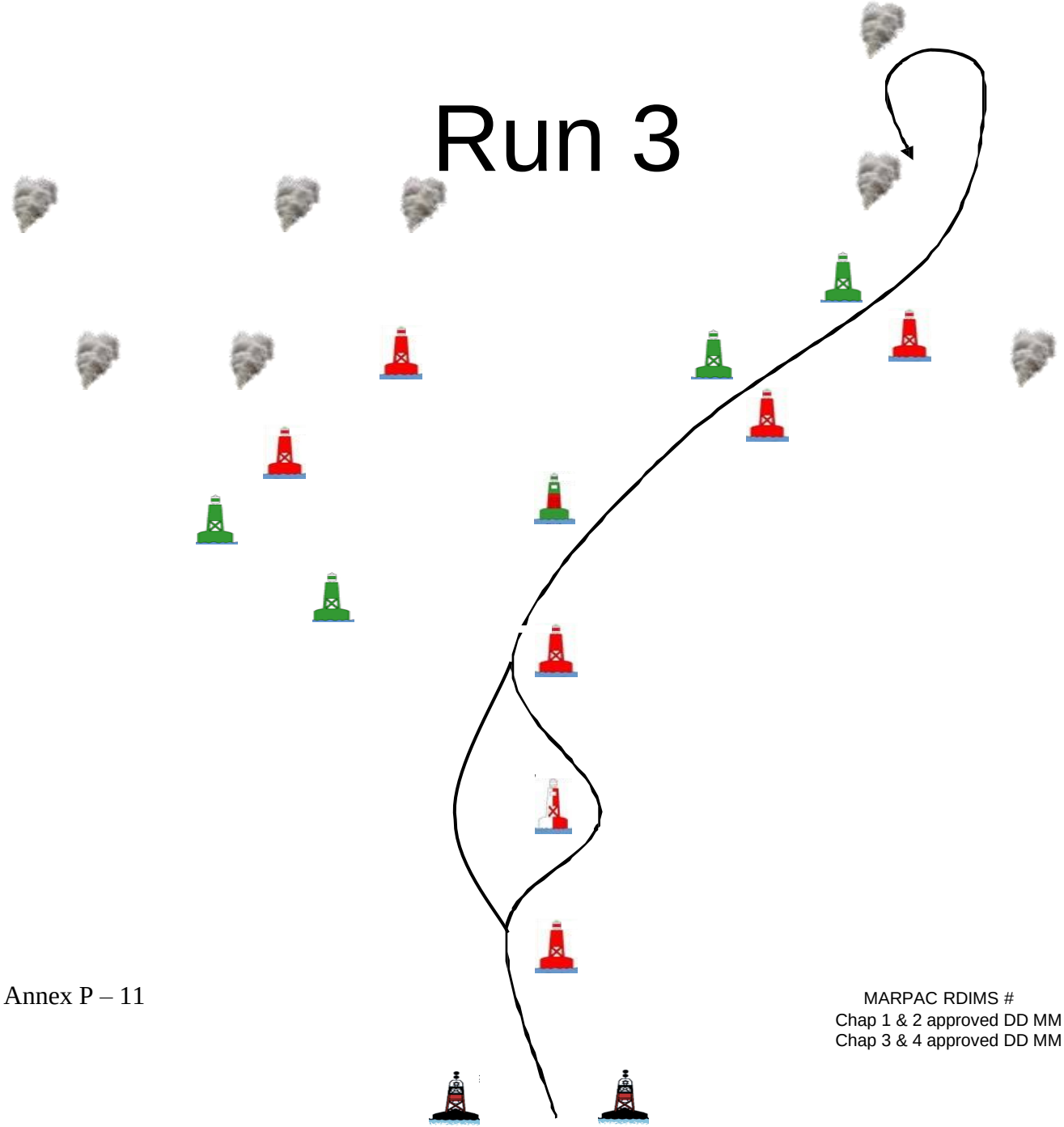
Run 2

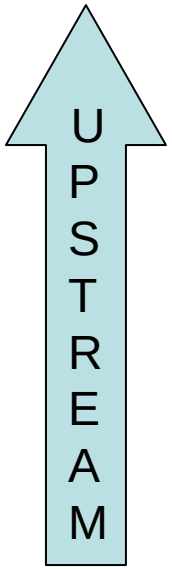


Ann 10

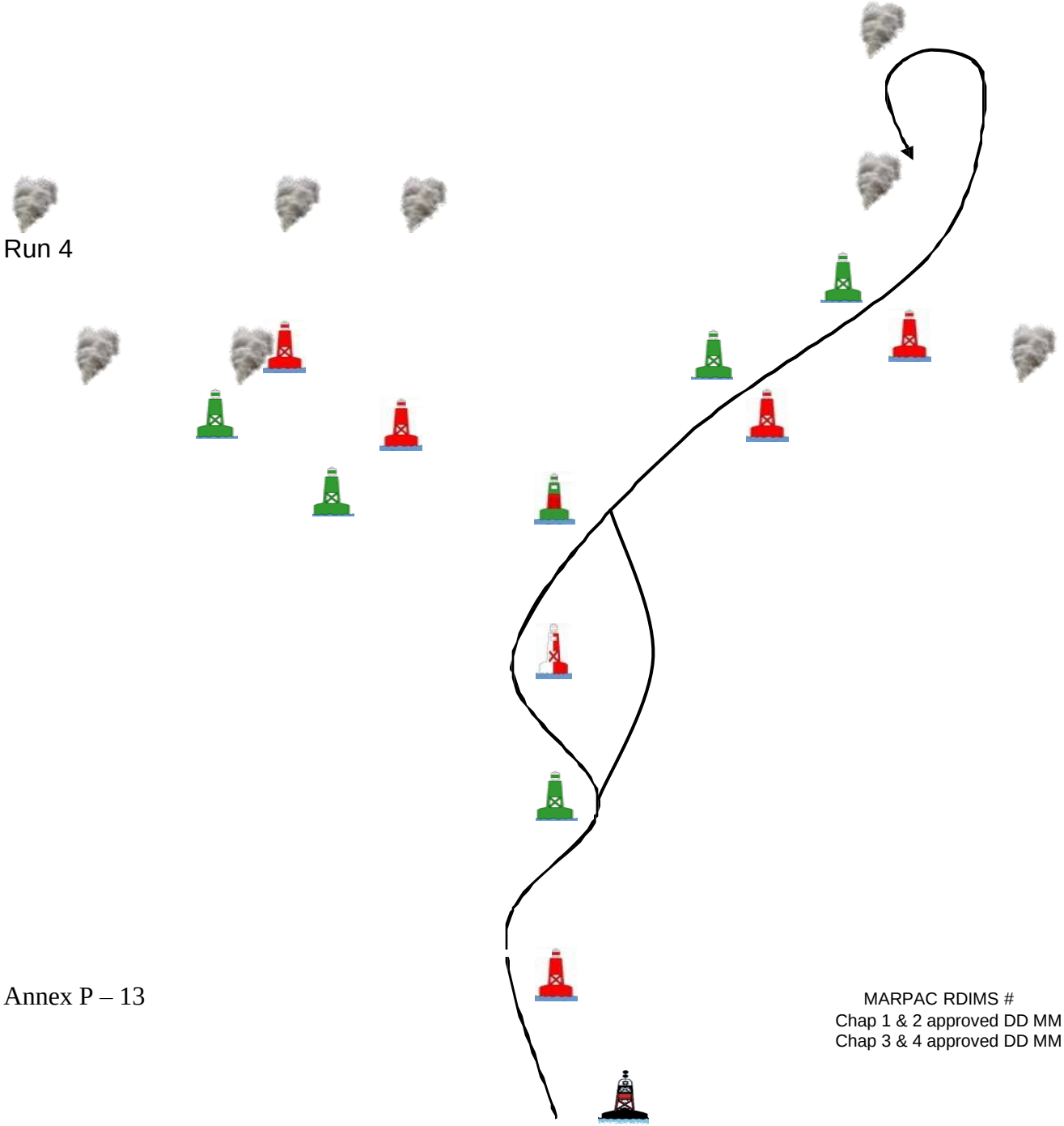
MARPAC RDIMS #
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

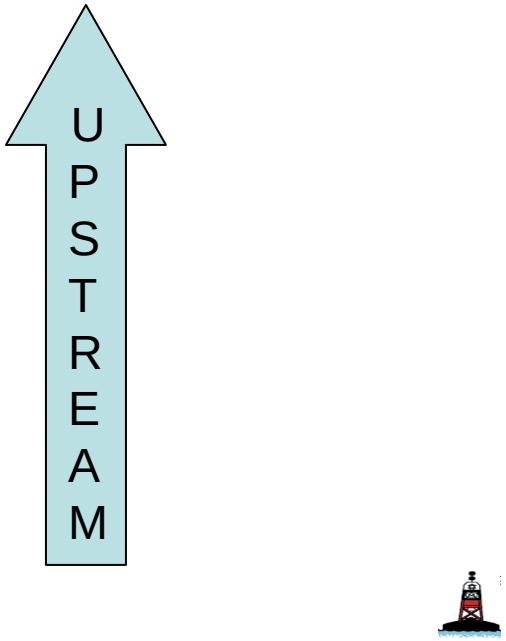
Run 3

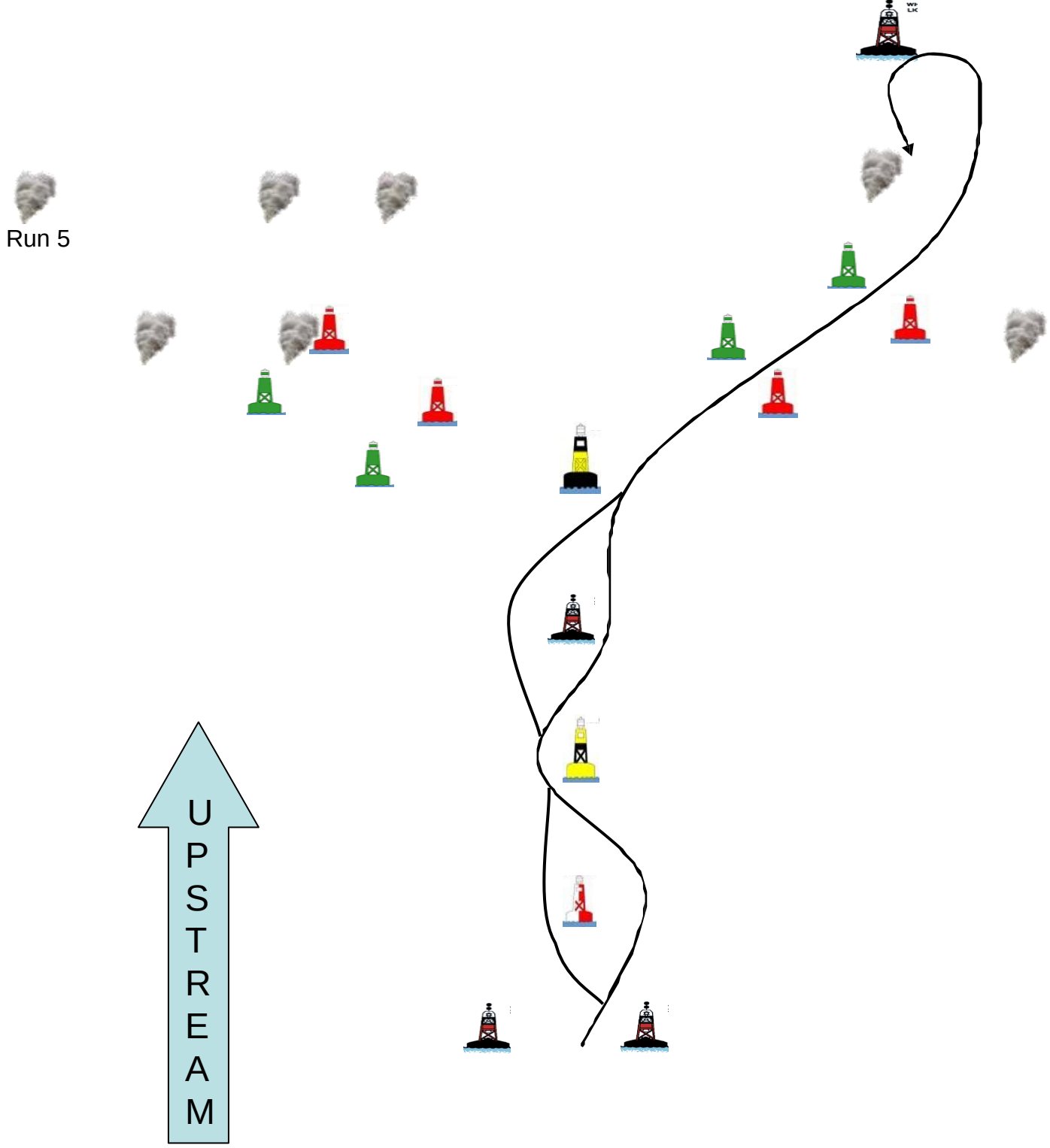



UPSTREAM

Run 4




UPSTREAM


Run 5
UPSTREAM